"It Takes Two to Tango": Joint Location Choices and Local Shocks*

Rosario Didier

University of Michigan

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Abstract

Most adults in the US live in couples, so decisions about where to live and whether to work are rarely made alone. I examine how joint decision-making shapes adjustment to local shocks and who bears them. Combining geocoded data from the Panel Study of Income Dynamics with individual-level measures of exposure to the China Trade Shock, I find that workers in exposed industries are more likely to leave their commuting zones than their non-exposed neighbors. Couples, however, move only when the husband is exposed. Wives, in turn, are more likely to participate in the labor force when their husbands are exposed and less likely when they are exposed themselves. Couples thus adjust through two margins: migration and wives' labor supply. I then extend a dynamic spatial and trade equilibrium model to include couples who jointly choose where to live, each spouse's sector, and whether the wife works. In the model, the shock raises aggregate employment and welfare in the long run. Compared with an economy of singles, couples change how the economy adjusts to it and who bears it, though not its size or location. First, workers respond less to the shock because wives decide with their husbands: without couples, the employment response would be 18% larger, and the migration response of exposed workers 17% larger. Second, all couples adjust the wife's labor supply; without this margin, the economy as a whole gains less. Third, couples move and lose together: without couples, 13% fewer people would relocate across state lines, and when the wife is exposed, both spouses would be better off as singles.

Keywords: Spatial equilibrium, Local labor markets, Household labor supply
JEL Codes: F16, R13, J12

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Contact: rosariod@umich.edu

I. Introduction

For most adults, the response to a local shock is a joint decision. According to the 2020 Census, roughly two-thirds of US households consisted of couples (Anderson et al., 2024). Unlike single individuals, couples must weigh two labor-market exposures against each other: a move that pays off for one spouse may not pay off for the other, making couples less likely to relocate in pursuit of better opportunities. At the same time, couples have an insurance margin that singles lack: the added-worker effect (Lundberg, 1985; Attanasio et al., 2005). When one spouse loses a job or earnings, the other spouse, usually the wife, may increase her labor supply to offset the household's lost income. In the presence of local shocks, therefore, couples have two margins of adjustment: they can move, although they are less mobile than singles, but when they do, they do it together; and they can adjust labor supply within the household, which singles cannot. How couples use these two margins shapes not only how the economy adjusts to a local shock, but also who bears its effects.

Yet spatial models of local shocks abstract from joint decision-making and its implications, because they assume that each worker decides alone. A substantial body of theoretical literature suggests that local labor markets should adjust to local economic shocks through worker mobility (Allen and Arkolakis, 2025), but empirical evidence generally indicates that workers migrate less in response to such shocks than the theory predicts (Autor et al., 2013; Dix-Carneiro and Kovak, 2017; Borusyak et al., 2026). While migration costs are the leading explanation for this disconnect (Allen and Arkolakis, 2025; Desmet and Parro, 2025), another potential factor remains underexplored: most workers do not make these decisions alone. If couples move less but insure more within the household, a model without couples may miss both how mobile workers are and who bears the cost of the shock.

In this paper, I examine how joint decision-making shapes adjustment to local shocks and who bears them, and I quantify the relative importance of the two margins through which couples adjust: migration and wives' labor supply. To do so, I combine new reduced-form evidence on how couples respond to local shocks in the US with a dynamic spatial and trade model in which couples decide together where to live and whether the wife works. My main finding is that couples change how the economy adjusts to a local shock and who bears it, even though they do not change its size or where it lands.

The first part of the paper documents novel empirical facts on how the China Trade Shock affected employment, out-migration, and within-household responses. I use geocoded data from the Panel Study of Income Dynamics (PSID), which tracks each household's composition, labor-market outcomes, and migration across commuting zones over time. The shock is the increase in import competition faced by US manufacturing after China joined

the WTO in 2001 (Autor et al., 2013, 2021), and I estimate its effects using a difference-in-differences design that compares exposed and non-exposed households living in the same commuting zone. Unlike previous work, which measured the shock at the commuting zone level, I measure exposure at the individual level and classify an individual as exposed if they were employed in 1999 in industries adversely affected by Chinese import competition between 2000 and 2012. This individual-level measure also allows me to distinguish couples in which the husband is exposed from couples in which the wife is.

Based on this design, I establish three empirical facts. First, within the same commuting zone, exposed individuals were 6.3 p.p. less likely to be employed during the shock and, on average, 4.7 p.p. more likely to leave their commuting zone than non-exposed individuals. Second, couples, who already migrate less than singles, move in response to the shock only when the husband is exposed: their out-migration rises by 5 p.p. when the husband is exposed, but does not change when the wife is exposed. Third, wives' labor force participation also responds asymmetrically: in relative terms, it rises by 5.4% when their husbands are exposed and falls by 14% when they themselves are exposed, with no offsetting response in husbands' labor supply. Together, the three facts suggest that couples mitigate the shock through a mix of spatial insurance (migration) and intra-household insurance (the added-worker effect), and that which margin they use depends on which spouse is hit.

To quantify what joint decision-making implies for the economy as a whole and the relative importance of migration and wives' labor supply, I extend an existing quantitative dynamic spatial and trade equilibrium model (Caliendo et al., 2019) by adding couples who jointly choose where to live and whether the wife works. Households consist of single men, single women, or couples and choose among location-sector bundles each period, subject to migration frictions and idiosyncratic preference shocks. Because location is the key choice couples must make jointly, I follow Rodríguez-Clare et al. (2026) and nest the preference shocks, allowing households to substitute across sectors within a location at a different rate than across locations. On the production side, each region contains multiple sectors that trade with one another, and firms combine male and female labor using a CES technology with sector-specific gender intensities. Thus, shocks can affect men and women differently, even within the same household.

In a couple, the decision of whether the wife works in the market or remains at home depends on the spouses' relative wages and a home-production productivity draw, after choosing the location and the spouses' sectors. The model thus embeds an endogenous participation margin and the economies of scale from marriage within a dynamic spatial and trade model. To account for the economies of scale of marriage and to compare welfare across household types, I develop a welfare measure in money-metric consumption-equivalent terms,

following Chiappori and Meghir (2015). As in Caliendo et al. (2019) (hereafter CDP), I solve the model using Dynamic Hat Algebra, which expresses the equilibrium without requiring the recovery of the levels of fundamentals, such as mobility costs.

To bring the model to the data, I align the quantitative environment with CDP: 50 US states, 37 foreign trade partners including China, 22 sectors, and non-employment, at a quarterly frequency for 200 periods starting in 2000Q1. I take the dispersion of preferences across sectors from the literature and, following Rodríguez-Clare et al. (2026), calibrate the dispersion across locations by indirect inference, so that the model reproduces the population response to import competition estimated by Autor et al. (2013). Of the four parameters that govern the couple's problem, I calibrate the wife's weight in the household objective and the public-good expenditure share using consumption expenditure data the shape of the home-production shock using the third empirical fact, and its lower bound to match married women's participation in 2000Q1. The migration facts are not targeted, and the model reproduces them.

In the quantitative exercise, the China Trade Shock reduces manufacturing employment but is, in the aggregate, a boom for US workers. In the long run, the decline in manufacturing jobs per capita is close to CDP's estimate (-0.21 vs. -0.23 p.p.), so adding couples changes neither where the shock lands nor its magnitude. Cheaper Chinese goods lower the consumption price index and increase the real return to market work relative to non-employment, and services absorb more workers than manufacturing sheds; consequently, the employment rate rises for everyone. This aggregate picture, however, masks substantial heterogeneity across household types. The decline in manufacturing employment is about twice as large for men as for women, and married men experience the largest decline in their manufacturing share of employment (-0.41 p.p., compared with -0.29 p.p. for single men) but the smallest employment gain, as they adjust mostly by switching sectors.

The average individual gains 0.18% in money-metric consumption-equivalent terms, and these gains differ sharply across household types. Single women gain the most (34% more than the average individual) because they are concentrated in expanding sectors and are free to move; single men gain the least (10% less than the average individual), and couples fall in between, closer to single men. Among couples, gains decline monotonically with exposure, and the husband's exposure determines their order: relative to couples in which neither spouse starts in manufacturing, gains are 10% lower when only the wife does, 22% lower when only the husband does, and 30% lower when both do. When the husband is exposed, the shock hits the household's main source of income; when only the wife is exposed, the couple can cushion part of her loss because she can leave the market and work at home.

Comparing the long-run results with those from an all-singles economy, in which I split

every couple into two singles and recalibrate the migration elasticity, and from economies in which I shut down migration, partially shut down wives' participation response, or both shows three reasons why couples matter for how the economy adjusts. First, workers respond less to the shock than they would in a model without couples. The two economies agree on the magnitude and location of the shock: manufacturing declines differ by less than 1%, aggregate welfare gains by about 2%, and state-level welfare gains correlate at 0.96. But the model without couples overestimates the shock's employment gains by 18%, driven by women's employment, and the migration response of exposed workers by 17%. This gap is one answer to the puzzle that workers migrate less than the theory predicts: in a model without couples, each worker decides alone, so everyone who moves because of the shock is an exposed worker, whereas with couples, exposed workers sometimes stay because their spouses are not exposed. A wife decides whether to work together with her husband, and, in the long run, the shock that raises her wage also raises his, making the household less reliant on her earnings. As a result, without couples, female labor supply would be 51% more elastic. Couples thus also change the gains of singles, because women's wages absorb more of the shock: without couples, single women would gain 17% less and single men 6% more.

Second, wives' participation insure every couple, regardless of exposure, and without it the economy as a whole gains less. Because couples decide jointly, a wife's labor supply responds not only to her own exposure to the shock but also to her husband's. Wives in every type of couple work more, because the shock raises women's wages relative to men's everywhere. Without this margin, the economy as a whole gains less: partially shutting down wives' participation response makes aggregate welfare gains 5% smaller; every type of couple gains less, including those in which neither spouse was exposed, and welfare gains fall in every state. Shutting down migration, by contrast, barely moves aggregate gains. A model without couples misses the opportunity cost of adjusting wives' participation: when wives enter the market, they give up home production. This cost is why the economy's aggregate gain is about 2% smaller with couples than without them.

Third, couples move and lose together. Without couples, more exposed workers move, but fewer people move overall, since unexposed spouses do not follow an exposed partner: 13% fewer people would relocate across states, and 27% fewer of the women who are wives in the baseline economy would relocate. Spouses also share each other's losses. When the wife is exposed, both spouses would be better off as singles, even though such couples fare better than those in which the husband is exposed, because the couple pools her lost earnings and she switches jobs, moves, and enters the labor market less than she would alone; otherwise, being in a couple shifts gains from the wife to the husband. Without migration, couples with

an exposed husband lose a larger share of their gain than either spouse would alone.

Taken together, these results show that couples change how the economy adjusts to a local shock and who bears it, even though they do not change its size or where it lands. Thus, understanding the incidence of local shocks requires treating the household, and not only the worker, as the unit that decides where to live and work.

Related Literature. This paper contributes to three strands of the literature. First, it contributes to the dynamic spatial general-equilibrium literature on trade and migration responses to local shocks (Redding and Rossi-Hansberg, 2017; Desmet et al., 2018; Monte et al., 2018; Allen and Donaldson, 2020, 2022; Donald et al., 2025; Desmet and Parro, 2025; Pellegrina and Sotelo, 2025). It is particularly closely related to Caliendo et al. (2019), which emphasizes the interaction between migration, productivity, and the spatial distribution of economic activity under the China Trade Shock, and to Rodríguez-Clare et al. (2026), who extend Caliendo et al. (2019) to separate mobility across sectors within a location from mobility across locations. My contribution is to bring the household into this class of models, with joint location choices and wives' labor supply decisions. Doing so leaves the size and location of the shock largely unchanged, but it changes how the economy adjusts to it: workers respond less, more people relocate across states, and couples absorb part of the shock at home, which changes who bears it, including singles.

Second, the paper contributes to the literature that studies how dual-earner households respond to shocks, highlighting the roles of risk sharing within couples (Attanasio et al., 2005; Pilossoph and Wee, 2021; Calvo et al., 2024) and co-location frictions (Guler et al., 2012; Flabbi and Mabli, 2018; Foerster and Ulbricht, 2023; Jayachandran et al., 2024; Yan, 2025).[1] I bring these insights to the study of local shocks. In an aggregate shock, couples can reallocate labor within the household, and because the shock is aggregate, the joint location friction does not bind. In the presence of a local shock, however, the joint location friction binds. They can adjust along two margins: couples move, though less than singles do, and they reallocate labor within the household. My contribution is to quantify these two margins in general equilibrium and to show what a model without them misses: households relocate together and share each other's losses, and wives' participation insures every couple, so that without it, the economy as a whole loses.

[1] There is an extensive literature on family migration and household adjustment. Starting with Mincer (1978), this literature has emphasized that migration is often a joint decision and that such decisions can have unequal consequences within the household. Subsequent work has documented the role of "trailing wives" and the labor market costs of family migration for women (Costa and Kahn, 2000; Cooke et al., 2009; Benson, 2014; Blau and Kahn, 2017; Fadlon et al., 2020; Braun et al., 2021; Le Barbanchon et al., 2021; Venator, 2026, 2024). Although beyond the scope of this paper, the outcomes of the "trailing wives" under a trade shock remain an open research question.

Third, the paper contributes to the literature on the China Trade Shock and its effects on US local labor markets (Autor et al., 2013, 2014, 2016, 2019, 2021, 2025; Acemoglu et al., 2016; Pierce and Schott, 2016; Greenland et al., 2019; Caliendo et al., 2019; Faber et al., 2022; Borusyak et al., 2026). This literature has documented substantial heterogeneity in the local effects of rising competition from Chinese imports.[2] My contribution to this literature is twofold. First, using the PSID, I provide new empirical results on the effects of the shock in the US by measuring individual-level exposure and comparing individuals within the same commuting zone. Although my analysis uses a different dataset, sample definition, and empirical design, these results are consistent with previous evidence that increased import competition from China harmed more exposed workers and places, with persistent employment effects on manufacturing workers, even though it improved aggregate welfare in the US economy (Autor et al., 2013, 2021; Caliendo et al., 2019). Second, to the best of my knowledge, this is the first paper to provide empirical evidence on within-household responses to the shock in the US. In this respect, the paper is closely related to Parey et al. (2025), who, using UK data, show that household responses to trade shocks can differ sharply by gender within the couple, finding an added-worker effect when wives are exposed. In the US, by contrast, wives increase their participation when their husbands are exposed, but husbands do not increase theirs when their wives are exposed.

The rest of the paper proceeds as follows. Section II presents the empirical analysis, detailing the approach and presenting the main empirical findings. Section III presents the theoretical model. Section IV brings the model to the data, and Section V presents the results of the quantitative exercise. Finally, Section VI concludes.

II. Empirical Analysis

This section presents the empirical analysis of how couples respond to the China Trade Shock. I first describe the data, define exposure at the individual and couple levels, and outline the empirical design. I then document three empirical facts concerning labor market outcomes, migration responses, and intra-household labor supply adjustments. Together, these findings motivate the theoretical framework that I develop in the next section.

[2] In particular, Autor et al. (2013) and subsequent papers document a significant decrease in manufacturing employment in US commuting zones with larger manufacturing shares, with long-lasting consequences that include increases in political polarization (Autor et al., 2020), child and adult poverty, and even mortality among young men (Autor et al., 2019), among several other topics.

II.1 Empirical Design

II.1.1 Data Source

The primary data source for the empirical analysis is the Panel Study of Income Dynamics (PSID), which follows individuals and their households over time. Most studies of the China Trade Shock use commuting zones as the unit of observation and rely on cross-sectional data. As a result, they cannot distinguish how exposure varies within households or track individuals across time and space. Because the PSID follows both spouses within the same household over time, it allows me to address both limitations.

The PSID has three additional advantages that make it particularly well suited to this analysis. First, the restricted geocoded version records household addresses at the census-tract level in every survey wave.[3] This information allows me to follow individuals across local labor markets, compare individuals within the same market, and study migration between markets. Following the previous literature, I define local labor markets as US commuting zones (CZs).

Second, the PSID allows me to construct marital status from household composition and each person's relationship to the household head in every period. This approach does not rely exclusively on current legal marital status, which may understate the number of couples by excluding cohabitation or overstate it by including individuals who are separated but not divorced. Given the PSID's sampling structure, I restrict the sample to three marital-status categories:[4]

1. "Married": Individuals in their first union who are legally married or cohabiting with a different-sex partner

2. "Divorced": Individuals who are legally divorced or separated, or whose cohabiting union ended

3. "Single": Individuals whom I never observe as "married" to another individual

I then pool single and divorced individuals into an unmarried category because neither group makes joint household decisions with a current partner.[5]

Third, the PSID contains detailed information on labor market outcomes, allowing me to examine employment, unemployment, and labor force participation. For employed in-

[3]The PSID staff has previously disclosed all results; access to the restricted data is under Contract 921.

[4]I exclude remarried individuals from the sample. I classify an individual who separated after a cohabitation as "divorced", even if their legal marital status is single.

[5]Assuming that divorced individuals do not make joint decisions follows previous literature studying life after divorce (Weiss and Willis, 1993; Del Boca and Flinn, 1995; Flinn, 2000; Reynoso, 2024; Didier and Lopez, 2026).

dividuals, I observe average weekly hours, hourly wages, annual labor income, and several additional outcomes. I also observe total household income for couples.[6]

I use all PSID waves conducted in odd-numbered years between 1991 and 2013.[7] I restrict the sample to individuals aged 20–55 who lived in the mainland US and were present in the 1999 wave. The final sample contains 4,731 individuals and 38,634 observations across 12 waves.

II.1.2 Individuals Exposed to the China Trade Shock

The China Trade Shock refers to the increase in competition from Chinese manufactured goods in the US following China's accession to the WTO in 2001 and continuing through 2012 (Autor et al., 2013, 2021). I identify exposed individuals in two steps. First, following Autor et al. (2021) (henceforth ADH), I define the growth in Chinese import competition faced by US manufacturing industry j between 2000 and 2012 as:

$$\Delta IC_j = 100 \times \frac{\Delta M_j^{\text{China,US}}}{Y_{j,2000} + M_{j,2000} - X_{j,2000}},$$

where $\Delta M_j^{\text{China,US}}$ is the change in US imports from China in industry j between 2000 and 2012, $Y_{j,2000}$ denotes industry j's shipments, and $M_{j,2000} - X_{j,2000}$ denotes its net imports in 2000. A positive value of ΔIC_j therefore indicates that US manufacturing industry j experienced increased competition from Chinese imports between 2000 and 2012.

To isolate the foreign supply-driven component of US imports from China, I follow ADH and measure industry-level growth in Chinese exports to eight other high-income countries.[8] I also use three-year-lagged values of Y_j, M_j, and X_j rather than base-year values. The resulting measure of import competition is

$$\widehat{\Delta IC_j} = 100 \times \frac{\Delta M_j^{\text{China,Other Countries}}}{Y_{j,1997} + M_{j,1997} - X_{j,1997}}.$$

A positive value of this measure indicates that industry j experienced increased competition from Chinese imports in other high-income countries. I therefore classify the corresponding US industry j as negatively exposed to the China Trade Shock.

Second, I identify the industry in which each individual i was employed in 1999 and

[6] I convert all monetary variables to 2017 dollars using the Personal Consumption Expenditures Price Index (PCE) from the US Bureau of Economic Analysis.

[7] Because the PSID was conducted annually until 1997 and biennially afterward, I consider only odd years to construct a biennial sample before and after 1997, which avoids double-counting migration.

[8] Autor et al. (2013) and their follow-up papers consider Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland. They select these countries based on the availability of data for the full sample period.

define the indicator Exposed_i as follows:

$$\text{Exposed}_i = \begin{cases} 1 & \text{if } i \text{ worked in 1999 in industry } j \text{ with } \widehat{\Delta IC_j} > 0 \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

For the analysis of within-household mechanisms, I further restrict the sample to couples. I divide couples into three groups: those in which neither spouse worked in an exposed industry in 1999, those in which the husband worked in an exposed industry, and those in which the wife worked in an exposed industry. I define couple h as exposed through the husband if

$$\text{Exposed Husband}_h = \begin{cases} 1 & \text{if he worked in 1999 in industry } j \text{ with } \widehat{\Delta IC_j} > 0 \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

and as exposed through the wife if

$$\text{Exposed Wife}_h = \begin{cases} 1 & \text{if she worked in 1999 in industry } j \text{ with } \widehat{\Delta IC_j} > 0 \\ 0 & \text{otherwise.} \end{cases} \tag{3}$$

II.1.3 Summary Statistics

Table 1 reports summary statistics for the pooled 1991–1999 sample. The first column describes the full sample; the second pools single and divorced individuals; and the third and fourth report statistics for husbands and wives, respectively. Married individuals constitute approximately 60% of the sample, single individuals 24%, and divorced individuals the remainder.

Unmarried individuals are slightly younger and less likely to be college graduates or white than married individuals. The average individual is 37 years old; two-fifths of the sample are women, one-third are college graduates, and 86% are white. Half of the sample have children under 18, including nearly one-quarter of unmarried individuals. The youngest child in the household is, on average, 4 years old and tends to be younger among unmarried individuals, consistent with the younger average age of unmarried individuals.

Married and unmarried individuals live in similarly exposed CZs but differ sharply in how often they migrate. According to ADH's measure, one in ten individuals lives in a CZ in the top two deciles of exposure to Chinese import competition. Exposure does not vary meaningfully by marital status, suggesting little spatial sorting along this dimension. The average migration rate is 8%, with unmarried individuals migrating at nearly twice the rate of married individuals.

In 1999, 15% of the sample worked in an industry exposed to competition from Chinese

Table 1: Summary Statistics Before the China Trade Shock

	(1) All	(2) Unmarried	(3) Couples	(4)
			Husbands	Wives
Panel A: Distribution by "Marital Status"				
Share of Single Individuals	0.24			
Share of Married Individuals	0.59			
Share of Divorced Individuals	0.16			
Panel B: Demographic Characteristics				
Age^{+}	36.86	34.69	38.92	37.19
Share of Female Individuals	0.42	0.55		
Share of College Graduates	0.33	0.29	0.37	0.32
Share of White Individuals	0.86	0.75	0.93	0.93
Share of Individuals with Kids	0.50	0.23	0.69	
Age of Youngest Kid^{+}	3.67	1.75	4.98	
Share in Highly Exposed CZs^{++}	0.11	0.12	0.11	
Migration Rate	0.08	0.11	0.06	
Panel C: Labor Market Outcomes				
Share of Exposed Individuals	0.15	0.14	0.19	0.10
Share in Routine Occupations	0.23	0.22	0.23	0.26
Share in Offshorable Occupations	0.12	0.14	0.08	0.16
Share Employed	0.91	0.89	0.97	0.85
Share Unemployed	0.03	0.06	0.02	0.02
Share Out of the Labor Force	0.05	0.05	0.01	0.13
Weekly Hours^{+}	40.80	39.89	45.34	33.60
Hourly Wage^{+}	19.54	16.98	23.70	16.45
Labor Income^{+}	42671.70	35953.36	57073.74	27617.31
Number of Observations	16,645	4,343	6,151	6,151

Notes. The unmarried category pools single and divorced individuals. Monetary values are expressed in 2017 USD using the PCE price index. $^{+}$: Indicates the sample mean. $^{++}$: Highly exposed CZs are those in the top 20% of exposure according to ADH.

imports, with substantial heterogeneity across groups. Exposure rates were 14% among unmarried individuals, 19% among husbands, and 10% among wives. At the household level, 20% of couples had at least one exposed spouse. As noted by Autor et al. (2013, 2021), the period of the China Trade Shock also featured two related economic forces: increas-

ing automation and broader globalization. Following Autor and Dorn (2013), I calculate routine-task intensity and offshorability indices for the occupations represented in the sample. Slightly less than one-quarter of observations correspond to individuals employed in routine-task-intensive occupations, while approximately one in ten workers is employed in an offshorable occupation. Offshorability varies substantially by gender, reflecting the occupational sorting of men and women. I control for both occupational characteristics in the empirical analysis.

Because the sample consists of prime-age individuals, employment and labor force participation rates are high, at 91% and 95%, but the composition of non-employment differs by marital status. Among non-employed individuals, unemployment is the most common status for unmarried individuals and husbands, whereas non-participation in the labor force is most common among wives. This pattern is consistent with unmarried individuals relying more heavily on reemployment after job loss, whereas married individuals can adjust labor supply within the household, thereby drawing on intra-household insurance.

The average individual works 40 hours per week, earns $19 per hour in 2017 USD, and receives approximately $42,000 in annual labor income. Husbands work the most hours, earn the highest hourly wages, and consequently receive the highest annual labor income; wives have the lowest values for all three outcomes. Married and unmarried individuals work similar hours on average, but married individuals earn approximately 20% higher hourly wages and therefore receive more labor income per capita. Although selection into marriage likely explains part of this gap, this pattern is also consistent with couples using intra-household labor supply adjustments to allocate resources more effectively.

II.1.4 Difference-in-Differences Design

To estimate the overall effect of exposure to the China Trade Shock, I use the following event-study specification for individual i in commuting zone c and year t:

$$\begin{aligned}\text{Outcome}_{i,c,t} =& \alpha \cdot \mathbb{1}[\text{Exposed}_i = 1] + \sum_{\tau \neq 1999} \beta_\tau \cdot \mathbb{1}[\text{Exposed}_i = 1] \cdot \mathbb{1}[t = \tau] \\ &+ X'\delta + \gamma_t + \gamma_c + \varepsilon_{i,c,t}\end{aligned} \tag{4}$$

The coefficients of interest, β_τ, measure the difference in outcomes between exposed and non-exposed individuals in year τ, relative to the corresponding difference in 1999. The control vector includes age and indicators for being female, holding a college degree, having children younger than 18, working in manufacturing in 1999, working in a routine occupation, and working in an offshorable occupation. The specification also includes commuting-zone and year fixed effects. I weight all regressions by PSID sampling weights and two-way cluster

standard errors at the individual and industry levels. To summarize the average effect during the shock, I also estimate a difference-in-differences version of equation (4) that replaces the year-specific interactions with a single interaction:

$$\begin{aligned}\text{Outcome}_{i,c,t} =&\beta_1 \cdot \mathbb{1}[\text{Exposed}_i = 1] + \beta_2 \cdot \mathbb{1}[\text{Exposed}_i = 1] \cdot \mathbb{1}[t \geq 2001] \\ &+ X'\delta + \gamma_c + \gamma_t + \varepsilon_{i,c,t}.\end{aligned} \tag{5}$$

To examine joint household decision-making, I estimate the following specifications for spouse s in couple h, commuting zone c, and year t. First, I focus on whether the husband is exposed:

$$\begin{aligned}\text{Outcome}_{s,h,c,t} =&\beta_1 \cdot \mathbb{1}[\text{Exposed Husband}_h = 1] + \beta_2 \cdot \mathbb{1}[\text{Exposed Husband}_h = 1] \cdot \mathbb{1}[t \geq 2001] \\ &+ X'\delta + \gamma_c + \gamma_t + \varepsilon_{s,h,c,t}\end{aligned} \tag{6}$$

and then on whether the wife is exposed:

$$\begin{aligned}\text{Outcome}_{s,h,c,t} =&\beta_1 \cdot \mathbb{1}[\text{Exposed Wife}_h = 1] + \beta_2 \cdot \mathbb{1}[\text{Exposed Wife}_h = 1] \cdot \mathbb{1}[t \geq 2001] \\ &+ X'\delta + \gamma_c + \gamma_t + \varepsilon_{s,h,c,t}.\end{aligned} \tag{7}$$

In these specifications, the coefficient of interest is β_2, which captures the post-2001 difference in outcomes between couples with an exposed spouse and those without, relative to the corresponding pre-2001 difference.[9] I estimate both specifications on the same sample of couples, so the comparison group in equation (6) includes couples in which only the wife is exposed, and vice versa. Both equations include the controls used in equation (4). For controls measured at the individual level, I include separate values for each spouse, such as the husband's and wife's ages.

I two-way cluster standard errors by the exposed spouse's industry and by household or individual, depending on how the outcome is measured. The industry dimension refers to the husband's industry in equation (6) and the wife's in equation (7). I cluster at the household level for outcomes measured at the household level, such as out-migration and household income, and at the individual level for outcomes measured for each spouse, such as employment and labor force participation.

[9]For this analysis, I drop 4% of couple-year observations across the full 1991–2013 panel in which both spouses were working in exposed industries in 1999, because I cannot measure whether one spouse's job was more important to the couple than the other's.

II.2 Empirical Facts

The three facts proceed from individual-level effects to within-household adjustment. The first establishes the labor market effects of the shock and examines whether exposed individuals respond by leaving their commuting zone. The second shows that couples migrate in response only when the husband is the exposed spouse. The third examines wives' labor supply responses, a margin of adjustment that couples can use alongside or instead of migration. Across all three analyses, I identify the effects by comparing individuals whose exposure differs by their 1999 industry, holding commuting zone and year fixed. The relevant comparison is therefore between exposed and non-exposed individuals facing the same local labor market, rather than between local labor markets with different levels of exposure.

Fact 1: Exposure to the China Trade Shock worsened labor market outcomes and increased out-migration.

Figure 1 presents estimates of equation (4) for employment, unemployment, labor force participation, weekly hours worked, hourly wages, and annual labor income. Table A.2 reports estimates from the corresponding difference-in-differences specification in equation (5), which summarizes the average effect during the China Trade Shock.

Panel A of Figure 1 shows that, during the first seven years of the shock, exposed individuals were less likely to be employed than non-exposed individuals. On average, exposure was associated with a 6.3-percentage-point (7%) decline in the probability of employment. After 2009, the point estimates remain close to their 2007 level, but the confidence intervals widen, and the estimates are no longer statistically significant. Most of the decline in employment between 2001 and 2007 reflects individuals leaving the labor force (Panel C) rather than entering unemployment (Panel B). Between 2001 and 2013, exposed individuals were, on average, 4% less likely to participate in the labor force than non-exposed individuals.

Weekly hours worked follow a pattern similar to employment (Panel D): individuals employed in exposed industries in 1999 worked 5% fewer hours per week during the shock than individuals employed in other industries that year. Hourly wages exhibit a different pattern (Panel E). Although the estimate is positive in 2001, the hourly wages of exposed individuals declined through the final year of the sample. On average, exposed individuals earned 10% lower hourly wages during the shock than non-exposed individuals. Combined with the reduction in hours worked, this decline resulted in 11% lower annual labor income among exposed individuals (Panel F).

My employment estimates are similar in magnitude to ADH's when I use their continuous measure of import competition. To compare my estimates more directly with the

previous literature, I replace the binary exposure measure in equation (1) with the continuous import-competition measure used by ADH (Table A.3).[10] The average decadalized growth in Chinese import competition implies a 0.5-percentage-point average decline in the probability of employment among exposed individuals relative to non-exposed individuals between 2001 and 2013. For comparison, converting ADH's cumulative estimate for the employment-to-population ratio between 2001 and 2019 into an evenly distributed effect and retaining only odd-numbered years, as in the PSID sample, implies an average decline of 0.55 percentage points.

These findings are consistent with previous evidence on the adverse effects of the China Trade Shock in the US (Autor et al., 2013, 2014, 2016, 2021; Acemoglu et al., 2016; Pierce and Schott, 2016). Despite using a different dataset, sample restrictions, and empirical approach, I similarly find a negative employment response among individuals working in exposed industries.

Given the deterioration in labor market outcomes among individuals employed in industries exposed to Chinese import competition, particularly the decline in employment during the first years of the shock, I next examine whether exposed individuals moved to other CZs in search of better opportunities. For each individual i, I define out-migration in year t as follows:

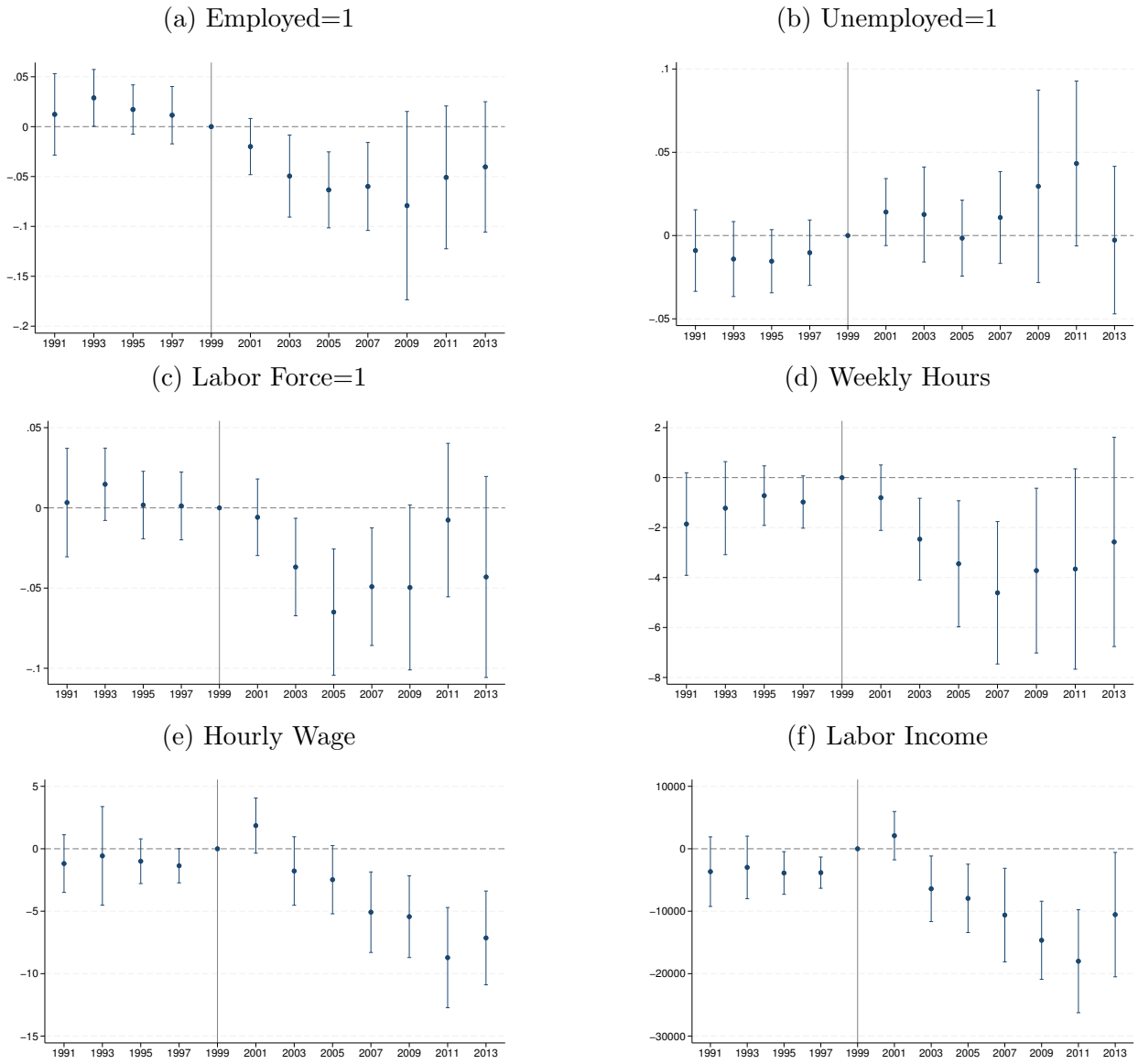
$$\text{Out-Migration}_{i,t} = \begin{cases} 1 & \text{if} \quad \text{CZ}_{i,t-2} \neq CZ_{i,t} \\ 0 & \text{if} \quad CZ_{i,t-2} = CZ_{i,t}. \end{cases}$$

During the shock, exposed individuals became increasingly likely to leave their CZ. Figure 2 presents estimates of equation (4) using out-migration as the outcome: the estimated difference rose by approximately 2 percentage points every two years, reaching 6 percentage points in 2013. Most of the response emerges with a lag of seven to ten years, consistent with Greenland et al. (2019)'s evidence on the effects of the China Trade Shock on internal migration in the US. On average, exposed individuals were 4.7 percentage points more likely to out-migrate than non-exposed individuals within the same commuting zone, as reported in Table A.4.

The out-migration response is consistent with ADH's evidence of a statistically significant decline in the population aged 25 to 39 and a statistically insignificant decline in the overall population of more exposed commuting zones. Moreover, Borusyak et al. (2026) argue that shift-share measures of industry-level shocks may attenuate estimated migration responses when they abstract from workers' outside options, as in ADH. Accordingly, the individual-level out-migration response I estimate should be larger than their aggregate estimate.

[10] One potential concern is that, given the small PSID sample, the particular set of CZs in my sample may drive the estimates. To address this concern, Appendix A.1 replicates ADH's findings using the same CZs included in my sample, ruling out this alternative explanation.

Figure 1: Exposure to the China Trade Shock and Labor Market Outcomes



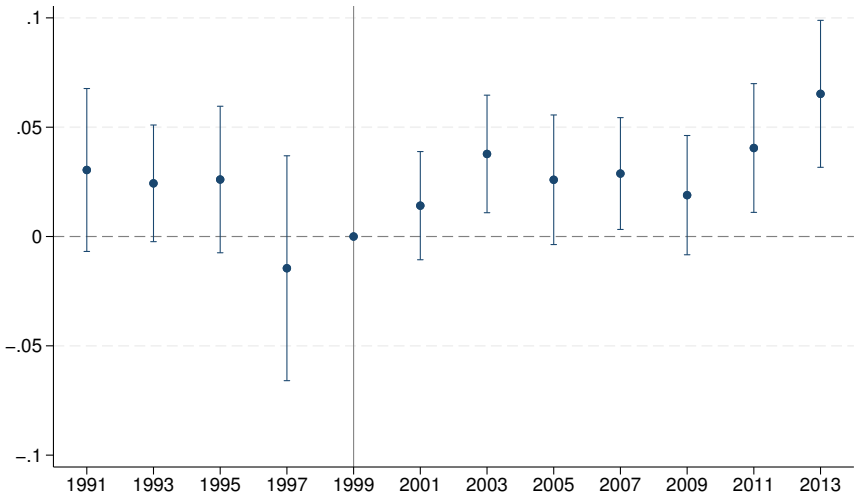
Notes. Each panel reports estimates from equation (4), using the variable named in the panel title as the outcome. Monetary outcomes are expressed in 2017 USD using the PCE price index and include zero values. Robust standard errors are two-way clustered at the industry and individual levels. The bars indicate 95% confidence intervals.

Fact 2: Couples out-migrate only when the husband is exposed.

To examine how couples' migration responses differ from those of individuals, I estimate Equations 6 and 7 using out-migration as the outcome at the household level.[11] Column 1 of

[11]The PSID records only the census tract of the household in which the interview takes place, which it always treats as the family's primary household. As a result, if only one spouse migrates, the PSID does not record the couple as migrating. Instead, it records the couple as "married, spouse absent," and the

Figure 2: Exposure to the China Trade Shock and Out-Migration Responses



Notes. Robust standard errors are two-way clustered by industry and individual. The bars indicate 95% confidence intervals.

Table 2 shows that couples in which the husband was exposed to the shock were 5 percentage points more likely to leave their commuting zone than couples in which the husband was not exposed.[12] By contrast, exposure of the wife produces no statistically significant migration response, and the point estimate in Column 2 is negative. Although this asymmetry may initially appear counterintuitive, it is consistent with evidence that husbands' employment plays a larger role in family migration decisions (Mincer, 1978; Costa and Kahn, 2000; Cooke et al., 2009; Benson, 2014; Blau and Kahn, 2017; Fadlon et al., 2020; Braun et al., 2021; Le Barbanchon et al., 2021; Venator, 2026, 2024).

I can rule out two alternative explanations for this asymmetry: changes in family composition and different migration responses among parents or among women. First, the asymmetry could be driven by changes in family composition. If, for example, couples divorce when the wife is exposed but not when the husband is exposed, this would bias my findings. In Appendix A.2, I study the effect of the trade shock on the probabilities of marriage, divorce, and fertility and find no effect on any of them. Second, the asymmetry could reflect nothing more than different migration responses among parents or among exposed women. In Appendix A.3, I expand equation (4) to include a triple interaction for having children

respondent reports for both spouses. According to PSID staff, this case is very unlikely. My estimates are therefore a lower bound on couples' migration responses, as they only account for couples who migrate together.

[12] The negative coefficient in the first row of Table 2 reflects the relatively favorable employment conditions in those manufacturing industries before the shock, which made couples less likely to migrate if the husband was employed there.

and for being a woman. In both cases, I cannot reject that parents or women responded in the same way as non-parents or men.

Table 2: Exposure to the China Trade Shock and Couples' Out-Migration Responses

	(1)	(2)
	Out-Migration=1	
Exposed Husband=1	-0.121***	
	(0.043)	
(Exposed Husband=1)x(Post=1)	0.050**	
	(0.022)	
Exposed Wife=1		0.064
		(0.074)
(Exposed Wife=1)x(Post=1)		-0.010
		(0.028)
Observations	13,695	13,695
R^2	0.10	0.09
Baseline Mean	0.05	0.05

Notes. Out-migration is measured at the household level. Robust standard errors are two-way clustered by household and the husband's industry in Column 1, and by household and the wife's industry in Column 2. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

Fact 3: Wives' labor force participation falls when they are directly exposed and rises when husbands are exposed.

Couples have an additional margin of adjustment, the so-called "added-worker effect" (Lundberg, 1985), which refers to a spouse who specializes in home production entering the labor market when the other spouse loses their job, to offset the household's income loss. I therefore study labor market responses to the shock, distinguishing between direct and indirect exposure. When wives are directly exposed to the shock, they are 13% less likely to be employed and 14% less likely to be in the labor force than non-exposed wives (Panel A of Table 3). However, when wives are indirectly exposed because their husbands worked in an exposed industry, they are 5.4% more likely to be in the labor force relative to the wives of non-exposed husbands (Panel B of Table 3). This provides suggestive evidence of

the added-worker effect among wives indirectly exposed to the China Trade Shock through their husbands.

Husbands, by contrast, show no added-worker effect. When husbands are indirectly exposed to the shock, they have the same probability of employment as non-exposed husbands, while earning a lower hourly wage (Table A.5), suggesting a lower reservation wage when indirectly exposed.

Table 3: Wife's Exposure to the China Trade Shock and Labor Market Outcomes

	Wife Employed=1	Wife Labor Force=1
Exposed Wife=1	0.041	0.016
	(0.099)	(0.110)
(Exposed Wife=1)x(Post=1)	-0.099**	-0.110**
	(0.045)	(0.048)
Exposed Husband=1	0.027	0.027
	(0.077)	(0.068)
(Exposed Husband=1)x(Post=1)	0.041***	0.042***
	(0.013)	(0.013)
Observations	14,083	14,083
Baseline Mean	0.77	0.78

Notes. Robust standard errors are two-way clustered at the wife's industry and individual levels in Panel A and at the husband's industry and individual levels in Panel B. * for p<.10, ** for p<0.05, *** for p<0.01.

The absence of an added-worker effect among husbands contrasts with the findings of Parey et al. (2025) for England and Wales. They show that when English husbands are indirectly exposed to the trade shock, older husbands increase their labor force participation and move into self-employment. They find a positive, though not statistically significant, response among indirectly exposed wives. The difference in the results for husbands could be driven by the fact that 93% of husbands in their sample are in the labor force, compared with 99% in my sample (Table 1). As a result, husbands in my sample face a ceiling effect that hinders identification of an added-worker effect for husbands.

Household income of exposed couples did not differ significantly from that of non-exposed couples during the shock, regardless of which spouse was exposed. To provide suggestive evidence on whether these asymmetric responses insure couples against the shock, I estimate Equations 6 and 7 using a measure of household income from the PSID. Columns 1 and 2

of Table A.6 report the estimates. This suggests that, when the husband is exposed, the combination of out-migration and the wife's labor force participation responses helps stabilize household income during the shock.

When the wife is exposed, stable household income reflects a smaller loss rather than insurance through migration or added labor supply. These couples do not migrate, the wife's employment falls, and the husband's employment does not change (Tables 3 and A.5). Before the shock, however, wives earned roughly half as much labor income as husbands (Table 1), so the loss of the wife's earnings is small relative to total household income. The point estimate in Column 2 of Table A.6 is negative, consistent with this smaller loss, but imprecisely estimated. The smaller loss also helps explain why these couples do not move: the income at stake is lower than it would be when the husband, the household's main earner, is exposed to the shock. Household income is also an incomplete measure of welfare, because it does not capture changes in home production, leisure, or the costs of moving. I quantify the welfare effects of both margins with the model developed in the next section.

The empirical evidence points to three central patterns that, together, suggest a trade-off for couples between moving and adjusting labor supply within the household. First, exposure to the China Trade Shock worsened labor market outcomes for directly exposed individuals, who responded by leaving their commuting zones. Second, among couples, this migration response was concentrated in households where the husband was exposed. Third, I find evidence of an added-worker effect during the trade shock, with wives' labor force participation responding differently depending on which spouse was directly exposed to the shock. On the one hand, joint location decisions make moving harder, so couples move only when the household's breadwinner is affected. On the other hand, they can reallocate labor within the household, making it more bearable to stay in the same location.

However, the empirical evidence cannot inform us about which margin of adjustment is more important, why, or what the general equilibrium effects of abstracting from couples are. In the next section, I develop a framework in which households jointly decide on location and labor supply, within a quantitative spatial and trade model.

III. Dynamic Spatial Model of Trade and Migration with Heterogeneous Households

This section develops a dynamic spatial and trade model to rationalize the three empirical facts documented in Section II and to quantify their welfare and spatial implications. The model extends the framework of Caliendo et al. (2019) (henceforth CDP) by incorporating

heterogeneous households and joint decision-making. In the model, households rather than individuals are the unit that moves: a couple jointly chooses where to live and each spouse's sector, and then decides whether the wife works in the market.

The economy consists of L locations, indexed by i as origins and j as destinations, populated by singles and couples. In each location, a population mass N is born at $t = 0$, comprising $N/2$ men and $N/2$ women. A fraction $\bar{s}_i$ of individuals are single, while the remaining fraction, $1 - \bar{s}_i$, belong to heterosexual couples. Time is discrete and indexed by $t \in \{0, 1, 2, \ldots\}$.

The model includes three household types, indexed by h: single men ($h = s_m$), single women ($h = s_f$), and couples ($h = c$). All households are forward-looking and respond to local conditions by choosing their place of residence and sector of employment. Couples make these choices jointly. After relocating, they draw a home-production productivity shock β, which, together with the spouses' relative wage, determines whether the wife participates in market work.

A central feature of the model is the interaction between migration and married women's labor force participation. Together, these two decisions shape how sectoral productivity shocks affect the spatial distribution of individuals and output, because men and women are exposed to different sectors.

III.1 Environment

The environment consists of three components: technology and trade, preferences and mobility, and the wife's participation decision. First, technology and trade determine equilibrium wages in each region-sector cell, and labor is a CES composite of male and female inputs, implying that sectoral shocks have differential effects on men's and women's wages. Second, preferences and mobility determine households' residential and employment choices. Couples jointly choose a location and a pair of sectors rather than making these decisions sequentially. Finally, conditional on relocation, the couple chooses whether the wife works in the market or in home production.

III.1.1 Technology

In each period t, every region i contains K productive sectors, indexed by $k, n \in \{1, \ldots, K\}$. Let $A_{i,k,t} > 0$ denote Hicks-neutral productivity in sector k of region i. Let $\psi_{k,m}, \psi_{k,f} > 0$ denote sector-specific gender-intensity parameters that are common across locations and satisfy $\psi_{k,m} + \psi_{k,f} = 1$.

A representative intermediate-good firm in sector k of origin i combines male labor, $M_{i,k,t}$,

and female labor, $F_{i,k,t}$, into an effective labor aggregate using a CES production function with elasticity of substitution $\sigma > 1$:

$$\mathcal{L}_{i,k,t} = \left(\psi_{k,m} M_{i,k,t}^{(\sigma-1)/\sigma} + \psi_{k,f} F_{i,k,t}^{(\sigma-1)/\sigma}\right)^{\sigma/(\sigma-1)}. \tag{8}$$

Output is then given by

$$Y_{i,k,t} = A_{i,k,t} \mathcal{L}_{i,k,t}.$$

The sectoral intermediate good is sold at producer price $p_{i,k,t}$. Shipping the good from origin-sector pair ik to destination-sector pair jn entails iceberg trade costs $\tau_{ik,jn} \geq 1$. Thus, the delivered price paid by sector n in destination region j is $p_{ik,jn,t} = p_{i,k,t}\tau_{ik,jn}$. Under perfect competition and constant returns to scale, the producer price equals the marginal cost of producing the sectoral intermediate good:

$$p_{i,k,t} = \frac{\bar{w}_{i,k,t}}{A_{i,k,t}} \implies \bar{w}_{i,k,t} = p_{i,k,t} A_{i,k,t}.$$

Here, $\bar{w}_{i,k,t}$ is the CES wage index, which gives the cost-minimizing unit cost of effective labor:

$$\bar{w}_{i,k,t} = \left(\psi_{k,m}^{\sigma} \left(w_{i,k,t}^{M}\right)^{1-\sigma} + \psi_{k,f}^{\sigma} \left(w_{i,k,t}^{F}\right)^{1-\sigma}\right)^{\frac{1}{1-\sigma}}.$$

Here, $w_{i,k,t}^{M}$ and $w_{i,k,t}^{F}$ denote the average wages paid to men and women, respectively, employed in sector k of location i at time t.

The first-order conditions of the cost-minimization problem yield optimal demands for male and female labor as functions of the effective labor aggregate and relative wages:

$$M_{i,k,t} = \psi_{k,m}^{\sigma} \left(\frac{w_{i,k,t}^{M}}{\bar{w}_{i,k,t}}\right)^{-\sigma} \mathcal{L}_{i,k,t}, \tag{9}$$

$$F_{i,k,t} = \psi_{k,f}^{\sigma} \left(\frac{w_{i,k,t}^{F}}{\bar{w}_{i,k,t}}\right)^{-\sigma} \mathcal{L}_{i,k,t}. \tag{10}$$

Inverting the labor demand functions in Equations 9 and 10 yields the equilibrium gender-specific factor prices within each region-sector pair:

$$w_{i,k,t}^{M} = \bar{w}_{i,k,t}\psi_{k,m} \left(\frac{\mathcal{L}_{i,k,t}}{M_{i,k,t}}\right)^{1/\sigma}, \qquad w_{i,k,t}^{F} = \bar{w}_{i,k,t}\psi_{k,f} \left(\frac{\mathcal{L}_{i,k,t}}{F_{i,k,t}}\right)^{1/\sigma}. \tag{11}$$

In each destination region j and sector n, a representative sectoral final-good firm combines goods from all origin-sector pairs into a sector-n composite using a CES aggregator with elasticity of substitution $\eta_n > 1$. Let $Q_{j,n,t}$ denote the quantity of this composite

produced at time t. The corresponding unit-cost function is

$$P_{j,n,t} = \left(\sum_{i=1}^{L} \sum_{k=1}^{K} p_{ik,jn,t}^{1-\eta_n} \right)^{1/(1-\eta_n)} . \tag{12}$$

The implied share of expenditure by sector n in destination region j on goods from sector k in origin region i is

$$\pi_{ik,jn,t} = \left(\frac{p_{ik,jn,t}}{P_{j,n,t}} \right)^{1-\eta_n} .$$

At the regional level, a representative final-good firm combines sectoral composites into a regional consumption good using a Cobb-Douglas aggregator with expenditure shares $\iota_k > 0$, where $\sum_{k=1}^{K} \iota_k = 1$:

$$Q_{j,t} = \prod_{k=1}^{K} \left(\frac{Q_{j,k,t}}{\iota_k} \right)^{\iota_k}$$

where $Q_{j,t}$ denotes the quantity of the regional final good produced in region j at time t. The associated regional price index is

$$P_{j,t} = \prod_{k=1}^{K} P_{j,k,t}^{\iota_k}. \tag{13}$$

Cobb-Douglas aggregation implies that expenditure on sector n is a constant share of total expenditure in region j: $P_{j,n,t} Q_{j,n,t} = \iota_n P_{j,t} Q_{j,t}$.

Each location j also contains a non-employment sector, indexed by 0, that is available to single men, single women, and married men. Married women do not choose sector 0; instead, they choose a productive sector and subsequently decide whether to work in the market or engage in home production. A wife who stays at home therefore keeps a potential sector, whose wage enters the couple's participation decision. The non-employment sector produces no output. An individual who chooses sector 0 in region j receives an exogenous nominal benefit b_j, which is constant over time:

$$w_{j,0,t}^{M} = w_{j,0,t}^{F} = b_j.$$

III.1.2 Preferences

Mobility decisions. The household's mobility problem is dynamic: each household $h \in \{s_m, s_f, c\}$ is forward-looking and discounts the future with discount factor $\delta \in (0, 1)$. At the start of every period, households begin in a particular location-sector position and must decide whether to switch sectors and whether to stay or move. Every household occupies a single location, so spouses always live together, but the sector component differs by household

type: for single individuals, it is one sector in $\{0, 1, \ldots, K\}$, while for couples, it is the pair formed by the husband's and wife's sectors. The husband's current and destination sectors satisfy $k_m, n_m \in \{0, 1, \ldots, K\}$, while the wife's current and destination market sectors satisfy $k_f, n_f \in \{1, \ldots, K\}$.

For convenience, I collapse a household's location-sector position into a single *bundle*. I write the origin (current) bundle as $\boldsymbol{o} \equiv (i, k_m, k_f)$, the destination (next-period) bundle as $\boldsymbol{d} \equiv (j, n_m, n_f)$, and a generic bundle summed over as $\boldsymbol{z} \equiv (\ell, q_m, q_f)$. For single men, a bundle is (i, k_m); for single women, (i, k_f); and for couples, (i, k_m, k_f). The same symbols $\boldsymbol{o}, \boldsymbol{d}, \boldsymbol{z}$ denote the bundle for every household type, and I unbundle them into components only where a formula refers to a particular location or sector.

Let $\mathcal{S}_h$ denote the set of feasible bundles for household type h, and define

$$\mathcal{S}_h(j) \equiv \{\boldsymbol{d} \in \mathcal{S}_h : \text{ the location of } \boldsymbol{d} \text{ is } j\}$$

as the set of feasible bundles available to type h that are located in j. For singles, $\mathcal{S}_h(j)$ contains the $K+1$ sectors available in j; for couples, it contains the $(K+1) \times K$ sector pairs available in j.

To decide their bundle for the next period, each household h in current bundle $\boldsymbol{o}$ draws two idiosyncratic preference shocks: a *location* shock $\varepsilon^{h,L}_{\boldsymbol{o}j,t}$ for each destination location j, and a *sector* shock $\varepsilon^{h,S}_{\boldsymbol{od},t}$ for each destination sector bundle within that location. The two enter additively, so the total preference shock attached to destination bundle $\boldsymbol{d}$, located in j, is

$$\varepsilon^{h}_{\boldsymbol{od},t} = \varepsilon^{h,L}_{\boldsymbol{o}j,t} + \varepsilon^{h,S}_{\boldsymbol{od},t}. \tag{14}$$

Following Rodríguez-Clare et al. (2026), I assume that the vector $\varepsilon^{h}_{\boldsymbol{o},t} \equiv (\varepsilon^{h}_{\boldsymbol{od},t})_{\boldsymbol{d} \in \mathcal{S}_h}$ is i.i.d. over time and households and has a nested Gumbel joint distribution,

$$F\left(\varepsilon^{h}_{\boldsymbol{o},t}\right) = \exp\left(-\sum_{j=1}^{L}\left(\sum_{\boldsymbol{d} \in \mathcal{S}_h(j)} \exp\left(-\varepsilon^{h}_{\boldsymbol{od},t}/\nu\right)\right)^{\nu/\theta}\right), \tag{15}$$

with $0 < \nu \leq \theta$.[13] The parameter ν governs the dispersion of preferences across sector bundles within a location, so $1/\nu$ determines substitution across sectors. Similarly, θ governs preference dispersion across locations, so $1/\theta$ determines substitution across locations. The restriction $\nu \leq \theta$ implies that the sectoral elasticity $1/\nu$ is at least as large as the migration

[13] Rodríguez-Clare et al. (2026) extend Caliendo et al. (2019) by allowing the elasticities of mobility across sectors within a region and across regions to differ, adding a margin that better matches the empirical evidence in Autor et al. (2013). In this setting, the extension distinguishes between mobility across sectors and mobility across regions, with the latter being the relevant margin for couples.

elasticity $1/\theta$. Thus, households substitute at least as readily across sectors within a location as they do across locations.[14]

A household moving from origin bundle $\boldsymbol{o}$ to destination bundle $\boldsymbol{d}$ incurs mobility cost $\mu_{\boldsymbol{od}} \geq 0$, with $\mu_{\boldsymbol{oo}} = 0$. The lifetime utility of a household of type h occupying bundle $\boldsymbol{o}$ at time t is

$$v^h_{\boldsymbol{o},t} = \log(W^h_{\boldsymbol{o},t}) + \max_{\boldsymbol{d}\in\mathcal{S}_h} \left\{ \delta \mathbb{E}\left(v^h_{\boldsymbol{d},t+1}\right) - \mu_{\boldsymbol{od}} + \varepsilon^h_{\boldsymbol{od},t} \right\},$$

where $W^h_{\boldsymbol{o},t}$ denotes the period-t indirect utility of household type h in current bundle $\boldsymbol{o}$. For singles $h \in \{s_m, s_f\}$, $W^{s_m}_{\boldsymbol{o},t} = w^M_{i,k,t}/P_{i,t}$ and $W^{s_f}_{\boldsymbol{o},t} = w^F_{i,k,t}/P_{i,t}$, as in CDP but allowing for gender-specific wages. For couples, $W^c_{\boldsymbol{o},t}$ also depends on the wife's market-work decision, as I describe below.

Let $V^h_{\boldsymbol{o},t} \equiv \mathbb{E}(v^h_{\boldsymbol{o},t})$. Then, using the properties of the nested Gumbel distribution (Appendix B.1), I can write the value function as

$$V^h_{\boldsymbol{o},t} = \log(W^h_{\boldsymbol{o},t}) + \theta \log\left(\sum_{\ell=1}^{L} \left(\sum_{\boldsymbol{z}\in\mathcal{S}_h(\ell)} \exp\left[\delta V^h_{\boldsymbol{z},t+1} - \mu_{\boldsymbol{oz}}\right]^{1/\nu} \right)^{\nu/\theta} \right). \tag{16}$$

I decompose the choice probability into a location choice and a conditional sector-bundle choice. Let $\Omega^h_{\boldsymbol{o}j,t} \equiv \sum_{\boldsymbol{z}\in\mathcal{S}_h(j)} \exp\left[\delta V^h_{\boldsymbol{z},t+1} - \mu_{\boldsymbol{oz}}\right]^{1/\nu}$. The share of type-$h$ households in bundle $\boldsymbol{o}$ that choose location j is

$$\lambda^h_{\boldsymbol{o}j,t} = \frac{\left(\Omega^h_{\boldsymbol{o}j,t}\right)^{\nu/\theta}}{\sum_{\ell=1}^{L}\left(\Omega^h_{\boldsymbol{o}\ell,t}\right)^{\nu/\theta}}. \tag{17}$$

Conditional on choosing location j, the share that selects destination bundle $\boldsymbol{d} \in \mathcal{S}_h(j)$ is

$$\lambda^h_{\boldsymbol{od}|j,t} = \frac{\exp\left[\delta V^h_{\boldsymbol{d},t+1} - \mu_{\boldsymbol{od}}\right]^{1/\nu}}{\Omega^h_{\boldsymbol{o}j,t}}. \tag{18}$$

The unconditional share of type-h households in bundle $\boldsymbol{o}$ that choose destination bundle $\boldsymbol{d} \in \mathcal{S}_h(j)$ is therefore

$$\lambda^h_{\boldsymbol{od},t} = \lambda^h_{\boldsymbol{od}|j,t}\, \lambda^h_{\boldsymbol{o}j,t}. \tag{19}$$

Thus, the location elasticity $1/\theta$ governs the response of location shares to each location's inclusive value, whereas the sectoral elasticity $1/\nu$ governs the response of within-location shares to the value of each sector bundle.

[14] When $\nu = \theta$, equation (15) reduces to the product of independent Gumbel distributions with scale θ.

Couple's problem. After relocating, a couple decides whether the wife works in the market or stays at home, based on the spouses' relative wage and a home-production productivity draw. Each couple draws an independently and identically distributed home-production productivity shock $\beta \sim \text{Pareto}(\chi, \kappa)$, with support $\beta \geq \chi > 0$ and shape parameter $\kappa > 0$.[15] The couple then jointly chooses the wife's labor-supply status $e_{\boldsymbol{d},t} \in \{0, 1\}$, where $e_{\boldsymbol{d},t} = 1$ indicates that the wife stays at home, along with private consumption levels $c_{\boldsymbol{d},f,t}$ and $c_{\boldsymbol{d},m,t}$ and public-good expenditure $q_{\boldsymbol{d},t}$. When the wife stays at home, the productivity shock increases the effective level of the household public good, $Q_{\boldsymbol{d},t} = \beta^{e_{\boldsymbol{d},t}} q_{\boldsymbol{d},t}$.

Conditional on choosing bundle $\boldsymbol{d}$, the couple therefore solves

$$\begin{aligned} \max_{c_{\boldsymbol{d},f,t}, c_{\boldsymbol{d},m,t}, q_{\boldsymbol{d},t}, e_{\boldsymbol{d},t} \in \{0,1\}} \quad & \left(Q_{\boldsymbol{d},t}^{\gamma} c_{\boldsymbol{d},f,t}^{1-\gamma}\right)^{\alpha} \left(Q_{\boldsymbol{d},t}^{\gamma} c_{\boldsymbol{d},m,t}^{1-\gamma}\right)^{1-\alpha} \\ \text{s.t.} \quad & P_{j,t}\left(c_{\boldsymbol{d},f,t} + c_{\boldsymbol{d},m,t} + q_{\boldsymbol{d},t}\right) = \underbrace{w_{j,n_m,t}^{M} + w_{j,n_f,t}^{F}(1 - e_{\boldsymbol{d},t})}_{I(e)}, \\ & Q_{\boldsymbol{d},t} = \beta^{e_{\boldsymbol{d},t}} q_{\boldsymbol{d},t}, \end{aligned} \tag{20}$$

with $0 < \alpha < 1$ and $0 < \gamma < 1$, where α is the wife's bargaining weight and γ is the expenditure share of the public good.

Following Chiappori and Mazzocco (2017), the optimal allocations conditional on $\boldsymbol{d}$ and $e_{\boldsymbol{d},t}$ are

$$\begin{aligned} c_{\boldsymbol{d},f,t}^{*}(e) &= \alpha(1-\gamma)\frac{I(e)}{P_{j,t}}, \\ c_{\boldsymbol{d},m,t}^{*}(e) &= (1-\alpha)(1-\gamma)\frac{I(e)}{P_{j,t}}, \\ q_{\boldsymbol{d},t}^{*}(e) &= \gamma\frac{I(e)}{P_{j,t}} \quad \Longrightarrow \quad Q_{\boldsymbol{d},t}^{*}(e) = \beta^{e}\gamma\frac{I(e)}{P_{j,t}} \end{aligned} \tag{21}$$

The wife works in the market when the gain in household income from her earnings outweighs the gain in the household public good from her home production. To find the decision rule for $e_{\boldsymbol{d},t}$, let $\mathcal{A} \equiv \alpha^{\alpha(1-\gamma)}(1-\alpha)^{(1-\alpha)(1-\gamma)}\gamma^{\gamma}(1-\gamma)^{1-\gamma}$, and define the sector-specific female-to-male wage ratio faced by a couple in $\boldsymbol{d}$ as

$$r_{\boldsymbol{d},t} \equiv \frac{w_{j,n_f,t}^{F}}{w_{j,n_m,t}^{M}}.$$

[15] The home-production productivity shock captures the non-monetary value generated when the wife engages in home production (Becker, 1965).

Then, the couple's static indirect utility conditional on choosing $\boldsymbol{d}$ and drawing β is

$$\mathcal{U}_{\boldsymbol{d},t}(\beta) = \mathcal{A}\frac{w^M_{j,n_m,t}}{P_{j,t}} \begin{cases} (1 + r_{\boldsymbol{d},t}) & \text{if} \quad e_{\boldsymbol{d},t} = 0 \\ \beta^\gamma & \text{if} \quad e_{\boldsymbol{d},t} = 1 \end{cases}$$

Hence, the wife works in the market if and only if

$$\beta \leq \beta^*_{\boldsymbol{d},t}, \qquad \beta^*_{\boldsymbol{d},t} = (1 + r_{\boldsymbol{d},t})^{1/\gamma}.$$

The market-work probability is therefore

$$\rho_{\boldsymbol{d},t} \equiv \Pr(e_{\boldsymbol{d},t} = 0) = 1 - \left(\frac{\chi}{\beta^*_{\boldsymbol{d},t}}\right)^\kappa = 1 - \chi^\kappa (1 + r_{\boldsymbol{d},t})^{-\kappa/\gamma}. \tag{22}$$

An interior participation margin in bundle $\boldsymbol{d}$ requires $0 < \rho_{\boldsymbol{d},t} < 1$. The upper bound, $\rho_{\boldsymbol{d},t} < 1$, holds for any $\chi > 0$, whereas the lower bound, $\rho_{\boldsymbol{d},t} > 0$, requires the lower support bound to lie below the market-work threshold: $\chi < (1 + r_{\boldsymbol{d},t})^{1/\gamma}$. Moreover, because the home-production productivity shock enters the couple's problem as $Q_{\boldsymbol{d},t} = \beta^{e_{\boldsymbol{d},t}} q_{\boldsymbol{d},t}$, the model imposes $\chi \geq 1$, ensuring that staying at home never reduces the effective level of the public good. For this condition to hold in every location-sector-time cell, the relevant restriction is $1 \leq \chi < \min_{\boldsymbol{d},t} (1 + r_{\boldsymbol{d},t})^{1/\gamma}$.

Given the wife's labor-supply rule, the ex ante expected utility from choosing $\boldsymbol{d}$ at time t is

$$\begin{aligned} W^c_{\boldsymbol{d},t} \equiv \mathbb{E}[\mathcal{U}_{\boldsymbol{d},t}(\beta)] &= \mathcal{A}\frac{w^M_{j,n_m,t}}{P_{j,t}} \left[(1 + r_{\boldsymbol{d},t}) F_\beta(\beta^*_{\boldsymbol{d},t}) + \int_{\beta^*_{\boldsymbol{d},t}}^{\infty} \beta^\gamma f_\beta(\beta)\, d\beta\right] \\ &= \frac{w^M_{j,n_m,t}}{P_{j,t}} \cdot \underbrace{\mathcal{A}\left[(1 + r_{\boldsymbol{d},t}) + \chi^\kappa \frac{\gamma}{\kappa - \gamma}(1 + r_{\boldsymbol{d},t})^{1-\kappa/\gamma}\right]}_{B_{\boldsymbol{d},t}}, \end{aligned}$$

where I can express $B_{\boldsymbol{d},t}$ entirely in terms of $\rho_{\boldsymbol{d},t}$ as $B_{\boldsymbol{d},t} = \mathcal{A}\,\chi^\gamma\,(1 - \rho_{\boldsymbol{d},t})^{-\gamma/\kappa}\,\frac{\kappa/\gamma - \rho_{\boldsymbol{d},t}}{\kappa/\gamma - 1}$. This term captures the quality-adjusted value of destination-sector bundle $\boldsymbol{d}$ for couples. Conditional on choosing $\boldsymbol{d}$, the couple's payoff combines the husband's sector-specific wage with the expected gain from the wife's endogenous labor-supply decision, summarized by $B_{\boldsymbol{d},t}$. Finiteness of the expected value requires $\kappa > \gamma$.

Comparative statics of the market-work probability. The probability that the wife works in the market in equation (22) depends only on the relative wage r, the parameters of the Pareto distribution, χ and κ, and the public-good expenditure share, γ. It does not depend on the level of either spouse's wage.

Participation rises with the wife's wage relative to her husband's:

$$\frac{\partial \rho}{\partial r} = \frac{\kappa}{\gamma} \cdot \frac{1-\rho}{1+r} > 0. \tag{23}$$

A higher relative wage raises the opportunity cost of staying home. Thus, the threshold β^* rises, and some draws that used to justify staying home no longer do, so more wives work.

Participation falls with the lower bound of home productivity, χ, and rises with the shape parameter, κ:

$$\frac{\partial \rho}{\partial \chi} = -\frac{\kappa}{\chi}(1-\rho) < 0, \qquad \frac{\partial \rho}{\partial \kappa} = -\frac{(1-\rho)\log(1-\rho)}{\kappa} > 0. \tag{24}$$

Raising the lower bound χ scales the whole distribution of home productivity upward while leaving the threshold unchanged, so every couple gains more when she stays at home. More couples draw above β^*, so participation falls. Because χ is a scale parameter, the effect is proportional: the elasticity of $1-\rho$ with respect to χ is κ. By contrast, a larger shape parameter concentrates the draws near χ and thins the right tail. Staying home takes an exceptionally good draw, one above β^*, and those draws become rarer, so more wives work.

Participation falls with the public-good share, γ:

$$\frac{\partial \rho}{\partial \gamma} = -\frac{\kappa(1-\rho)}{\gamma^2}\log(1+r) < 0. \tag{25}$$

A higher public-good share raises the utility from any given β, since β^γ grows with γ when $\beta > 1$. The threshold $(1+r)^{1/\gamma}$ therefore falls: a smaller home-productivity draw now matches the income gain from working. More couples choose home production. The effect is proportional to $\log(1+r)$ because $\log \beta^* = \log(1+r)/\gamma$, so a change in γ moves the threshold more when the income gain from working is larger.

Indirect utility. The indirect utility relevant for household h's dynamic migration problem in equation (16) is[16]

$$W^h_{\boldsymbol{o},t} = \begin{cases} \frac{w^M_{i,k,t}}{P_{i,t}} & \text{if} \quad h = s_m \\ \frac{w^F_{i,k,t}}{P_{i,t}} & \text{if} \quad h = s_f \\ \frac{B_{\boldsymbol{o},t} w^M_{i,k_m,t}}{P_{i,t}} & \text{if} \quad h = c \end{cases}$$

Here, $P_{i,t}$ denotes the regional aggregate consumption price index defined in equation (13).

[16] Singles have the same preferences over private consumption and the household good as spouses, $\gamma \log Q + (1-\gamma)\log c$, but do not share Q with a spouse. Their indirect utility is therefore $\log(w/P)$ up to the constant $\gamma \log \gamma + (1-\gamma)\log(1-\gamma)$, which cancels from all welfare changes. This formulation assumes that welfare is comparable across marital statuses; the money-metric index below makes this comparison explicit.

III.2 Equilibrium

The economic geography at time t, denoted by Γ_t, consists of the time-varying exogenous sectoral productivities and the time-invariant shares of single individuals born in each region, migration costs, iceberg trade costs, and non-employment benefits: $\Gamma_t = \{A_{i,k,t}, \bar{s}_i, \mu_{\boldsymbol{od}}, \tau_{ik,jn}, b_j\}_{i,j=1,\,k,n=1}^{L,K}$. I collect the remaining time-invariant parameters in the vector $\bar{\Gamma} = \{N, L, K, \delta, \chi, \kappa, \alpha, \gamma, \sigma, \nu, \theta, \{\psi_{k,m}, \psi_{k,f}, \eta_k, \iota_k\}_{k=1}^{K}\}$.

Let $N_{\boldsymbol{d},t}^{s_m}$ and $N_{\boldsymbol{d},t}^{s_f}$ denote, respectively, the masses of single men and single women living in location j and working in sector $n \in \{0, \ldots, K\}$ at time t. Let $N_{\boldsymbol{d},t}^{c}$ denote the mass of couples living in location j in which the husband is assigned to sector $n_m \in \{0, \ldots, K\}$ and the wife is assigned to productive market sector $n_f \in \{1, \ldots, K\}$. The supplies of male and female labor to productive sectors $n = 1, \ldots, K$ are therefore

$$M_{j,n,t} = N_{j,n,t}^{s_m} + \sum_{n_f=1}^{K} N_{j,n,n_f,t}^{c}, \qquad F_{j,n,t} = N_{j,n,t}^{s_f} + \sum_{n_m=0}^{K} \rho_{j,n_m,n,t} N_{j,n_m,n,t}^{c}. \tag{26}$$

These expressions make the components of a couple's bundle explicit. The husband's sector is the second argument, and the wife's sector is the third. Accordingly, male labor supply in sector n sums over the wife's sector, whereas female labor supply in sector n sums over the husband's sector and weights each cell by the wife's probability of participation. Thus, female labor supply responds to wages through both mobility and participation, whereas male labor supply responds only through mobility.

Let $N_t = \left\{\{N_{\boldsymbol{d},t}^{s_m}\},\ \{N_{\boldsymbol{d},t}^{s_f}\},\ \{N_{\boldsymbol{d},t}^{c}\}\right\}$ denote the complete distribution of household masses across locations and sectoral bundles at time t. Because the problems of intermediate-good firms, final-good firms, and married women's labor supply are static, I define the *temporary equilibrium* as follows.

Definition 1. *Given $(N_t, \Gamma_t, \bar{\Gamma})$, a **temporary equilibrium** is a vector of productive-sector wages $\{w_{j,n,t}^{M}, w_{j,n,t}^{F}\}$, sectoral prices $\{p_{j,n,t}, P_{j,n,t}\}$, regional price indices $\{P_{j,t}\}$, and allocations $\{\rho_{\boldsymbol{d},t}, M_{j,n,t}, F_{j,n,t}, Y_{j,n,t}, Q_{j,n,t}, Q_{j,t}, \pi_{ik,jn,t}\}$ such that, for every productive region-sector pair jn and period t:*

(i) Intermediate-good firms maximize profits subject to the CES production technology, and wages satisfy (11);

(ii) Final-good firms maximize profits subject to sectoral CES and regional Cobb-Douglas aggregation. Sectoral price indices satisfy (12), and regional price indices satisfy (13);

(iii) Couples optimally choose the wife's labor supply, and participation probabilities satisfy (22) for every sector pair (n_m, n_f);

(iv) Labor markets clear: $p_{j,n,t}Y_{j,n,t} = w^M_{j,n,t}M_{j,n,t} + w^F_{j,n,t}F_{j,n,t}$;

(v) Goods markets clear under balanced trade: $p_{i,k,t}Y_{i,k,t} = \sum_{j=1}^{L}\sum_{n=1}^{K}\pi_{ik,jn,t}\,\iota_n P_{j,t}Q_{j,t}$.

Given the path of exogenous fundamentals and an initial distribution of households across location-sector bundles, denoted by $N_0 = \left\{\{N^{s_m}_{\boldsymbol{o},0}\},\ \{N^{s_f}_{\boldsymbol{o},0}\},\ \{N^{c}_{\boldsymbol{o},0}\}\right\}$, I can define the model's *sequential competitive equilibrium*. Using the two-stage choice probabilities in Equations 17 and 18, the population of each household type evolves according to

$$N^h_{\boldsymbol{d},t+1} = \sum_{\boldsymbol{o}\in\mathcal{S}_h} \lambda^h_{\boldsymbol{od}|j,t}\,\lambda^h_{\boldsymbol{o}j,t}\,N^h_{\boldsymbol{o},t}, \qquad \boldsymbol{d}\in\mathcal{S}_h(j), \quad \forall h. \tag{27}$$

Definition 2. *Given* $(N_0, \Gamma_t, \bar{\Gamma})$*, a* ***sequential competitive equilibrium*** *is a sequence of productive-sector wages* $\{w^M_{j,n,t}, w^F_{j,n,t}\}_{t=0}^{\infty}$*, prices* $\{p_{j,n,t}, P_{j,n,t}, P_{j,t}\}_{t=0}^{\infty}$*, values* $\{V^h_{\boldsymbol{o},t}\}_{t=0}^{\infty}$*, and allocations* $\{\rho_{\boldsymbol{d},t}, \pi_{ik,jn,t}, \lambda^h_{\boldsymbol{o}j,t}, \lambda^h_{\boldsymbol{od}|j,t}, N^h_{\boldsymbol{o},t}, M_{j,n,t}, F_{j,n,t}, Y_{j,n,t}, Q_{j,n,t}, Q_{j,t}\}_{t=0}^{\infty}$ *for every household type* h*, such that, in each period* t*:*

(i) The temporary equilibrium holds;

(ii) Couples and single individuals solve their respective migration problems: value functions satisfy (16), while location and conditional sectoral choice probabilities satisfy (17) and (18);

(iii) The population of each household type evolves according to (27);

(iv) Productive-sector male and female labor supplies satisfy (26).

III.3 Welfare Analysis of a Sectoral Productivity Shock

To quantify welfare, I combine equation (16) with $\mu_{\boldsymbol{oo}} = 0$ to express the expected lifetime utility of household type h in bundle $\boldsymbol{o}$ at time t as

$$\begin{aligned} V^h_{\boldsymbol{o},t} &= \log(W^h_{\boldsymbol{o},t}) + \delta V^h_{\boldsymbol{o},t+1} - \theta\log\left(\lambda^h_{\boldsymbol{o}i,t}\right) - \nu\log\left(\lambda^h_{\boldsymbol{oo}|i,t}\right) \\ &= \sum_{s=t}^{\infty}\delta^{s-t}\log\left(\frac{W^h_{\boldsymbol{o},s}}{\left(\lambda^h_{\boldsymbol{o}i,s}\right)^{\theta}\left(\lambda^h_{\boldsymbol{oo}|i,s}\right)^{\nu}}\right). \end{aligned} \tag{28}$$

Here, $\lambda^h_{\boldsymbol{o}i,t}$ denotes the probability of remaining in the origin location, and $\lambda^h_{\boldsymbol{oo}|i,t}$ denotes the probability of remaining in the origin sector bundle, conditional on remaining in the origin location. Appendix B.1 derives this recursion.

I define the no-shock economy by an alternative sequence of sectoral productivities $\{A_{j,n,t}\}_{t\geq 1}$. Following CDP's timing assumption, the change is unanticipated at $t = 0$, and

households fully learn the new productivity path at $t = 1$. Let $\hat{x}_{t+1} = \dot{x}'_{t+1}/\dot{x}_{t+1}$ denote the ratio of changes over time in the no-shock and with-shock equilibria, where $\dot{x}_{t+1} \equiv x_{t+1}/x_t$, and let $\tilde{x}_t \equiv x'_t/x_t = \prod_{s=1}^{t} \hat{x}_s$ denote the corresponding ratio of levels, under the normalization that the value of x is the same in both economies in $t = 0$, such that $x'_0 = x_0$.

I use Dynamic Hat Algebra[17] to compute expected lifetime utility in the with-shock economy, $V^h_{\boldsymbol{o},0}$, and in the no-shock economy, $V'^h_{\boldsymbol{o},0}$. I define the consumption compensating variation for household type h in bundle $\boldsymbol{o}$ at time $t = 0$, denoted by $\zeta^h_{\boldsymbol{o}}$, implicitly by

$$V'^h_{\boldsymbol{o},0} = V^h_{\boldsymbol{o},0} + \sum_{s=0}^{\infty} \delta^s \log(\zeta^h_{\boldsymbol{o}}) \iff \log(\zeta^h_{\boldsymbol{o}}) = (1-\delta)(V'^h_{\boldsymbol{o},0} - V^h_{\boldsymbol{o},0}).$$

Because the temporary equilibrium and migration shares are identical across the two economies at $t = 0$, the welfare change in consumption-equivalent units is

$$Z^h_{\boldsymbol{o}} \equiv \log(\zeta^h_{\boldsymbol{o}}) = \sum_{s=1}^{\infty} \delta^s \log\left(\frac{\hat{W}^h_{\boldsymbol{o},s}}{\left(\hat{\lambda}^h_{\boldsymbol{o}i,s}\right)^{\theta} \left(\hat{\lambda}^h_{\boldsymbol{o}\boldsymbol{o}|i,s}\right)^{\nu}} \right), \tag{29}$$

where

$$\hat{W}^h_{\boldsymbol{o},s} = \begin{cases} \hat{w}^M_{i,k_m,s}/\hat{P}_{i,s} & \text{if} \quad \boldsymbol{o} = (i,k_m),\ h = s_m, \\ \hat{w}^F_{i,k_f,s}/\hat{P}_{i,s} & \text{if} \quad \boldsymbol{o} = (i,k_f),\ h = s_f, \\ \hat{B}_{\boldsymbol{o},s}\hat{w}^M_{i,k_m,s}/\hat{P}_{i,s} & \text{if} \quad \boldsymbol{o} = (i,k_m,k_f),\ h = c. \end{cases}$$

equation (29) decomposes the welfare change into changes in real income, which for couples include the change in the value of the wife's participation option, $\hat{B}_{\boldsymbol{o},s}$, and changes in the option value of moving, captured by the staying probabilities. Conditional on the observed staying shares, a larger θ, corresponding to a lower migration elasticity, increases the welfare weight assigned to changes in the probability of remaining in the origin location. Similarly, a larger ν increases the weight assigned to changes in the probability of remaining in the origin sector bundle conditional on staying in that location. Thus, declines in these probabilities enter the welfare measure with weights θ and ν, respectively, where $\theta \geq \nu$.

I aggregate welfare changes across initial location-sector bundles to obtain the aggregate welfare effect of a sectoral productivity shock:

$$Z = \underbrace{\sum_{i=1}^{L}\sum_{k=0}^{K} N^{s_m}_{\boldsymbol{o},0} Z^{s_m}_{\boldsymbol{o}}}_{Z^{sm}} + \underbrace{\sum_{i=1}^{L}\sum_{k=0}^{K} N^{s_f}_{\boldsymbol{o},0} Z^{s_f}_{\boldsymbol{o}}}_{Z^{sf}} + \underbrace{\sum_{i=1}^{L}\sum_{k_m=0}^{K}\sum_{k_f=1}^{K} N^{c}_{\boldsymbol{o},0} Z^{c}_{\boldsymbol{o}}}_{Z^{c}}. \tag{30}$$

[17] See Appendix C for more details.

Money Metric Welfare Index. The aggregate in equation (30) adds welfare changes measured in utility units, which are not directly comparable between singles and spouses. For couples, \$1 of market expenditure can generate more than \$1 in consumption-equivalent value due to economies of scale within marriage. Thus, a 1% gain for a spouse and a 1% gain for a single person are percentages of different bases. Following Chiappori and Meghir (2015), I construct a Money Metric Welfare Index (MMWI) for singles and couples to obtain a comparable money-metric measure of welfare changes for the quantitative analysis.

The MMWI measures each person's welfare as the income they would need to attain their utility level if they had to purchase the household public good at its full market price. For a single, this income is the real wage, w/P, because a single already pays the full price of the public good. Thus, the MMWI for a spouse can be understood as the income they would need to attain the utility level they get as married if they were single.

Constructing the MMWI for a spouse requires three intermediate objects. I first define the Conditional Sharing Rule (CSR), denoted by $\tilde{\varsigma}$, which allocates the income remaining after expenditure on the public good between the spouses. Let α_g represent the bargaining weight of the spouse of gender g, such that $\alpha_f = \alpha$ and $\alpha_m = 1-\alpha$. Given the functional form of the couple's utility, the CSR corresponds to a fixed share of income after paying for the public good, where the share reflects each spouse's bargaining weight, so that $\tilde{\varsigma}_g = \alpha_g(1-\gamma)I(e)$. Then, I define the Lindahl prices, which correspond to the share of the cost of one unit of Q, $P_Q \equiv P\beta^{-e}$, paid by each spouse. Because both spouses value Q identically, each bears a share of its cost equal to their bargaining weight, $P_g = \alpha_g P_Q$. Next, I combine the CSR and Lindahl prices to compute the Generalized Sharing Rule (GSR), which adds to each spouse's private expenditure the value of the public good at their Lindahl price, $\varsigma_g^* = \tilde{\varsigma}_g + P_g Q^* = \alpha_g I(e)$. Thus, a wife who remains at home does not earn a wage, but is allocated resources equivalent to αw^M, a share of her husband's wage.

The MMWI of each spouse rescales their GSR by the ratio of the public good's market price to their Lindahl price, raised to the power γ. Let $\bar{P}_Q = P$ denote the market price of the public good, that is, its price without the boost from home production. Then, the MMWI of spouse g, denoted by m_g, corresponds to

$$m_g = \underbrace{\varsigma_g^*}_{\alpha_g I(e)} \cdot \underbrace{\bar{P}_Q^{\gamma}}_{P^{\gamma}} \cdot \underbrace{P_g^{-\gamma}}_{(\alpha_g P\beta^{-e})^{-\gamma}} = \alpha_g^{1-\gamma}\beta^{e\gamma} I(e). \tag{31}$$

Because couples choose their bundle before β is realized, I weight welfare changes below by the ex ante index, $\bar{m}_g = \alpha_g^{1-\gamma}\,\mathbb{E}[\beta^{e\gamma} I(e)]$, which moves with the wife's participation probability.

Summing across spouses, $m_f + m_m = \left(\alpha^{1-\gamma} + (1-\alpha)^{1-\gamma}\right)\beta^{e\gamma} I(e)$, where $\alpha^{1-\gamma} + (1-$

$\alpha)^{1-\gamma} > 1$. Thus, the spouses would jointly require more income than the household possesses if each had to purchase the public good separately, and $\alpha^{1-\gamma} + (1-\alpha)^{1-\gamma}$ measures the economies of scale within marriage in money terms. Hence, I now measure welfare for couples in the same units as wages and prices, so I can compare it with that of singles.

I aggregate welfare changes across household types by weighting each person's consumption-equivalent change by their money metric in the initial period, $\bar{Z}^{\$} = \bar{Z} + \text{Cov}(m, Z)/\bar{m}$, where $\bar{Z}$ is the unweighted per-person average and the covariance is taken across individuals. To a first-order approximation, mZ is the income that compensates a person for the shock, so $\bar{Z}^{\$}$ is the aggregate money gain as a share of aggregate money-metric income. The covariance term is positive when the gains accrue to individuals with larger money metrics, such as spouses, and negative when they accrue to those with smaller money metrics, such as singles. Because each spouse's expected utility is a constant multiple of the couple's (equation (B.1)), both spouses inherit the couple's consumption-equivalent change, $Z^c_{\boldsymbol{o}}$; only their weights differ. Appendix B.2 derives the index step by step.

The model's mechanisms map directly onto the three empirical facts. Sectoral productivity shocks transmitted through trade lower wages in exposed region-sector cells and lead workers to leave them, consistent with Fact 1. Because men and women are distributed differently across sectors, these wage changes differ by gender. Couples jointly choose a single location, but spouses enter this decision asymmetrically: the husband's wage $w^M_{j,n_m,t}$ scales the couple's entire payoff, whereas the wife's wage enters only through the relative wage $r_{\boldsymbol{d},t}$ in $B_{\boldsymbol{d},t}$. This asymmetry provides the mechanism underlying Fact 2. Finally, the participation threshold $\beta^*_{\boldsymbol{d},t}$ generates the adjustment margin in Fact 3, with the direction observed in the data: a decline in the husband's sectoral wage raises $r_{\boldsymbol{d},t}$, increases the threshold, and induces more married women to enter market work, whereas a decline in the wife's own sectoral wage lowers $r_{\boldsymbol{d},t}$ and has the opposite effect. Section IV disciplines the model using the data.

IV. Bringing the Model to the Data

This section applies the model developed above to the data to quantify the China Trade Shock in an economy with heterogeneous households. It first describes the quantitative environment, distinguishing parameters drawn from the literature from inputs observed directly in the data. It then presents the calibration of preference dispersion across sectors and locations, as well as the couple-specific parameters, followed by the method for quantifying the China Trade Shock.

IV.1 Quantitative Environment

The quantitative environment follows CDP. The economy comprises 87 locations: the 50 US states, with the District of Columbia included in Virginia, and 37 foreign trade partners, consisting of China, the 35 largest US trading partners, and the Rest of the World. It includes 22 productive sectors, comprising 12 manufacturing sectors, retail, construction, and eight service sectors, as well as a non-employment sector. Each model period corresponds to one quarter, and I solve the model for 200 quarters beginning in 2000Q1. I choose to adopt CDP's quantitative environment because it provides a trade block that has already been disciplined by the data, ensuring that the productivity and trade fundamentals are not constructed specifically for this analysis. One caveat is the difference in timing: the empirical analysis in Section II follows ADH's 2001–2012 window (Autor et al., 2021), whereas the quantitative exercise retains the earlier 2000–2007 window (Autor et al., 2013).

Because the trade side is taken directly from CDP, the quantitative exercise uses their full production block rather than the simplified version presented in Section III. Intermediate-good firms use both labor and intermediate inputs from other sectors, and total factor productivity combines the Hicks-neutral component introduced in Section III with a variety-specific component. Implementing the household extension therefore requires replacing the labor input demanded by intermediate-good firms with the effective labor aggregate in equation (8). I take all other inputs to the trade block directly from CDP.[18]

δ and σ from previous literature. I set the discount factor to $\delta = 0.99$, following CDP, and take the within-sector elasticity of substitution between male and female labor, σ, from Acemoglu et al. (2004) rather than estimating it directly. This choice is motivated by the identification requirements implied by the CES technology in Section III:

$$\log\left(\frac{w^M_{i,k,t}}{w^F_{i,k,t}}\right) = \log\left(\frac{\psi_{k,m}}{\psi_{k,f}}\right) - \frac{1}{\sigma}\log\left(\frac{M_{i,k,t}}{F_{i,k,t}}\right).$$

Identifying σ requires exogenous variation in the relative supply of male and female labor; without such variation, the estimated coefficient would be biased. Acemoglu et al. (2004) estimate an analogous relationship (their Equation 11), using the mobilization of men during World War II as an exogenous shift in women's relative labor supply. Accordingly, I set σ to 1.4, the lower bound of their estimated range. This conservative choice reflects the fact that the cohorts examined in their study were almost entirely retired by 2000Q1, when

[18] Sectoral trade elasticities $\{\eta_n\}$ are from Caliendo and Parro (2015); bilateral trade shares $\{\pi_{ik,jn,t}\}$, region-sector output and expenditure, and the path of sectoral productivity changes in China are from CDP; final-good expenditure shares $\{\iota_n\}$ come from the Bureau of Economic Analysis (BEA) input-output tables.

the quantitative model begins. Since then, as more women entered the labor force, they specialized in occupations where they had a comparative advantage, so men and women sort differently across occupations and are less substitutable within a sector (Ngai and Petrongolo, 2017; Cortes et al., 2023; Bhalotra et al., 2025).

Directly observed inputs. I construct household-side inputs from US microdata. I measure the initial distribution of households across states and sector pairs, $\{N^{s,m}_{\boldsymbol{o},0}, N^{s,f}_{\boldsymbol{o},0}, N^{c}_{\boldsymbol{o},0}\}$,[19] and the state-level share of singles, $\bar{s}_i$, using the 5% IPUMS Census 2000. I measure male and female employment, wages, and wives' labor-force participation using the 2000 Census and the American Community Survey (ACS) for 2001–2007. I recover the sector-specific gender-intensity parameters $\{\psi_{k,m}, \psi_{k,f}\}$ from the hours worked by gender in each sector. I measure regional non-employment income, b_j, using state unemployment-insurance benefits from the US Department of Labor and hold it fixed at its 2000 value. Table 4 summarizes these inputs and their sources.[20]

One discrepancy between the model and the data remains: the model assigns every wife to a productive sector, but the data do not reveal the potential sector of a non-employed wife. In the model, couples first choose the wife's productive sector and then determine her participation status. A naive approach would assign non-employed wives to productive sectors in proportion to the sectoral distribution of employed wives. This approach creates a selection problem: sectors employing larger shares of wives would also receive larger shares of non-employed wives, even though wives assigned to those sectors are precisely the least likely to be out of the labor force. To recover each non-employed wife's potential sector, I use the model to derive a selection correction.[21] The implied mass of non-employed married women in state j and sector n is

$$NonF^{c}_{j,n,t} = \underbrace{F^{c}_{j,n,t}}_{\text{observed}} \cdot \underbrace{\frac{1-\rho_{j,n,t}}{\rho_{j,n,t}}}_{\text{the model's odds of non-employment}}.$$

Scaling observed employment by $1/\rho_{j,n,t}$ corrects for selection and assigns non-employed wives to their potential productive sectors.

[19] The quantitative exercise holds household composition fixed: marriage, divorce, fertility, and population growth are excluded, so couples remain together throughout the transition. Figure A.4 provides supporting evidence for this assumption.

[20] Two conventions determine the interpretation of the participation margin. First, non-employment refers to being *out of the labor force*, rather than unemployed. Including unemployed individuals would roughly double the measured flows into non-employment and would confound the wife's participation decision with the husband's job loss. Second, following the convention in this literature, an unemployed worker retains the sector of their most recent job. Thus, a positive sector index does not necessarily indicate current employment, and I always measure participation using an explicit labor-force indicator.

[21] See Appendix D.1 for details.

Table 4: Observed Quantitative Inputs

Object	Description	Source
$\bar{s}_i$	State-level share of singles	IPUMS Census 2000
N_i	State-level population	IPUMS Census 2000 and ACS 2001–2007
$N^{s,m}_{\boldsymbol{o},0}, N^{s,f}_{\boldsymbol{o},0}$	Initial populations of single men and women by state × sector	IPUMS Census/ACS
$N^{c}_{\boldsymbol{o},0}$	Initial population of couples by state × husband-wife sector pair	IPUMS Census/ACS
$M_{\boldsymbol{o},t}, F_{\boldsymbol{o},t}$	Male and female employment by state × sector	IPUMS Census/ACS
$\psi_{k,m}, \psi_{k,f}$	Sector-specific gender intensities	Hours worked by gender, IPUMS Census/ACS
b_j	State-level non-employment income	US Department of Labor unemployment-insurance benefits, fixed at the 2000 value

Notes. The sample follows CDP and includes individuals aged 25–65 across 50 states, excluding widows. Non-employment refers to being out of the labor force rather than being unemployed. Sectors are classified according to CDP's 22-sector partition.

The final data inputs are the initial migration matrices. Because I solve the model using Dynamic Hat Algebra, the time-invariant mobility costs $\mu_{\boldsymbol{od}}$ cancel from the recursion governing the choice probabilities (Proposition C.2). The recursion therefore requires only two initial objects for each household type: the location shares $\lambda^{h}_{\boldsymbol{o}j,0}$ defined in equation (17) and the within-location shares $\lambda^{h}_{\boldsymbol{od}|j,0}$ defined in equation (18).

Following CDP, I construct the initial unconditional transition probability $\lambda^{h}_{\boldsymbol{od},0}$ from equation (19) as the product of an interstate migration probability from the ACS and a sectoral transition probability from the Current Population Survey (CPS). For interstate movers, I impute the sectoral transition probability using the destination state's sectoral transition matrix:

$$\lambda^{h}_{(i,k)(j,n),0} = \underbrace{f^{h}_{ij}}_{\text{ACS}} \times \underbrace{P^{h}_{j}(n \mid k)}_{\text{CPS}}.$$

This construction presents two challenges: the surveys differ in frequency, and CPS cells become sparse once split by household type. First, the ACS is annual, whereas the CPS is monthly. I address this mismatch following CDP. Second, to address the sparse cells, I

temporarily aggregate the 12 manufacturing sectors into a single sector, and do the same for the service sectors. I then disaggregate the resulting transition probabilities using national gender-specific sorting patterns. Appendix D.2 documents each step in the construction of the migration matrices.

IV.2 Dispersion of Preferences across Sectors and Locations

The model features two dispersion parameters: ν, taken from the literature, and θ, which I calibrate. The parameter ν governs substitution across sectors within a location (equation (18)), whereas θ governs substitution across locations (equation (17)).

CDP's dispersion parameter maps directly onto ν, because it is estimated only from sectoral transitions within a location. In fact, following Artuc et al. (2010), they estimate their dispersion parameter using sectoral transitions among men in the March Current Population Survey (CPS) from 1975 to 2000. Because the CPS follows addresses, respondents who change residence are removed from the sample. The matched panel therefore records sectoral transitions only among individuals who remain in the same location and contains no transitions across locations. Consequently, the estimated parameter corresponds to applying Artuc et al. (2010)'s method to the within-location choice in equation (18), which is precisely the role of ν in the present model. The only distinction is that continuation values V^h are household-type specific, reflecting the economies of scale associated with marriage. Because the model does not impose differential rates of sectoral substitution across household types, I set ν to CDP's quarterly value of 5.34 for all household types.

I calibrate θ by indirect inference to match the population response to import competition in Autor et al. (2013), following Rodríguez-Clare et al. (2026). Autor et al. (2013) regress the change in the log of the working-age population of a commuting zone on its exposure to Chinese imports and obtain a coefficient of -0.050. I estimate the same regression using model-generated state-level data. The dependent variable is the percentage change in state population between 2000 and 2007, scaled to a decade as in Autor et al. (2013). The regressor is the state's exposure to Chinese imports over 2000–2007, predicted using the Autor et al. (2013) instrument and rescaled to the mean of their commuting-zone measure so that the model and empirical coefficients have the same units.[22] The calibrated value, $\theta = 10.67$, generates a coefficient of -0.051, compared with the target of -0.050. Because $\nu < \theta$, households substitute more readily across sectors within a location than across locations.

Accounting for couples increases the dispersion parameter needed to match the same

[22] I restrict the model sample to the 48 states in the Autor et al. (2013) estimation sample, excluding Alaska and Hawaii. Using first-stage fitted exposure makes the ordinary least squares coefficient in the model the counterpart of the two-stage least squares coefficient in the data.

population response. Calibrating θ in an otherwise identical economy composed entirely of single individuals, in which each individual independently chooses a location and sector, yields a $\hat{\theta}$ that is 5% lower under the same target and value of ν. Given the model's solution timing, this difference provides a lower bound on the quantitative importance of accounting for couples.[23]

IV.3 Calibrating the Couple's Maximization Problem

Four parameters govern the couple's maximization problem and have no counterpart in CDP because CDP's economy has no couples. They are the wife's weight in the household objective α, the public-good expenditure share γ, and the lower bound and shape of the home-production shock, χ and κ. Because I solve the couple's problem following the three-stage formulation in Chiappori and Mazzocco (2017), I identify them *sequentially*, each from a distinct piece of variation: α and γ from household expenditure shares, the slope κ/γ from the China Trade Shock, and χ from the level of married women's participation. Table 5 summarizes the resulting values; Appendix D.3 describes the construction in detail.

I calibrate α and γ to exactly match observed expenditure shares. Because the couple's utility function in equation (20) is Cobb–Douglas, α corresponds to the wife's share of total private consumption,[24] while γ corresponds to the share of public-good expenditure in total household expenditure:

$$\alpha = \frac{c_f^*}{c_f^* + c_m^*}, \qquad \gamma = \frac{q^*}{q^* + c_f^* + c_m^*}.$$

I measure both expenditure shares using the 2000–2007 Consumer Expenditure Interview Survey. I measure the public-good share γ directly as the share of total expenditure devoted to market inputs into home production.[25] Estimating the bargaining weight α requires an additional step because the observed assignable private good, adult apparel, represents only a subset of private consumption. I therefore recover the spouses' resource shares using the method of Browning et al. (2013), which relies on expenditures by *singles*. The resulting

[23] At the model's θ, the all-agent economy delivers a 3.6% weaker response. This may seem counterintuitive at first glance, because singles are more mobile than couples. If I had matched the long-run response, the $\hat{\theta}$ in the all-singles economy would be higher, as the intuition predicts.

[24] The quantification holds α fixed, thereby ruling out shock-induced renegotiation of intrahousehold bargaining power. It also abstracts from the intensive margin of the wife's labor supply: the wife either participates in market work or does not. The latter restriction appears empirically innocuous because, alongside the participation responses documented in Fact 3, wives' hours exhibit no statistically significant response to either spouse's exposure. These estimates are available upon request.

[25] The home-production goods comprise shelter, utilities, food at home, and market substitutes for home time, such as domestic services and appliance repair.

values are $\alpha = 0.47$ and $\gamma = 0.49$.[26]

I calibrate κ and χ using the wife's participation rule in equation (22), which is log-linear in the relative wage. Expressing the equation in terms of the non-participation probability and taking logs yields

$$\log\big(1-\rho(r)\big) = \underbrace{\kappa \log \chi}_{\text{intercept}} - \underbrace{(\kappa/\gamma)}_{\text{slope}} \log(1+r). \tag{32}$$

Direct estimation of equation (32) is complicated by endogeneity and sample selection. First, the relative wage r faced by a couple is an equilibrium outcome. Second, I observe relative wage changes only for couples in which the wife participates in market work, creating a truncation problem.

To address the endogeneity of r, I use the empirical results in Section II to provide exogenous variation through the China Trade Shock. When the husband is exposed, the shock directly affects his wage but not his wife's wage, influencing her participation decision through the relative wage r, as required by the exclusion restriction underlying equation (32). The same shock simulated in the quantitative exercise therefore disciplines the model's participation margin. To address truncation, the change in $1+r$ is evaluated at the relative wage of all couples, not only those in which the wife works (Appendix D.3.2).

Because equation (32) is linear, I identify its slope using the following Wald ratio:

$$\frac{\kappa}{\gamma} = -\frac{\Delta \log(1-\rho)}{\Delta \log(1+r)},$$

where I measure both changes within the same empirical experiment. Wives of exposed husbands increase their labor force participation by 4.2 percentage points from a baseline of 0.78 (Panel b of Table 3), implying $\Delta \log(1-\rho) = -0.212$. The husband's exposure increases the relative wage r by 12.2 log points, corresponding to a 4.8 log-point increase in $1+r$ (Table D.1). Together, these estimates imply

$$\frac{\kappa}{\gamma} = -\frac{-0.212}{0.048} = 4.44 \quad \Longrightarrow \quad \kappa = \gamma \times \frac{\kappa}{\gamma} = 2.17.$$

The resulting calibration satisfies the model's theoretical restriction, $\kappa > \gamma$.

Conditional on the estimated slope, I calibrate χ to match the *level* of married women's labor force participation. Specifically, I choose χ so that the model's mass-weighted average participation rate equals its empirical counterpart in $t_0 = 2000$Q1. The resulting value, $\chi = 1.92$, satisfies the support restriction $\chi \geq 1$ and yields an interior participation margin in every cell. The calibration therefore uses two participation rates, each for a different purpose:

[26] Appendix D compares these estimates with those in the previous literature and discusses potential sources of discrepancy.

I identify the slope from the PSID experiment using that sample's baseline participation rate of 0.78, whereas χ anchors the level to the model population's mass-weighted participation rate of 0.621.

The calibrated parameters imply that married women's labor force participation responds roughly one-for-one to the female wage. Differentiating $\rho = 1 - \chi^{\kappa}(1+r)^{-\kappa/\gamma}$ with respect to the female wage and evaluating the resulting expression at the model's anchor, $\bar{\rho} = 0.621$ and $r_0 = 0.607$, yields

$$e \;=\; \frac{\partial \log \rho}{\partial \log w_f} \;=\; \frac{\kappa}{\gamma} \cdot \frac{1-\bar{\rho}}{\bar{\rho}} \cdot \frac{r_0}{1+r_0} \;=\; 1.02$$

Thus, a 1% increase in the female wage raises married women's labor force participation by 1.02%. This elasticity lies within the range for married women surveyed by Blundell and MaCurdy (1999) and exceeds the ≈ 0.25–0.30 aggregate extensive-margin elasticity preferred by Chetty et al. (2011), which is not specific to married women.

Table 5: Calibrated Household Parameters

Object	Name	Value
α	Wife's weight in the household objective	0.47
γ	Public-good expenditure share	0.49
κ/γ	Slope of the participation rule	4.44
κ	Pareto shape parameter for home productivity	2.17
χ	Pareto lower bound for home productivity	1.92

Notes. See Appendix D for details.

IV.4 Quantifying the China Trade Shock

To quantify the effects of the China Trade Shock, I need to define the shock within the model. Following CDP, I represent the China Trade Shock as an increase in total factor productivity in *Chinese* manufacturing sectors and calibrate the path of Chinese sectoral productivity so that the resulting growth in US imports from China reproduces the first stage in Autor et al. (2013).[27] The model therefore defines the shock at the industry level, consistent with the reduced-form evidence in Section II.

I quantify the shock by comparing the two economies defined for productivity shocks in Section III.3. As in CDP, the *with-shock* economy includes the observed growth in Chinese

[27]The shock is a foreign supply shock that is transmitted through the trade block as a demand shock to US producers: cheaper Chinese imports reduce both the prices and quantities at which US manufacturers sell in domestic and third-country markets.

productivity, whereas the *no-shock* economy eliminates this growth. Formally, $\dot{A}'_{\text{China},k,t} = 1$ for every manufacturing sector k and period t in the no-shock economy.

All results reported below are differences between the with-shock and no-shock economies. Accordingly, a positive employment effect indicates that the China Trade Shock increased employment relative to a world without the shock, while a positive welfare effect indicates that the relevant household type was better off.

V. Quantitative Consequences of the Shock Under Joint Decision-Making

In this section, I quantify how the China Trade Shock affected each household type and what we learn about the economy's adjustment to it when we include couples. First, I examine the effects of the China Trade Shock on manufacturing employment, total employment, population changes, participation responses, and welfare. Second, I compare the baseline economy with an all-singles economy in which every couple is split into two singles, and with economies in which I shut down migration, partially shut down wives' participation response, or both. These comparisons show three things we learn when we include couples: a model without couples overstates how much workers respond to the shock; wives' participation insures every couple, so that without it, the economy as a whole gains less; and couples move and lose together.

V.1 Quantitative Effects of the China Trade Shock

The China Trade Shock, consistent with reduced-form evidence, reduces aggregate manufacturing employment and lowers it for every group (Figure 3a).[28] The decline observed over the first 15 years does not reverse in the long run (Panel A of Table 6). There are important gender differences in the manufacturing decline: women's decline is about half that of men's and varies little by marital status. Married men experience the largest manufacturing decline by $t = 199$, 44% larger than that of single men. Less than half of this gap comes from married men being more concentrated in manufacturing in 2000Q1: relative to their initial manufacturing share, married men's decline is still 22% larger than single men's, and married women's is 19% larger than single women's.

Rather than pushing workers into non-employment, the shock raises the employment rate of every group, with a particularly large increase for single women (Figure 3b). The

[28] When manufacturing decline is measured as manufacturing jobs per capita, a metric comparable to CDP, both quantification exercises yield similar declines (-0.21 vs -0.23 p.p.; Table E.1).

shock is in fact a boom for workers: the increase in productivity in Chinese manufactured goods implies a cheaper consumption basket in the US, thereby reducing the regional price index. The relative real wage differential between working in productive sectors and non-employment then increases, incentivizing single women and all men to work more. Services benefit from the boom: services gain 18% more jobs per capita than manufacturing loses (Figure 3c).

For married women, the employment gain is the added-worker effect, because her participation is a couple's decision: the shock hits male-intensive manufacturing, raising female wages relative to male wages and drawing wives into the labor force, as shown in equation (23). Wives respond the most when only their husband starts in manufacturing (Panel B of Table 6). Married men experience the largest manufacturing losses yet gain the least in employment: because most of them are already working, most of their adjustment comes from reallocation across sectors rather than exiting the workforce. In fact, by $t = 199$, married men account for 53% of the non-manufacturing jobs created by the shock, although they hold only 33% of non-manufacturing jobs in 2000Q1.

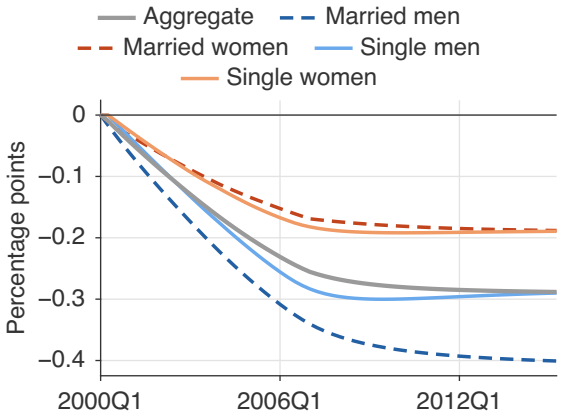
Lower prices, together with increases in the employment rate and participation responses, translate into welfare gains for all groups, with the average individual in the US experiencing a 0.18% increase in welfare, as measured by the MMWI.[29] Most of these gains come from real income, through a cheaper consumption basket. These gains, however, differ across household types: single women gain the most (34% more than the average individual) and single men the least (10% less; Panel A of Table 6). Single women are concentrated in expanding sectors, are underrepresented in the 12 manufacturing sectors affected by the shock, and are free to move. Couples fall in between the two single groups, but their gains are much closer to those of single men, because the husband's wage drives the household's gain, while the wife's contribution depends on her participation.

Within couples, welfare gains decrease monotonically with exposure and are ordered by the husband's exposure (Panel B of Table 6). When neither spouse starts in manufacturing, couples have the highest welfare gain and are comparable to the average individual. Couples in which only the wife starts in manufacturing gain 10% less than couples in which neither starts in manufacturing, followed by couples in which only the husband does (22% less), and finally couples in which both do (30% less). Since his wage enters utility directly and hers depends on her productivity at home, his exposure reduces household income, whereas hers affects the option value of her working.

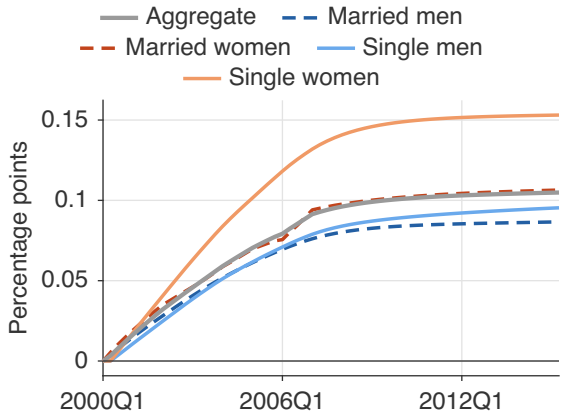
[29] One potential concern would be that considering couples changes the trade equilibrium. In Figure E.5, I compare the welfare gains for 35 countries, excluding China, and the rest of the world, in my model and in CDP. The two series correlate at 0.997, and the same 30 of 36 regions gain in both. Welfare in the rest of the world is also the same in every economy I study below (Table E.1).

Figure 3: Employment Effects of the China Trade Shock in the Short Run

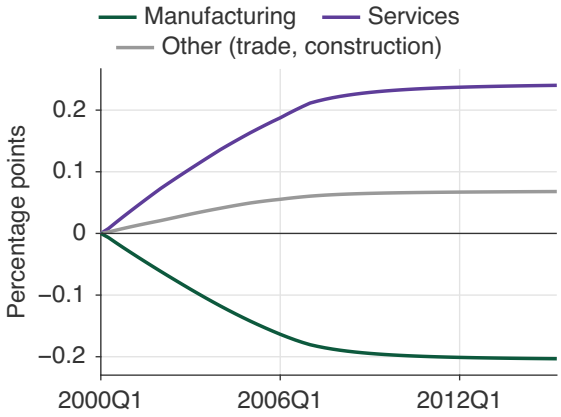
(a) Manufacturing share of employment



(b) Employment rate



(c) Employment per capita, by sector



Notes. Responses to the shock (with-shock minus no-shock economy) in the baseline economy over the first 62 quarters (2000Q1–2015Q2). Panel (a) is the manufacturing share of employment, in p.p. of employment; panel (b) is the employment rate, in p.p. of population; panel (c) is employment per capita by sector, in p.p. of population, where manufacturing covers the 12 manufacturing sectors and other covers wholesale and retail trade and construction.

The states where manufacturing was concentrated tend to lose population and to see larger increases in married women's participation. In Figure 4a, I plot the distribution of manufacturing employment in 2000Q1 across states, and I compare it against the change in population (Figure 4b), the change in married women's labor force participation (Figure 4c), and welfare across states (Figure 4d).[30] The correlation between population change and the 2000Q1 manufacturing share is −0.41, indicating that the states most exposed to the

[30] In Figure E.6, I plot the change in population, the change in married women's labor force participation, and welfare across states.

Table 6: Long-Run Effects of the China Trade Shock

	Share (%)	Manuf. share of employment (p.p.)	Employment rate (p.p.)	Married women's participation (p.p.)	Migration response (p.p.)	Welfare (% CE)	Welfare relative (%)
Panel A. By household type							
Aggregate	100.0	−0.293	0.112		+0.0166	0.175	—
Single men	11.7	−0.287	0.106		+0.0206	0.157	−10.1
Single women	15.0	−0.188	0.158		+0.0115	0.234	+33.9
Married men	36.7	−0.412	0.096		+0.0259	0.170	−2.9
Married women	36.7	−0.192	0.111	0.101	+0.0053	0.170	−2.9
Panel B. Couples by exposure at $t = 1$							
All couples	100.0			0.101	−0.0066	0.170	−5.5
Neither	76.6			0.100	−0.0109	0.180	—
Wife only	6.6			0.101	−0.0062	0.161	−10.3
Husband only	13.7			0.105	+0.0141	0.140	−22.1
Both in manuf.	3.1			0.103	+0.0078	0.125	−30.4

Notes. Responses at $t = 199$ (with-shock minus no-shock economy), except shares (2000Q1) and welfare. The manufacturing share is in p.p. of employment, and the employment rate is in p.p. of population. The participation response is the change in married women's quarterly participation rate, in p.p.; a non-working wife is tagged by her potential sector. In Panel A, the migration response is the difference between the shares of manufacturing and non-manufacturing workers who have changed state at least once by $t = 199$, in p.p., counting each person once; individuals are tagged by their sector at $t = 1$, married men by the husband's sector, and married women by the wife's (potential) sector. In Panel B, couples are tagged by both spouses' sectors at $t = 1$, and the migration response is the change in the share of couples in the cell that have changed state at least once by $t = 199$, in p.p. Welfare is the money-metric consumption-equivalent gain, weighting each person by their money metric at 2000Q1. Welfare relative is $100 \times (G_g/G - 1)$ relative to the aggregate in Panel A and to couples in which neither spouse starts in manufacturing in Panel B. Blank cells are not defined for the group.

shock tend to be also the ones that lose population. The correlation between the 2000Q1 manufacturing share and married women's participation response is +0.43: wives go to work where their husbands' jobs were. At the same time, the correlation between participation response and population change is −0.44, implying that wives work more in states where people leave.

Welfare gains are smallest in states where people leave (Figures E.6a and E.6c); in particular, Louisiana, Kentucky, Indiana, and Mississippi are the four states furthest below their national average and the four that lose the most population. Across states, welfare relative to the national gain correlates with population change at +0.97 and with the initial

manufacturing share at -0.50 (Figure 4d). The geography of gains is nearly identical across household types (Figure E.7), but couples experience the greatest spatial inequality in welfare, with a coefficient of variation of 0.31, compared with 0.23 for single men and 0.16 for single women. This reflects the joint-location friction: a couple cannot take one job in one state and the other in another.

V.2 Why Couples Matter for How the Economy Adjusts

To quantify what we learn when we include couples, I compare the previous results with the effect of the shock in a counterfactual world in which everyone is single, which I call the all-singles economy, and with economies in which I shut down the two margins through which couples adjust to the shock: migration across states and wives' participation.

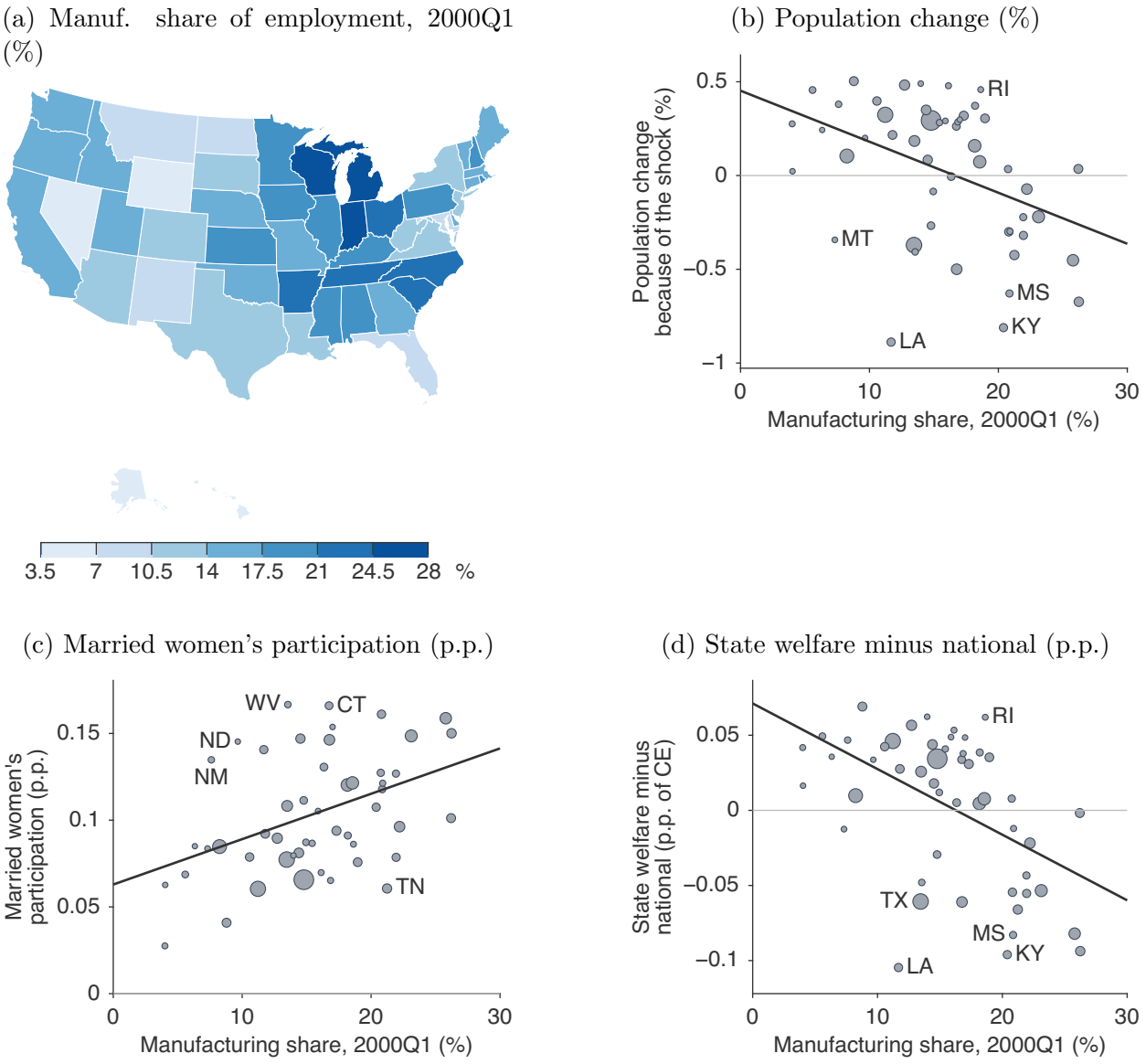
To build the all-singles economy, I split every couple into two singles, reconstruct all inputs from the data, and recalibrate the migration elasticity to match the same moment as in the baseline economy. I then solve the model again under this structure. I compare the same people across the two economies: a married woman in the baseline economy is, in the all-singles economy, a single woman who chooses her location and sector on her own, and both economies weight each person by their baseline money metric at 2000Q1.

I then shut down the two margins through which couples adjust to the shock, migration and wives' participation, one at a time and together. In the first one, I set $1/\theta \to 0$ to shut down migration responses. Because $1/\theta$ is the migration elasticity in the model, this shuts down migration across states, while switching sectors within a state is still allowed. In the second one, I set $\kappa/\gamma \to 1$ and recalibrate χ accordingly to partially shut down participation responses.[31] In the third one, I make both changes at once. In Table 7, I summarize each exercise. Each exercise is a full re-solve of the model with the corresponding parameter changed, while all other parameters (except χ in the second exercise) are held at their baseline values. I also shut down migration in the all-singles economy, so that the economy without migration has a counterpart without couples.

I report the main responses to the shock for every economy in Table 8 and the welfare gains in Table 9. Tables E.3 and E.4 report results for every group. All tables report levels, so each economy can be compared with any other. Unless stated otherwise, every percentage change in the text is relative to the baseline economy. In economies with couples, married

[31] The participation response depends on κ/γ, because the derivative of ρ with respect to $\ln(1+r)$ equals $\kappa/\gamma(1-\rho)$. So, once the participation *level* is held fixed, κ/γ is the only free multiplier on the response. The model requires, however, that $\kappa > \gamma$, so the most the response can be damped is $\kappa_0/\gamma = 4.44\times$. For computational reasons, I set $\kappa = 0.65$, so the response is damped by a factor of $3.39\times$. I also recalibrate χ to keep the anchor participation rate unchanged. As a result, in this counterfactual, the wives' participation response to the shock is only partially shut down (it is 59% smaller).

Figure 4: The China Trade Shock across States, by Initial Manufacturing Share



Notes. Panel (a) is the manufacturing share of the state's employment in 2000Q1. In panels (b) to (d), each dot is a state, sized by its population in 2000Q1. The horizontal axis is the state's manufacturing share of employment in 2000Q1, in %. All outcomes are long-run responses ($t = 199$) in the baseline economy, defined as: the change in the state's population as a percentage of its no-shock level, which sums to zero across states because the national population is fixed; the change in married women's participation rate, in p.p., averaged over the couples living in the state; and the state's money-metric welfare gain minus the national gain (0.175), in p.p., for the people living in the state in 2000Q1. Lines are OLS fits; correlations, unweighted across the 50 states, are −0.41, +0.43, and −0.50. Labeled states are the five with the largest residuals.

Table 7: Economies Compared in the Quantitative Exercise

Economy	Couples	Migration	Participation	Change relative to the baseline economy
Baseline	✓	✓	✓	—
All-singles	—	✓	—	Couples split into two singles; θ recalibrated
All-singles, no migration	—	—	—	All-singles economy with $1/\theta \to 0$
No migration	✓	—	✓	$1/\theta \to 0$
No participation	✓	✓	damped	$\kappa/\gamma \to 1$
No migration & no participation	✓	—	damped	Both changes

Notes. A check mark indicates that the margin responds to the shock as in the baseline economy; a dash indicates that it is shut down or does not exist. To set $1/\theta \to 0$ to shut down migration responses, I set $\theta = 10^6$. To set $\kappa/\gamma \to 1$, given the restrictions driven by the data, I set $\kappa = 0.65$ and recalibrate χ to keep the value of the participation rate for married women in 2000Q1 constant. This implies that I only partially shut down the participation response in those counterfactuals.

men and women carry the couple's welfare; in all-singles economies, they are the same people as singles.

The shock is the same in every economy: couples change how the economy absorbs the shock, but they do not change the extent to which manufacturing employment falls. Manufacturing jobs per capita fall within 4% of the baseline in every economy, and welfare in the rest of the world does not move: the mean gain of the other 36 regions differs by less than 2% across economies, and their gains correlate with those in CDP at 0.997 in every economy (Table E.1).

Couples matter for how the economy adjusts to the shock for three reasons. First, workers respond less to the shock than they would in a model without couples. Second, wives' participation matters: it insures every couple, regardless of exposure, and without it, the economy as a whole gains less. Third, couples move and lose together.

Result 1: Workers respond less to the shock than they would in a model without couples.

Even though the manufacturing decline is the same in models with and without couples (the two differ by less than 1%), the model without couples overstates the employment gains of the shock. The all-singles economy generates 18% larger employment gains than

Table 8: Responses to the China Trade Shock across Economies

	Baseline	All-singles	All-singles, no mig.	No migration	No participation	No mig. & no part.
Panel A. Employment						
Manuf. share (p.p.)	−0.293	−0.291	−0.286	−0.287	−0.285	−0.279
Employment rate (p.p.)	0.112	0.132	0.119	0.099	0.085	0.073
Female emp. rate (p.p.)	0.124	0.170	0.161	0.117	0.080	0.075
Panel B. Migration						
Exposed (p.p.)	+0.0166	+0.0194	—	—	+0.0170	—
Share living in a different state because of the shock (%)						
Aggregate	0.150	0.130	—	—	0.148	—
Married women	0.159	0.116	—	—	0.158	—
Panel C. Women's labor supply						
Wives' particip. (p.p.)	0.101			0.097	0.041	0.040
Elasticity	0.452	0.682	0.644	0.426	0.276	0.258

Notes. Responses at $t = 199$ (with-shock minus no-shock economy), in levels. Panel A is in p.p. of employment (manufacturing share) and of population (employment rates). In Panel B, the migration response of the exposed is the difference between the shares of manufacturing and non-manufacturing workers who have changed state at least once by $t = 199$, in p.p., counting each person once. In Panel C, married women's participation is the change in their quarterly participation rate (in p.p.), and the female labor-supply elasticity is the log female employment response to the log female real-wage response. In all-singles economies, married women are the same people as singles. A dash marks a margin that is shut down; blank cells are not defined in that economy.

the baseline economy (Table 8), primarily because women's employment responds 36.5% more.[32] This difference arises because female labor supply is 51% more elastic in the all-singles economy. A wife's labor supply decision depends on the relative wage $r = w_f/w_m$, rather than on her own wage alone, so her employment response is smaller whenever her husband's wage moves with hers, and the shock raises his wage too in the long run. Within the baseline economy, married women's labor supply is 21% less elastic than single women's, even though their wages respond by the same amount: by $t = 199$, the husband's wage gain offsets 61% of the wife's, so her relative wage rises much less than her own wage.

A model without couples also overstates the migration response of exposed individuals. The difference between the shares of manufacturing and non-manufacturing workers who have moved to another state by $t = 199$ is 17% larger in the all-singles economy (Panel B of Table 8). Exposed wives are tied to their husbands' location: when only the wife is

[32] Women account for 35% of the gain in non-manufacturing employment in the baseline economy and 41% in the all-singles economy, mostly in other services and in accommodation and food.

Table 9: Welfare Effects of the China Trade Shock across Economies

	Baseline economy		Counterfactual economies				
	Home prod.	Valued as singles	All-singles	All-singles, no mig.	No mig.	No part.	No mig. & no part.
Panel A. Welfare gain (% CE)							
Aggregate	0.175	0.194	0.178	0.180	0.176	0.167	0.169
Single men	0.157	0.157	0.166	0.170	0.162	0.153	0.158
Single women	0.234	0.234	0.194	0.195	0.236	0.255	0.257
Married men	0.170	0.193	0.164	0.167	0.171	0.157	0.159
Married women	0.170	0.193	0.193	0.194	0.171	0.157	0.159
Panel B. Spatial							
CV of state welfare	0.278		0.204	0.327	0.448	0.289	0.462

Notes. Money-metric consumption-equivalent gains (%), summed over $t = 2, \ldots, 199$. Each row represents the same people across all economies, and each economy weights each person by their baseline money metric in 2000Q1. In the economies with couples, married men and women carry the couple's welfare; in the all-singles economies, they are the same people as singles. The first two columns are the baseline economy: the first values a non-working wife's time as home production, as in the model, and the second values it as a single, with the non-employment benefit b_j, with every decision unchanged. In Panel B, CV is the population-weighted coefficient of variation of state welfare gains.

exposed, the same woman would respond almost four times as much as a single, whereas an exposed husband responds about as much in the couple as he would as a single. This is the asymmetry in the second empirical fact: couples move when the husband is exposed, but not when the wife is.

The economy's gains are roughly the same in both models (the gains are 2% larger in the all-singles economy), but they are distributed differently. Comparing the gains of married individuals to those of their counterfactual singles, I find that in the all-singles economy, married women gain 14% more than in the baseline economy, whereas married men gain 3% less (Table 9). Because a wife's decision to participate depends on her wage relative to her husband's, she responds to the shock partly through his exposure and cannot fully capture the expansion in services; moreover, both spouses share a single household-level gain. Couples also redistribute gains among singles, who make no joint decisions: in the all-singles economy, single women gain 17% less and single men 6% more than in the baseline economy. Because female employment responds 36.5% more in the all-singles economy, wages absorb less of the shock here. Specifically, in the all-singles economy, the female wage response is 11% smaller, the male wage response is 9% larger, and the relative wage rises by 39% less

(Panel E of Table E.3). In the baseline economy, single women therefore supply labor to a tighter market, while single men supply labor to a weaker one.

Joint decision-making leaves the geography of welfare gains almost unchanged but makes it more unequal, which is consistent with couples having the greatest spatial inequality in welfare in the baseline economy (Figure E.7c). The spatial distribution of welfare gains is nearly identical across the two economies (Figure E.8), with a cross-state correlation of +0.96. However, welfare gains are 27% less dispersed in the all-singles economy (Panel B of Table 9): as singles, the same men's welfare would be 13% less dispersed across states and the same women's 48% less, so living as a couple ties a wife's welfare to where the couple lives.

Result 2: Wives' participation insures every couple, and without it, the economy as a whole gains less.

Couples have an additional margin of adjustment that singles lack: adjusting the wife's labor supply, and all couples use it in the long run. Because couples make a joint labor-supply decision, a wife's labor supply responds not only to her own exposure to the shock but also to her husband's. In the short run, wives respond to the shock as in the reduced-form evidence in Table 3. Figure E.3 plots married women's participation response by the couple's exposure until 2015. Wives whose husbands are exposed enter the market at the start of the shock, whereas wives who are exposed themselves leave the market. Wives in unexposed couples enter gradually, because the shock raises women's relative wage everywhere: by $t = 199$, it rises in all 50 states, and so does married women's participation. In the long run, every couple, regardless of exposure, uses this margin of adjustment. By $t = 199$, the response is nearly the same in every type of couple: it is largest when only the husband is exposed, but only 5% larger than when neither spouse is (Panel B of Table 6).

When I partially shut down participation, the economy as a whole gain less. Aggregate welfare gains are 5% lower (Tables 9 and E.2), and couples gain 7% less and single men 2% less. Single women are the only group who benefit from shutting down participation: they gain 9% more. Their gain works through the same mechanism as the all-singles comparison, pushed further. Damping κ/γ takes married women further out of the labor market; female labor gets scarcer still, and the single women who are free to respond capture the wage.[33] Every state loses: welfare falls in all 50 states (Figure E.4).

Among couples, the losses reflect the added-worker mechanism: they are largest when only the husband is exposed (-10%; Table E.2), because a wife whose husband is exposed

[33]The female wage response is 9% larger and the male wage response 3% smaller than in the baseline economy; because men and women are complements in production, single men lose with them.

but who is not herself exposed would enter the market but cannot. Couples in which only the wife is exposed lose too (-5%), because the same margin also lets her leave the market and enjoy the cushion of home production. Even unexposed couples gain 7% less. The added-worker margin is insurance every couple holds. Without the added-worker effect, couples also move differently, although only slightly: couples with an exposed husband move relatively more, and all other couples move less (Panel D of Table E.3).

Adjusting the wife's participation is not free for couples: wives who enter the market give up home production, and only a model with couples can properly account for this opportunity cost. As shown above, aggregate gains in the all-singles economy are 2% larger than in the baseline economy (Table 9). But in the baseline economy, a wife's time at home is valued as home production, and when the shock draws her into the market, the couple sacrifices some of that time. If instead I value a non-working wife's time as a single's, i.e., at the non-employment benefit b_j, and keep every decision unchanged, the baseline economy's aggregate gain is 8.5% larger than in the all-singles economy (Appendix E.3). This valuation also redistributes gains from couples in which only the wife is exposed to couples in which only the husband is exposed, because home production insures the couple when her wage falls.

Result 3: Couples move and lose together.

Although exposed workers respond less to the shock in the baseline economy, more people end up in a different state. By $t = 199$, without couples, 13% fewer people end up in a different state (Panel B of Table 8). Wives account for most of the difference. In the baseline economy, they live in a different state as often as their husbands do, whereas, as singles, 27% fewer of them do. Wives follow their husbands: a couple's location responds to the husband's exposure, as in the second empirical fact, whereas, as singles, the same women respond to their own wages, which the shock raises, so they stay.

Within couples, spouses share each other's losses. In Panel A of Table 10, I compare each couple's welfare gain with the gains of the same husband and wife as singles in the all-singles economy. When the wife is exposed, both spouses would be better off as singles: couples in which only the wife starts in manufacturing gain 4% less than the same husband and 10% less than the same wife as singles, and couples in which both do gain 7% and 29% less, respectively. An exposed wife adjusts less in every margin than she would alone: her manufacturing employment falls by the same amount, but as a single, the same woman would move 24% more into other sectors, raise her participation by 68%, and respond almost four times as much by migrating. Otherwise, the couple's gain falls between the two spouses' gains as singles, so the couple redistributes welfare from the wife to the husband: when only

the husband is exposed, the couple gains 5% more than he would as a single and 26% less than she would as a single.

Migration matters more for couples with an exposed husband than it would for the same spouses as singles. Panel B of Table 10 reports the change in welfare from shutting down migration, relative to each economy's own gain. Couples in which only the husband starts in manufacturing lose 3.4% of their gain, compared with 2.8% for the same husband and 0.3% for the same wife as singles; when both spouses start in manufacturing, the couple loses 2.7%, compared with 2.1% and 0.5%. Across all couples, removing migration shifts gains from husbands to wives: the couple gains 0.7%, compared with 1.7% for the same husbands and less than 0.5% for the same wives as singles. Appendix E.2 reports how shutting down migration redistributes welfare across states and why wives do not work more when couples cannot move.

Table 10: Couples and the Same Spouses as Singles

	Couple (Baseline)	Same husband as a single	Same wife as a single	Couple, participation damped
Panel A. Welfare gain (% CE), with migration				
All couples	0.170	0.164	0.193	
Neither	0.180	0.172	0.195	
Wife only	0.161	0.168	0.179	
Husband only	0.140	0.134	0.190	
Both in manuf.	0.125	0.135	0.177	
Panel B. Change in welfare from removing migration (% of each economy's own gain)				
All couples	+0.7	+1.7	+0.3	+1.1
Neither	+1.6	+2.7	+0.6	+2.0
Wife only	+0.7	+1.4	+0.1	+1.0
Husband only	−3.4	−2.8	−0.3	−3.4
Both in manuf.	−2.7	−2.1	−0.5	−2.5

Notes. Couples are tagged by both spouses' sectors at $t = 1$; a non-working wife is tagged by her potential sector. The couple's welfare is based on the baseline economy; the same husband and wife as singles are the two former spouses in the all-singles economy. Panel A: money-metric consumption-equivalent gains (%), weighting both economies by the baseline money metric at 2000Q1. Panel B: change in welfare from shutting down migration, as a percentage of each economy's own gain: no migration relative to the baseline economy for the couple, all-singles economy without migration relative to the all-singles economy for the spouses as singles, and no migration & no participation relative to no participation for the couple with participation damped.

Without adjusting the wife's participation in a world without migration, couples fall further behind their counterfactual singles. The economy without either margin leaves married men 5% and married women 18% worse off (Table 9), and the aggregate gap to the all-singles economy widens from 2% to 6%. Shutting down migration alone barely moves this gap, so the widening comes from losing the added-worker effect, which an all-singles economy has no counterpart to. The two margins are weak complements: for every group, shutting down both costs slightly less than the sum of shutting down each one alone (Table E.2).

Taken together, the three reasons presented above show that couples do not change the size or location of the China Trade Shock, but they matter for how the economy adjusts to it. In particular, workers respond less to the shock than they would in a model without couples, because of how couples make their decisions. All couples adjust the wife's participation, and without it, welfare gains fall in every state. Finally, couples matter for how the economy adjusts to the shock because couples move and lose together.

VI. Conclusion

Most adults respond to local shocks as a couple, which means they must weigh two labor-market exposures against each other, making them less likely to move. At the same time, they have an insurance margin that singles lack: when one spouse's earnings fall, the other can work more. I show that couples change how the economy adjusts to a local shock and who bears it, even though they do not change its size or location.

Using PSID data to compare individuals who were and were not exposed to the China Trade Shock within the same commuting zone, I document three facts. First, exposed individuals experienced worse labor-market outcomes and were more likely to leave their commuting zones. Second, couples responded asymmetrically: out-migration increased when the husband was exposed, but not when the wife was exposed. Third, wives' labor supply exhibited a similar asymmetry: rising when the husband was exposed (an added-worker effect) and falling sharply when the wife herself was exposed. Thus, couples mitigate the shock through both spatial and intra-household insurance, and which margin they use depends on which spouse is hit.

To quantify what joint decision-making implies for the economy as a whole, I extend Caliendo et al. (2019) to incorporate heterogeneous households: single men, single women, and couples. Couples jointly choose a location and a pair of sectors, with substitution across sectors and locations occurring at different rates, as in Rodríguez-Clare et al. (2026). After choosing where to live and work, they decide whether the wife works in the market or at home. The couple's problem therefore introduces two margins that do not exist for singles:

joint location choice makes moving more difficult, and endogenous labor-force participation makes staying more bearable.

The quantitative exercise shows that the China Trade Shock is, overall, a boom for US workers: cheaper imports lower the price index, employment rates rise across all household types, and the average individual gains 0.18% in consumption-equivalent terms. Manufacturing's share of employment falls by 0.29 p.p., and households adjust through both migration and increased labor-force participation. The states where manufacturing was concentrated also lose the most population and exhibit the largest increases in the labor supply of married women. Moreover, participation increases most in the states with the largest population losses. The gains differ across household types: single women gain the most, single men the least, and couples fall in between, with gains ordered by the husband's exposure because his earnings are the household's main source of income.

Comparing these results with those from an all-singles economy, in which I split every couple into two singles, and from economies in which I shut down migration, partially shut down wives' participation response, or both, shows three reasons why couples matter for how the economy adjusts to a local shock. First, workers respond less to the shock than they would in a model without couples. A wife decides whether to work together with her husband, and, in the long run, the shock that raises her wage also raises his, so fewer women enter the labor market when couples are present: without couples, the employment response would be 18% larger and the migration response of exposed workers 17% larger. Without couples, more exposed workers move, but fewer people move overall, since unexposed spouses do not follow an exposed partner. Because a model without couples treats every mover as an exposed worker, it overstates how much exposed workers move, which is one reason why workers migrate less than the theory predicts. The gains would also be shared differently: because women's employment would respond more and women's wages would absorb less of the shock, single women would gain 17% less, single men 6% more, and gains would be 27% less dispersed across states.

Second, adjusting the wife's labor supply insures every couple, regardless of exposure, and without it, the economy as a whole gains less, and welfare gains fall in every state. This margin of adjustment has an opportunity cost: wives who enter the market give up home production, which is why aggregate gains are about 2% smaller with couples than without them. Third, couples move and lose together. Because a couple's location responds to the husband's exposure and spouses move together, more people relocate across state lines when couples are involved, and spouses share each other's losses: when the wife is exposed, both spouses would be better off as singles, because she adjusts less on every margin than she would alone.

The policy implication is that aggregate gains, and even their geography, provide an incomplete picture of who benefits from the shock. Trade adjustment assistance is designed around individuals, targeting workers or industries that are exposed to trade shocks. Yet the binding mobility constraint operates at the household level, and a transfer that replaces a displaced husband's earnings does not restore the option his family has lost. Policies that reduce the cost of moving jointly or increase the destination's value to the spouse who is not driving the move would help alleviate this constraint. Moreover, because couples change how women's labor supply responds, the shock's incidence reaches single workers who were never employed in an exposed industry.

The direction of the gender asymmetry in my results is a feature of *this* shock rather than a general property of the model. Because the China Trade Shock affects manufacturing, a male-intensive sector, it exposes husbands more than wives and draws wives into the labor market. A shock to a female-intensive sector would reverse the asymmetry, although its welfare consequences need not be a mirror image. Thus, this paper opens a broader research agenda on how the economy adjusts to local shocks and who bears them when the household, not only the worker, decides where to live and work.

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Appendix

"It Takes Two to Tango": Joint Location Choices and Local Shocks

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A. Robustness Exercises

A.1 The China Trade Shock across Sample Commuting Zones

Given the PSID sample restrictions and the size of the PSID, I include 375 commuting zones out of the 722 that ADH use in their analysis. My results could therefore be driven by which zones are included in the PSID rather than by differential exposure to the China Trade Shock. To assess this concern, I replicate ADH's main results using the 375 CZs represented in the sample.[34][35]

Table A.1 first compares the CZs in the sample with the full set used by ADH. Because the restricted sample is a subset of the full sample, the two groups are not independent, and a conventional difference-in-means test is therefore ill-defined. The final column instead reports the standardized difference, defined as the difference in means between the restricted and full samples divided by the square root of the average of their variances. This measure is expressed in standard-deviation units, so it is comparable across variables measured on different scales. Following Imbens and Rubin (2015), absolute standardized differences below 0.25 indicate comparability. All standardized differences in Table A.1 are 0.13 or smaller, including the difference for the change in import penetration. By this criterion, the CZs in the PSID sample are comparable to the full ADH sample with respect to both the control variables and the China Trade Shock itself.

Next, I estimate ADH's Equation (2) for the CZs in the sample:

$$\Delta Y_{i,t+h} = \alpha_t + \beta_{1,h}\Delta IP_i^{China,US} + X'_{i,t}\beta_2 + \varepsilon_{i,t+h},$$

where $\Delta Y_{i,t+h}$ denotes the change in outcome Y for CZ i between $t = 2000$ and year $t + h$. As in ADH, $\Delta IP_i^{China,US}$ is instrumented with $\Delta IP_i^{China,Others}$, following the identification logic described in Section II. The analysis considers ADH's three main outcome categories: employment, income, and population headcount.

Estimated on the CZs in the PSID sample, ADH's specification yields responses that closely track ADH's full-sample estimates, except for the changes in population of younger age groups. Figure A.1 replicates ADH's employment results for the manufacturing employment-to-population ratio in panel a, the non-manufacturing employment-to-population ratio in panel b, and the total wage-and-salary employment-to-population ratio in panel c. Across all three outcomes, the estimates based on the full set of CZs closely align with those based only on the CZs in the PSID sample.

[34] Because the analysis uses the geocoded restricted version of the PSID, the identities of the commuting zones in the sample cannot be disclosed, nor can a map be provided at any geographic level below the state.

[35] This exercise uses the replication package provided by David Dorn on his website.

Table A.1: Comparing CZs in the Sample with All CZs

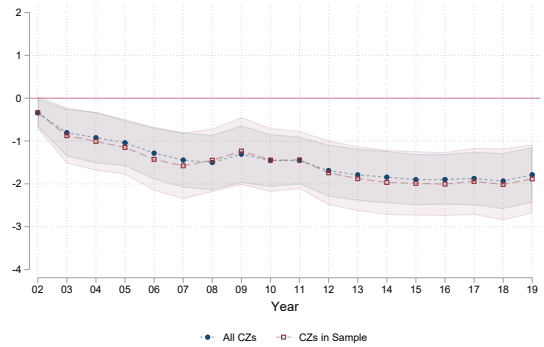
	Full		Restricted			Standardized
	Mean	SD	Mean	SD	Diff	Difference
Control variables (in 2000)						
Manufacturing emp./total emp.	16.18	7.46	15.68	6.91	-0.49	-0.07
Female emp./total emp.	64.38	5.51	64.54	5.21	0.16	0.03
Routine occupation emp./total emp.	31.90	2.37	32.20	2.16	0.30	0.13
Offshorability index	0.00	0.51	0.06	0.48	0.07	0.13
College educated/total pop.	53.61	7.47	54.41	6.88	0.80	0.11
Foreign born/total pop.	14.76	12.81	15.74	12.89	0.97	0.08
Non-white population/total pop.	18.18	10.95	18.81	10.36	0.63	0.06
Population 65+/total pop.	12.37	2.91	12.15	2.81	-0.21	-0.07
Population 40-64/total pop.	30.10	1.87	30.02	1.79	-0.08	-0.04
Population 0-17/total pop.	25.63	2.22	25.65	2.14	0.02	0.01
Δ log population 1970-1990/2	12.31	12.25	12.77	12.35	0.47	0.04
Change in CZ import penetration (2000 to 2012)						
Imports by US	0.89	0.59	0.88	0.55	-0.01	-0.02
Imports by other high-inc. countries	1.22	0.69	1.21	0.64	-0.01	-0.02
Number of Commuting Zones	722		375			

he income responses in the sample CZs also closely track ADH's full-sample estimates. Figure A.2 reports the corresponding results for income: personal income per capita in panel a, government transfers per capita in panel b, and total labor compensation per worker in panel c. The only visible divergence is a somewhat larger increase in government transfers among the sample CZs, although the coefficients are not statistically distinguishable.

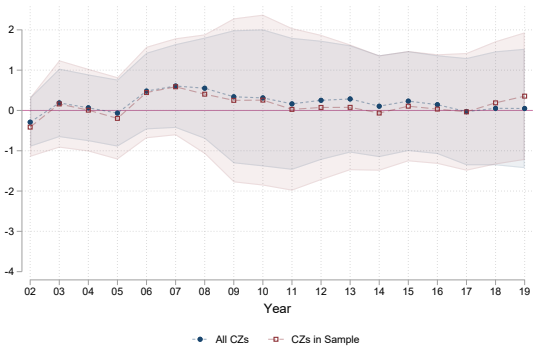
Finally, population headcounts, which ADH use to measure migration responses, are the only outcomes for which the two sets of estimates diverge, and only among younger age groups (Figure A.3). The estimates are similar for the total population and for individuals ages 40–64. However, the population decline among individuals ages 25–39 is larger in the sample CZs, and the estimates indicate a statistically significant decline among individuals ages 18–24 that is absent from ADH's full-sample results. The difference among individuals aged 25–39 is relevant for my sample, whose average age is 37: it suggests that young adults left the sample CZs somewhat more than they left the average CZ. My main estimates, however, compare exposed and non-exposed individuals, so a population decline common to both groups within a CZ does not, by itself, drive the estimated differences.

Figure A.1: China Trade Shock Impact on Employment

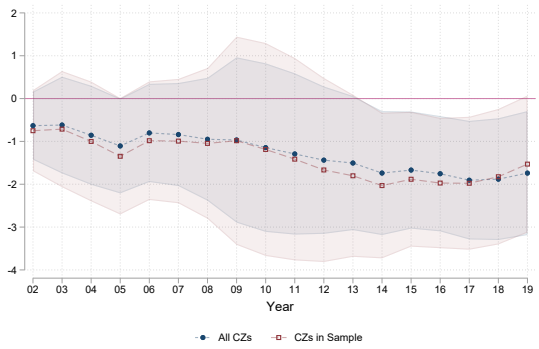
(a) Manufacturing Emp./Working-age pop.



(b) Non-manufacturing Emp./Working-age pop.



(c) Total Wage and Salary Emp./Working-age pop.



Notes. In each panel, I plot the original ADH result for that outcome variable in blue and, in red, the result when I consider only the CZs in the sample. A CZ is considered in the sample if I can estimate its fixed effect when estimating equation (5).

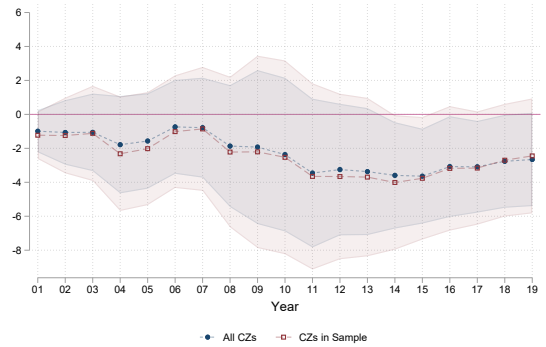
Taken together, the two exercises indicate that, although the PSID sample restrictions reduce geographic coverage, they do not appear to select local labor markets in which the China Trade Shock had systematically different effects. The CZs represented in the PSID are comparable to the full ADH sample in terms of observable characteristics, including exposure to the China Trade Shock, and estimating ADH's specification on these CZs produces similar coefficients.

A.2 The China Trade Shock and Household Composition

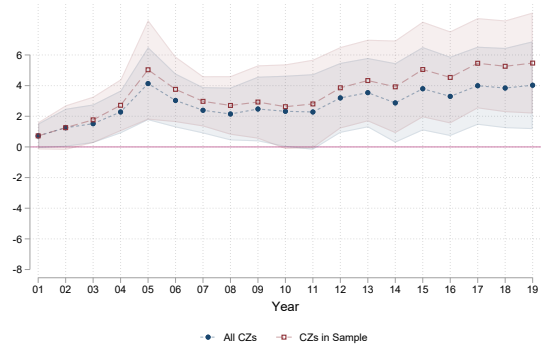
One potential concern is that the China Trade Shock alters household structure. Such changes could bias the results through two channels: the shock may affect marriage, divorce, or fertility decisions that subsequently shape out-migration responses, and couples who re-

Figure A.2: China Trade Shock Impact on Log Income

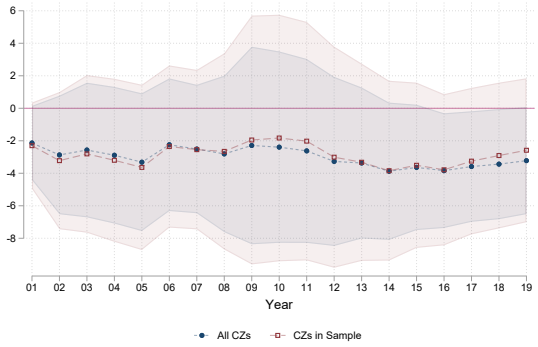
(a) Log(Personal Income per capita)



(b) Log(Government Transfers per capita)



(c) Log(Total Labor Compensation per worker)



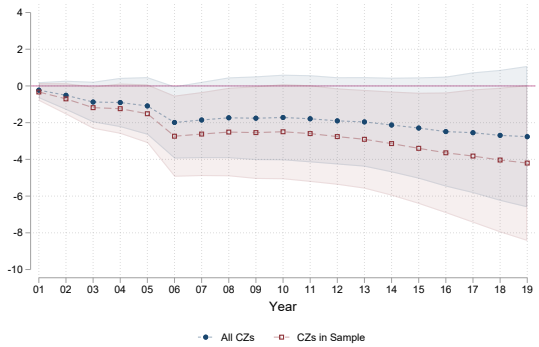
Notes. In each panel, I plot the original ADH result for that outcome variable in blue and, in red, the result when I consider only the CZs in the sample. A CZ is considered in the sample if I can estimate its fixed effect when estimating equation (5).

main together after the shock may differ systematically from those who divorce. To assess this concern, I use the full sample of individuals and estimate equation (4) for the probabilities of marrying, divorcing, and having a new child. Figure A.4 shows that, between 2001 and 2013, exposed individuals were no more likely than non-exposed individuals to marry, divorce, or have a child.

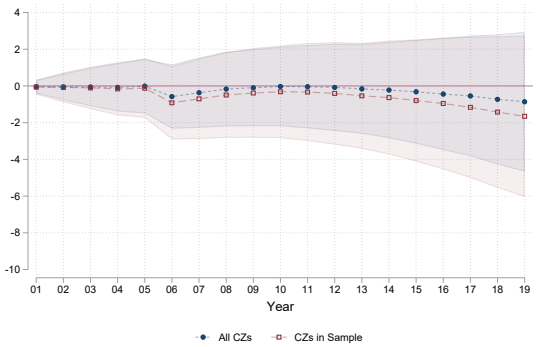
At first glance, this result may appear inconsistent with Autor et al. (2019), who find that a one-unit trade shock reduces the share of previously married women by 0.21 p.p. and increases the share of never-married women by 1.2 p.p.. The apparent discrepancy reflects differences in how marital status is measured. In the ACS, marital status records legal status and does not separately identify cohabiting women. Consequently, women who transition from single to cohabiting are classified as married in my data, whereas they are classified as single in Autor et al. (2019). In addition, because the ACS is a repeated cross-section, a

Figure A.3: China Trade Shock Impact on Log Population Headcounts

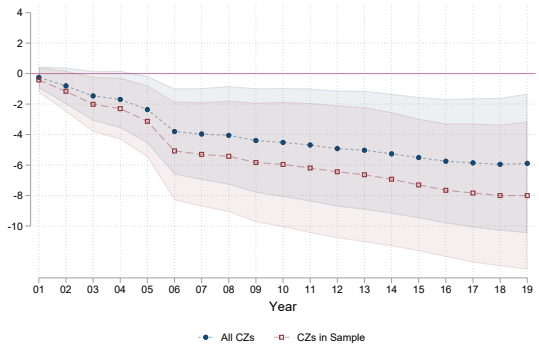
(a) Log(Population, 18–64)



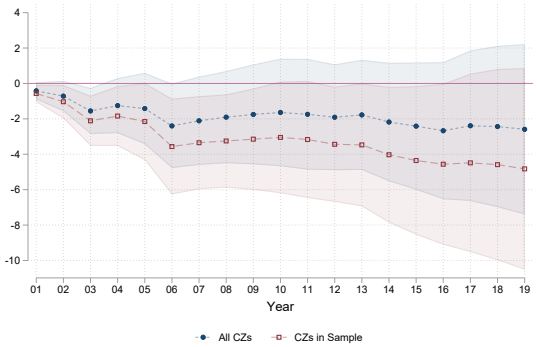
(b) Log(Population, 40–64)



(c) Log(Population, 25–39)



(d) Log(Population, 18–24)



Notes. In each panel, I plot the original ADH result for that outcome variable in blue and, in red, the result when I consider only the CZs in the sample. A CZ is considered in the sample if I can estimate its fixed effect when estimating equation (5).

woman's prior marital status cannot be recovered once she is observed as married. Women who transition from divorced to married are therefore classified simply as married in their analysis, whereas they are excluded from my sample.

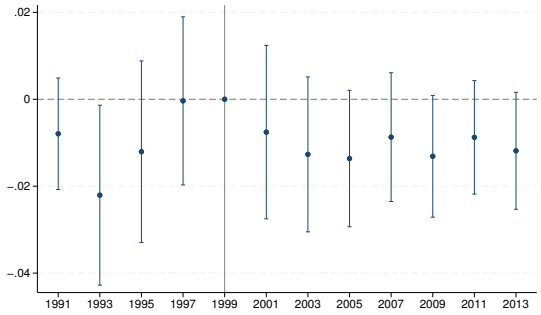
I also find no robust effect of trade shocks on fertility, in contrast to Autor et al. (2019). One likely explanation is that I allow both men and women to be exposed, whereas they focus exclusively on women. My estimates therefore capture a broader set of household shocks and adjustment margins.

A.3 Other Migration Drivers

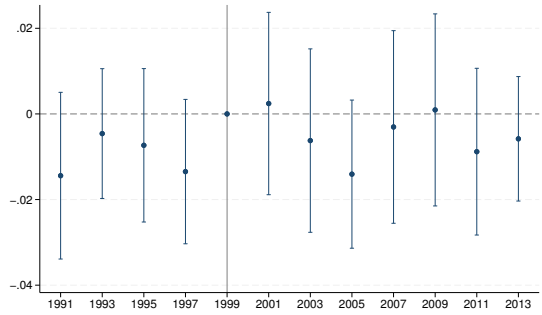
The asymmetric migration response within couples could instead reflect two other drivers: the presence of children or gender differences in responses to the shock. First, the behavioral differences I attribute to joint decision-making may reflect the presence of children rather than couple status itself. To address this concern, I extend equation (4) to a triple-difference

Figure A.4: Exposure to the China Trade Shock and Family Outcomes

(a) Marriage=1



(b) Divorce=1



(c) Fertility=1

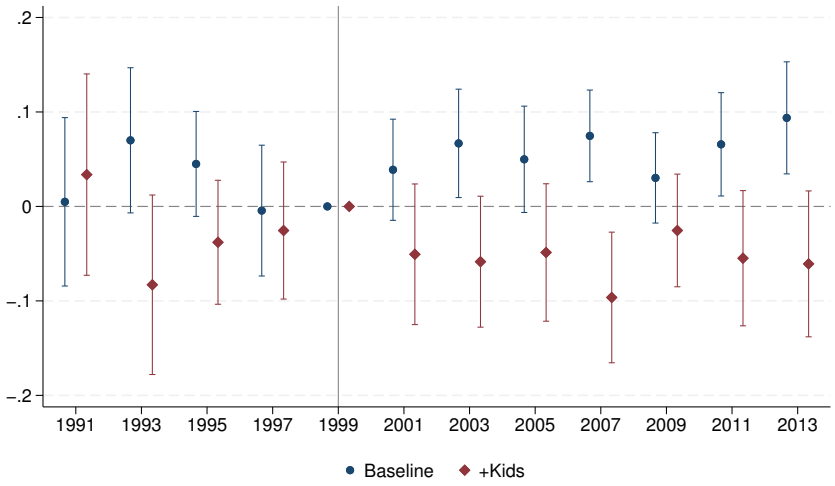


Notes. Each panel plots the estimates of equation (4) for the outcome named in its title. Robust standard errors are two-way clustered at the industry and individual levels. Confidence intervals are constructed at the 95% level. In the equivalent static difference-in-differences model, the coefficient of interest is not statistically significant for any of the three outcomes.

specification that accounts for whether an individual has a child younger than 18. If individuals with children respond differently from those without children, the coefficient on the triple interaction should be statistically different from zero. Figure A.5 shows that this coefficient is not statistically significant in any year except 2007. Thus, exposure does not generate systematically different out-migration responses among individuals with and without children. Individuals with children nevertheless have a baseline out-migration probability that is 4.2 p.p. lower than that of individuals without children. Therefore, the two groups differ in their intercepts, even though their responses to exposure are similar.

Second, the estimated differences could reflect gender-specific responses to the shock. To address this concern, I extend equation (4) to a triple-difference specification that accounts for whether the individual is a woman. If women and men respond differently to the shock, the coefficient on the triple interaction should be statistically different from zero. Figure A.6 shows that this coefficient is not statistically significant in any year during the shock. Thus, exposed women do not exhibit a systematically different out-migration response from

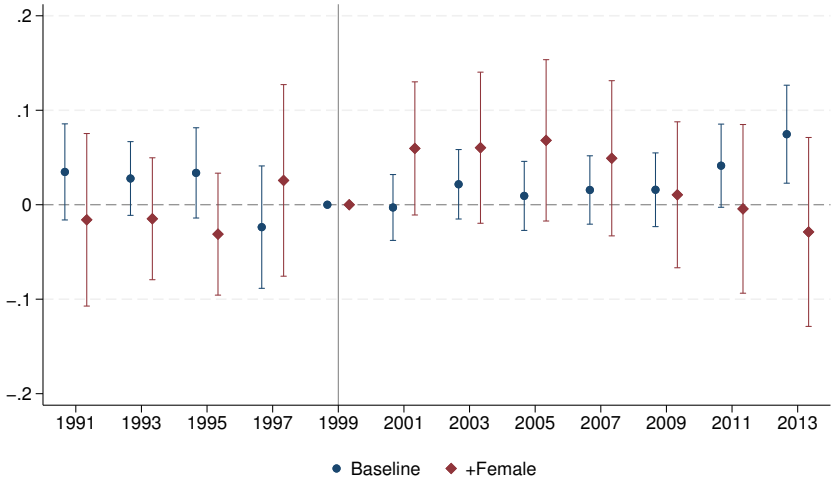
Figure A.5: Exposure to the China Trade Shock and Out-Migration Responses, by Parental Status



Notes. Robust standard errors are two-way clustered at the industry and individual levels. Confidence intervals are constructed at the 95% level. In blue, I plot the coefficient on the interaction between exposure and year. In red, I plot the coefficient on the triple interaction among exposure, year, and an indicator for having a child younger than 18.

exposed men.

Figure A.6: Exposure to the China Trade Shock and Out-Migration Responses, by Gender



Notes. Robust standard errors are two-way clustered at the industry and individual levels. Confidence intervals are constructed at the 95% level. In blue, I plot the coefficient on the interaction between exposure and year. In red, I plot the coefficient on the triple interaction among exposure, year, and an indicator for being a woman.

Taken together, these results indicate that the asymmetric migration response among

couples, depending on which partner is exposed, is driven neither by the presence of children in the household nor by general gender differences in responses to the shock.

A.4 Additional Tables

Table A.2: Exposure to the China Trade Shock and Labor Market Outcomes

	(1) Employed=1	(2) Unemployed=1	(3) Lab Force=1
Exposed=1	-0.007	-0.002	-0.008
	(0.019)	(0.008)	(0.017)
(Exposed=1)x(Post=1)	-0.063***	0.023**	-0.039***
	(0.017)	(0.010)	(0.012)
Baseline Mean	0.90	0.04	0.94
	(1) Weekly Hours	(2) Hourly Wage	(3) Labor Income
Exposed=1	-0.454	-0.677	1,978.003
	(1.644)	(4.011)	(8,565.422)
(Exposed=1)x(Post=1)	-2.058**	-2.581**	-5595.397**
	(0.889)	(1.160)	(2,273.988)
Baseline Mean	40.62	24.26	50,236.60
Observations	38,634	38,634	38,634

Notes. Robust standard errors are two-way clustered at the industry and individual levels. * for p<0.10, ** for p<0.05, *** for p<0.01.

Table A.3: Binary versus Continuous Exposure to the China Trade Shock

	(1) Employed=1	(2) Unemployed=1	(3) Labor Force=1
Exposed=1	-0.007	-0.002	-0.008
	(0.019)	(0.008)	(0.017)
(Exposed=1)x(Post=1)	-0.063***	0.023**	-0.039***
	(0.017)	(0.010)	(0.012)
Shock	0.003*	-0.000	0.003**
	(0.002)	(0.001)	(0.001)
(Shock)x(Post=1)	-0.007***	0.003***	-0.003***
	(0.002)	(0.001)	(0.001)
Observations	38,634	38,634	38,634
R2	0.08	0.05	0.06
Baseline Mean	0.90	0.04	0.94

Notes. Robust standard errors are two-way clustered at the industry and individual levels. * for p<0.10, ** for p<0.05, *** for p<0.01.

Table A.4: Exposure to the China Trade Shock and Out-Migration Responses

	(1) Out-Migration=1
Exposed=1	-0.064
	(0.042)
(Exposed=1)x(Post=1)	0.047**
	(0.021)
Observations	37,443
R^2	0.09
Baseline Mean	0.07

Notes. Robust standard errors are two-way clustered at the industry and individual levels. $*$ for p<0.10, $**$ for p<0.05, $***$ for p<0.01.

Table A.5: Wife's Exposure to the China Trade Shock and Husband's Labor Market Outcomes

	(1) Husband Employed=1	(2) Husband's Hourly Wage
Exposed Wife=1	-0.021	-1.527
	(0.020)	(1.355)
(Exposed Wife=1)x(Post=1)	-0.030	-9.349***
	(0.026)	(1.557)
Observations	14,083	14,083
R^2	0.10	0.33
Baseline Mean	0.95	30.42

Notes. All monetary values are in 2017 USD, using the PCE index, and include zeros. Robust standard errors are two-way clustered at the wife's industry and individual levels. * for p<0.10, ** for p<0.05, *** for p<0.01.

Table A.6: Couple's Exposure to the China Trade Shock and Household Income Responses

	(1)	(2)
	Household Income	
Exposed Husband=1	7,357.215	
	(11692.540)	
(Exposed Husband=1)x(Post=1)	3,930.551	
	(6,330.419)	
Exposed Wife=1		529.105
		(12003.901)
(Exposed Wife=1)x(Post=1)		-2348.701
		(4,316.531)
Observations	14,083	14,083
R^2	0.35	0.35
Baseline Mean	120226.29	120600.92

Notes. All monetary values are in 2017 USD, using the PCE index, and include zeros. Robust standard errors are two-way clustered at the husband's industry and household levels in Column 1 and the wife's industry and household levels in Column 2. * for p<0.10, ** for p<0.05, *** for p<0.01.

B. Model Derivations

This appendix derives the choice probabilities and value function under the nested Gumbel shock and then develops the money-metric measure used to compare welfare across singles and couples.

B.1 Choice Probabilities and Expected Values

The derivation follows Appendix B.2 of Rodríguez-Clare et al. (2026). Fix h, $\boldsymbol{o}$, and t, and suppress these indices whenever doing so creates no ambiguity. Define $x_{\boldsymbol{d}} \equiv \delta V_{\boldsymbol{d},t+1} - \mu_{\boldsymbol{od}}$. The household chooses $\boldsymbol{d}$ if and only if $x_{\boldsymbol{d}} + \varepsilon_{\boldsymbol{d}} \geq x_{\boldsymbol{z}} + \varepsilon_{\boldsymbol{z}}$ for every $\boldsymbol{z}$.

Choice probabilities. Let $G_{\boldsymbol{d}}(\cdot)$ denote the partial derivative of F in equation (15) with respect to its $\boldsymbol{d}$-th argument. For any $\boldsymbol{d} \in \mathcal{S}_h(j)$,

$$G_{\boldsymbol{d}}(\varepsilon) = \frac{1}{\theta}\left(\sum_{\boldsymbol{z}\in\mathcal{S}_h(j)} e^{-\varepsilon_{\boldsymbol{z}}/\nu}\right)^{\nu/\theta-1} e^{-\varepsilon_{\boldsymbol{d}}/\nu}\, F(\varepsilon).$$

The choice probability for $\boldsymbol{d}$ is $\int G_{\boldsymbol{d}}(\varepsilon_{\boldsymbol{d}}, \{\varepsilon_{\boldsymbol{d}} + x_{\boldsymbol{d}} - x_{\boldsymbol{z}}\}_{\boldsymbol{z}\neq\boldsymbol{d}})\, d\varepsilon_{\boldsymbol{d}}$. Evaluating each remaining argument at $\varepsilon_{\boldsymbol{z}} = \varepsilon_{\boldsymbol{d}} + x_{\boldsymbol{d}} - x_{\boldsymbol{z}}$ yields $e^{-\varepsilon_{\boldsymbol{z}}/\nu} = e^{-\varepsilon_{\boldsymbol{d}}/\nu} e^{(x_{\boldsymbol{z}} - x_{\boldsymbol{d}})/\nu}$ and therefore

$$\sum_{\boldsymbol{z}\in\mathcal{S}_h(\ell)} e^{-\varepsilon_{\boldsymbol{z}}/\nu} = e^{-\varepsilon_{\boldsymbol{d}}/\nu} e^{-x_{\boldsymbol{d}}/\nu}\,\Omega_\ell, \qquad \Omega_\ell \equiv \sum_{\boldsymbol{z}\in\mathcal{S}_h(\ell)} e^{x_{\boldsymbol{z}}/\nu},$$

so Ω_ℓ is shorthand for the inclusive value $\Omega^h_{\boldsymbol{o}\ell,t}$, where $\Omega^h_{\boldsymbol{o}j,t} \equiv \sum_{\boldsymbol{z}\in\mathcal{S}_h(j)} \exp\left[\delta V^h_{\boldsymbol{z},t+1} - \mu_{\boldsymbol{oz}}\right]^{1/\nu}$. Substituting, and writing $T \equiv e^{-x_{\boldsymbol{d}}/\theta} \sum_\ell \Omega_\ell^{\nu/\theta}$,

$$\Pr(\boldsymbol{d}) = \frac{e^{x_{\boldsymbol{d}}/\nu}}{\Omega_j} \cdot \frac{\Omega_j^{\nu/\theta}}{\sum_\ell \Omega_\ell^{\nu/\theta}} \int_{-\infty}^{\infty} \frac{1}{\theta}\, T\, e^{-\varepsilon_{\boldsymbol{d}}/\theta} \exp\left(-T e^{-\varepsilon_{\boldsymbol{d}}/\theta}\right) d\varepsilon_{\boldsymbol{d}}.$$

The integrand is the density of a univariate Gumbel distribution with scale θ, so the integral equals one. The first fraction corresponds to $\lambda^h_{\boldsymbol{od}|j,t}$ in equation (18), while the second corresponds to $\lambda^h_{\boldsymbol{o}j,t}$ in equation (17).

Expected value. The joint probability that $x_{\boldsymbol{d}} + \varepsilon_{\boldsymbol{d}} \leq a$ and that $\boldsymbol{d}$ is chosen is obtained by integrating the same expression up to $a - x_{\boldsymbol{d}}$. This yields $\lambda^h_{\boldsymbol{od},t} \exp\left(-\tilde{T} e^{-a/\theta}\right)$, where $\tilde{T} \equiv \sum_\ell \Omega_\ell^{\nu/\theta}$. Conditional on $\boldsymbol{d}$ being chosen, the maximized payoff therefore follows a Gumbel distribution with location $\theta \log \tilde{T}$ and scale θ, implying a mean of $\theta \log \tilde{T} + \theta\, \bar{e}$,

where $\bar{e}$ is the Euler–Mascheroni constant. Because this expression does not depend on $\boldsymbol{d}$,

$$\mathbb{E}\left[\max_{\boldsymbol{d}}\{x_{\boldsymbol{d}}+\varepsilon_{\boldsymbol{d}}\}\right]=\theta\log\left(\sum_{\ell=1}^{L}\Omega_{\boldsymbol{o}\ell,t}^{\nu/\theta}\right)+\theta\,\bar{e},$$

which, after dropping the constant, delivers equation (16).

Welfare recursion. From equation (17) evaluated at the origin location i, $\sum_{\ell}\Omega_{\boldsymbol{o}\ell,t}^{\nu/\theta}=(\lambda_{\boldsymbol{o}i,t}^{h})^{-1}\Omega_{\boldsymbol{o}i,t}^{\nu/\theta}$. From equation (18) evaluated at the origin bundle and $\mu_{\boldsymbol{oo}}=0$, $\Omega_{\boldsymbol{o}i,t}=(\lambda_{\boldsymbol{oo}|i,t}^{h})^{-1}\exp(\delta V_{\boldsymbol{o},t+1}^{h})^{1/\nu}$. Combining,

$$\theta\log\left(\sum_{\ell}\Omega_{\boldsymbol{o}\ell,t}^{\nu/\theta}\right)=\delta V_{\boldsymbol{o},t+1}^{h}-\theta\log\left(\lambda_{\boldsymbol{o}i,t}^{h}\right)-\nu\log\left(\lambda_{\boldsymbol{oo}|i,t}^{h}\right),$$

which, substituted into equation (16) and iterated forward, gives equation (28).

B.2 The Money-Metric Welfare Index

I follow Chiappori and Meghir (2015) to construct a Money-Metric Welfare Index (MMWI), defined as the income each spouse would require to attain their utility level if they faced the full price of public consumption. For notational simplicity, I suppress the indices $\boldsymbol{d}$ and t.

Let $u_g\equiv\gamma\log Q+(1-\gamma)\log c_g$, for $g\in\{f,m\}$, denote the log utility of spouse g. Substituting equation (21) into u_g, I obtain

$$\begin{aligned}u_f&=\log\frac{I(e)}{P}+e\gamma\log\beta+K+(1-\gamma)\log\alpha,\\ u_m&=\log\frac{I(e)}{P}+e\gamma\log\beta+K+(1-\gamma)\log(1-\alpha),\end{aligned}$$

where $K\equiv\gamma\log\gamma+(1-\gamma)\log(1-\gamma)$. In levels, $U_g=\mathcal{A}_g\beta^{e\gamma}I(e)/P$ with $\mathcal{A}_f=e^{K}\alpha^{1-\gamma}$ and $\mathcal{A}_m=e^{K}(1-\alpha)^{1-\gamma}$. Ex ante utility corresponds to

$$\mathbb{E}[U_g]=(\mathcal{A}_g/\mathcal{A})\,\mathbb{E}[\mathcal{U}]. \tag{B.1}$$

Thus, each spouse's expected utility is a fixed multiple of the couple's expected utility. The ratio of private consumption is $c_f^*/c_m^*=\alpha/(1-\alpha)$ and is independent of wages, β, and e. Because the model contains no distribution factors, an increase in the wife's wage raises household income without changing her share of it.

1. Conditional sharing rule (CSR). The CSR allocates between the spouses the income remaining after expenditure on the public good:

$$\tilde{\varsigma}_f = Pc_f^* = \alpha(1-\gamma)I, \qquad \tilde{\varsigma}_m = Pc_m^* = (1-\alpha)(1-\gamma)I, \qquad \tilde{\varsigma}_f + \tilde{\varsigma}_m = (1-\gamma)I.$$

The CSR therefore excludes both the γI spent on the public good and the increase in public-good production generated by home production.

2. Lindahl prices. One unit of Q costs the couple $P_Q \equiv P\beta^{-e}$. Thus, when the wife remains at home, home production reduces the effective price of the public good. Each spouse's personalized price equals their marginal willingness to pay for Q, expressed in monetary units:

$$P_g = P\,\frac{\partial u_g/\partial Q}{\partial u_g/\partial c_g} = \frac{\gamma}{1-\gamma}\cdot\frac{Pc_g^*}{Q^*} \quad \Longrightarrow \quad P_f = \alpha P_Q, \qquad P_m = (1-\alpha)P_Q.$$

These personalized prices satisfy the Samuelson condition $P_f + P_m = P_Q$. Because both spouses value Q identically, each bears a share of its cost equal to their bargaining weight.

3. Generalized sharing rule (GSR). The GSR adds each spouse's valuation of public expenditure at their personalized price to their private expenditure:

$$\varsigma_f^* = Pc_f^* + P_f Q^* = \alpha(1-\gamma)I + \alpha\gamma I = \alpha I, \qquad \varsigma_m^* = (1-\alpha)I.$$

A wife who remains at home ($e = 1$, $I = w^M$) has no labor earnings but is allocated resources equivalent to αw^M. The home-production shock does not directly affect the GSR; instead, it operates through the Lindahl price.

4. Money-Metric Welfare Index (MMWI). The MMWI of spouse g, denoted by m_g, is the income that spouse g would need to attain utility u_g when facing a reference price vector rather than personalized prices. An individual with preferences $Q^\gamma c^{1-\gamma}$ and income m, facing price P for private consumption and $\bar{P}_Q$ for the public good, attains

$$v(m) = \log m - (1-\gamma)\log P - \gamma \log \bar{P}_Q + K.$$

Equating $v(m_g)$ with u_g and solving for m_g yields the general formula $m_g = \varsigma_g^*\,(\bar{P}_Q/P_g)^\gamma$.

The remaining question is which reference price $\bar{P}_Q$ to use. If I use the couple's own effective price, $\bar{P}_Q = P\beta^{-e}$, then $m_f = \alpha^{1-\gamma}I$ and $m_m = (1-\alpha)^{1-\gamma}I$. Under this reference price, the index does not capture home production: for a wife who stays at home, it equals $\alpha^{1-\gamma}w^M$ regardless of her draw of β. I therefore use the market price, $\bar{P}_Q = P$, so that the index explicitly accounts for home production. In this case, $v(m) = \log(m/P) + K$, and

setting $v(m_f) = u_f$ yields

$$m_f = \alpha^{1-\gamma}\beta^{e\gamma} I, \qquad m_m = (1-\alpha)^{1-\gamma}\beta^{e\gamma} I. \tag{B.2}$$

Rewriting the wife's index as $m_f = \alpha I \cdot \alpha^{-\gamma} \cdot \beta^{e\gamma}$ separates three forces. The first term, αI, is her GSR share. The second term, $\alpha^{-\gamma}$, corresponds to the Lindahl-price adjustment. The third, $\beta^{e\gamma}$, captures the value of home production for spouses.

Couples solve the mobility problem in equation (16) before β is realized. The relevant ex ante index is therefore the certain income $\bar{m}_g$ that allows a person with the same preferences to attain $\mathbb{E}[U_g]$. Because income m yields level utility $e^K m/P$, and because $\mathbb{E}[U_g] = e^K \alpha_g^{1-\gamma}\, \mathbb{E}[\beta^{e\gamma} I]/P$, with $\alpha_f \equiv \alpha$ and $\alpha_m \equiv 1-\alpha$,

$$\bar{m}_f = \alpha^{1-\gamma}\, \Phi\, w^M, \qquad \bar{m}_m = (1-\alpha)^{1-\gamma}\, \Phi\, w^M. \tag{B.3}$$

where $\Phi \equiv \mathbb{E}[\beta^{e\gamma} I]/w^M$.

Summing equation (B.2) across spouses, I can express the gains from the public nature of Q as

$$\frac{m_f + m_m}{\beta^{e\gamma} I} = G(\alpha, \gamma) \equiv \alpha^{1-\gamma} + (1-\alpha)^{1-\gamma} > 1 \quad \text{whenever } 0 < \gamma < 1.$$

Thus, the spouses would jointly require more income than the household possesses if each had to purchase the public good separately at its full market price.

5. MMWI for singles. The central assumption is that singles have the same preferences as spouses but cannot share the public good with another household member. A single of gender g with wage w^g therefore solves

$$\max \gamma \log Q^g + (1-\gamma) \log c^g \quad \text{s.t.} \quad P(c^g + Q^g) = w^g.$$

This problem implies $c^g = (1-\gamma) w^g/P$ and $Q^g = \gamma w^g/P$, with indirect utility $u_s^g = \log \frac{w^g}{P} + K$. The constant K does not affect choices and cancels out in every change in welfare. Thus, the model's $W^{s_g} = w^g/P$ is an exact representation of these preferences, and no quantitative result changes. What a single lacks is not the public good itself but its *publicness*: a single person must pay its full price. Accordingly, the MMWI for a single of gender g is

$$m^g = \frac{w^g}{P}.$$

6. Money-Metric-Weighted Aggregate Welfare. Starting from equation (30), I can express the per-person consumption-equivalent change in utility as

$$\bar{Z} = \frac{\sum N^{s_m} Z^{s_m} + \sum N^{s_f} Z^{s_f} + \sum N^{c} Z^{c}}{\sum N^{s_m} + \sum N^{s_f} + \sum N^{c}}.$$

The measure $\bar{Z}$ averages percentage changes defined relative to different bases: for a single, the base is w/P; for a spouse, it is a share of the household's full income, together with a shared public good. To express the per-person consumption-equivalent change in monetary units, I therefore construct a money-metric-weighted measure that aggregates across singles and couples.

Because each spouse's expected utility is a constant multiple of the couple's expected utility, the consumption-equivalent change of each spouse is identical to that of the couple when expressed in hat notation. The money-metric-weighted measure is therefore

$$\bar{Z}^{\$} = \frac{\sum m_i \, Z_i}{\sum m_i} = \bar{Z} + \frac{\mathrm{Cov}(m_i, Z_i)}{\bar{m}}.$$

The covariance term captures whether the gains from the shock accrue to individuals with larger money metrics, such as spouses, or to those with smaller money metrics, such as singles.

C. Dynamic Hat Algebra

This appendix develops the Dynamic Hat Algebra used to solve and simulate the model. It expresses the equilibrium entirely in terms of changes over time and ratios across the with-shock and no-shock economies, so that I never have to recover the levels of the fundamentals. The appendix proceeds in two parts. I first state the two propositions that characterize the with-shock economy: one for the temporary equilibrium at a given point in time and another for the full sequential equilibrium, together with the algorithm used to solve them. I then derive the corresponding results for the no-shock economy, defined by an alternative path of sectoral productivities.

C.1 Solving the Model

Notation. Let $\dot{x}_{t+1} = x_{t+1}/x_t$ denote the proportional change in any variable x between t and $t+1$ in the with-shock economy. Let $\hat{x}_{t+1} = \dot{x}'_{t+1}/\dot{x}_{t+1}$, where $\dot{x}'_{t+1} = x'_{t+1}/x'_t$, denote the ratio of the changes in the no-shock economy to those in the with-shock economy. Primes denote levels in the no-shock economy throughout. For every feasible bundle $\boldsymbol{o}$, define $u^h_{\boldsymbol{o},t} = \exp(V^h_{\boldsymbol{o},t})$, and define regional expenditure as $E_{j,t} \equiv P_{j,t}Q_{j,t}$. Because local sectoral productivities $\{A_{j,n,t}\}$ are the only fundamentals that vary over time, a quantitative application begins with a path of productivity changes $\{\dot{A}_{j,n,t}\}$ and solves the equilibrium system below.

For couples, define the quality-adjusted value of destination $\boldsymbol{d}$ as

$$B_{\boldsymbol{d},t} = \mathcal{A}\left[(1+r_{\boldsymbol{d},t}) + \chi^{\kappa}\frac{\gamma}{\kappa-\gamma}\left(1+r_{\boldsymbol{d},t}\right)^{1-\kappa/\gamma}\right].$$

Because $\mathcal{A}$ is constant, it cancels from all proportional changes and cross-economy ratios of B. I retain it in the level expressions below only for completeness. The proportional change in B is

$$\dot{B}_{\boldsymbol{d},t+1} \equiv \frac{B_{\boldsymbol{d},t+1}}{B_{\boldsymbol{d},t}} = \frac{(1+r_{\boldsymbol{d},t+1}) + \chi^{\kappa}\frac{\gamma}{\kappa-\gamma}\left(1+r_{\boldsymbol{d},t+1}\right)^{1-\kappa/\gamma}}{(1+r_{\boldsymbol{d},t}) + \chi^{\kappa}\frac{\gamma}{\kappa-\gamma}\left(1+r_{\boldsymbol{d},t}\right)^{1-\kappa/\gamma}}. \tag{C.1}$$

Proposition C.1. *Given the temporary-equilibrium allocation at t, the temporary equilibrium at $t+1$ can be computed without knowing the levels of the fundamentals. It is characterized by the following system of equations:*

$$\begin{aligned}
\dot{w}^M_{j,n,t+1} &= \dot{p}_{j,n,t+1}\dot{A}^{(\sigma-1)/\sigma}_{j,n,t+1}\dot{M}^{-1/\sigma}_{j,n,t+1}\dot{Y}^{1/\sigma}_{j,n,t+1},\\
\dot{w}^F_{j,n,t+1} &= \dot{p}_{j,n,t+1}\dot{A}^{(\sigma-1)/\sigma}_{j,n,t+1}\dot{F}^{-1/\sigma}_{j,n,t+1}\dot{Y}^{1/\sigma}_{j,n,t+1},\\
\dot{Y}_{j,n,t+1} &= \dot{A}_{j,n,t+1}\left(s_{j,n,m,t}\dot{M}^{(\sigma-1)/\sigma}_{j,n,t+1} + s_{j,n,f,t}\dot{F}^{(\sigma-1)/\sigma}_{j,n,t+1}\right)^{\sigma/(\sigma-1)},
\end{aligned}$$

$$\dot{P}_{j,n,t+1} = \left(\sum_{i=1}^{L} \sum_{k=1}^{K} \pi_{ik,jn,t} \left(\dot{p}_{i,k,t+1} \right)^{1-\eta_n} \right)^{1/(1-\eta_n)},$$

$$\pi_{ik,jn,t+1} = \pi_{ik,jn,t} \left(\frac{\dot{p}_{i,k,t+1}}{\dot{P}_{j,n,t+1}} \right)^{1-\eta_n},$$

$$\dot{P}_{j,t+1} = \prod_{n=1}^{K} \dot{P}_{j,n,t+1}^{\iota_n}, \qquad \dot{E}_{j,t+1} = \dot{P}_{j,t+1} \dot{Q}_{j,t+1},$$

$$\dot{p}_{i,k,t+1} \dot{Y}_{i,k,t+1} = \sum_{j=1}^{L} \sum_{n=1}^{K} \varphi_{ik,jn,t} \frac{\pi_{ik,jn,t+1}}{\pi_{ik,jn,t}} \dot{E}_{j,t+1},$$

for all region-sector pairs (j,n)*, where* $s_{j,n,m,t} \equiv \frac{w^M_{j,n,t} M_{j,n,t}}{p_{j,n,t} Y_{j,n,t}}$ *and* $s_{j,n,f,t} \equiv \frac{w^F_{j,n,t} F_{j,n,t}}{p_{j,n,t} Y_{j,n,t}}$ *are the male and female shares of sectoral revenue, and* $\varphi_{ik,jn,t} \equiv \frac{\pi_{ik,jn,t}\, \iota_n P_{j,t} Q_{j,t}}{p_{i,k,t} Y_{i,k,t}}$ *is the share of origin-sector revenue from* (i,k) *sold to destination sector* (j,n)*. Moreover, for each destination bundle relevant to couples,* $\dot{r}_{\boldsymbol{d},t+1} = \dot{w}^F_{j,n_f,t+1} / \dot{w}^M_{j,n_m,t+1}$*,* $r_{\boldsymbol{d},t+1} = r_{\boldsymbol{d},t}\, \dot{r}_{\boldsymbol{d},t+1}$*,* $\rho_{\boldsymbol{d},t+1} = 1 - \chi^{\kappa} \left(1 + r_{\boldsymbol{d},t+1}\right)^{-\kappa/\gamma}$*, and* $\dot{B}_{\boldsymbol{d},t+1}$ *follows from equation* (C.1)*.*

Building on Proposition C.1, I next characterize the sequential equilibrium in time differences. equation (18) implies that the ratio of within-location shares between $t+1$ and t is

$$\frac{\lambda^h_{\boldsymbol{od}|j,t+1}}{\lambda^h_{\boldsymbol{od}|j,t}} = \frac{\exp\left[\delta \left(V^h_{\boldsymbol{d},t+2} - V^h_{\boldsymbol{d},t+1}\right)\right]^{1/\nu}}{\Omega^h_{\boldsymbol{o}j,t+1} / \Omega^h_{\boldsymbol{o}j,t}}, \qquad \frac{\Omega^h_{\boldsymbol{o}j,t+1}}{\Omega^h_{\boldsymbol{o}j,t}} = \sum_{\boldsymbol{z} \in \mathcal{S}_h(j)} \lambda^h_{\boldsymbol{oz}|j,t} \left(\dot{u}^h_{\boldsymbol{z},t+2}\right)^{\delta/\nu},$$

where the second equality follows from the time invariance of $\mu_{\boldsymbol{oz}}$. Define the proportional change in the inclusive value of location j as

$$D^h_{\boldsymbol{o}j,t} \equiv \sum_{\boldsymbol{z} \in \mathcal{S}_h(j)} \lambda^h_{\boldsymbol{oz}|j,t} \left(\dot{u}^h_{\boldsymbol{z},t+2}\right)^{\delta/\nu}. \tag{C.2}$$

The location shares in equation (17) and the value function in equation (16) depend on continuation values only through the inclusive values. Their time differences can therefore be expressed entirely in terms of $D^h_{\boldsymbol{o}j,t}$.

Proposition C.2. *Given the economy's initial allocation, including the initial location shares* $\lambda^h_{\boldsymbol{o}j,0}$ *and within-location shares* $\lambda^h_{\boldsymbol{od}|j,0}$*, together with an anticipated convergent sequence of productivity changes, the sequential equilibrium in time differences can be solved without recovering the levels of the fundamentals. The solution satisfies the following system of*

nonlinear equations:

$$\lambda^h_{\boldsymbol{od}|j,t+1} = \frac{\lambda^h_{\boldsymbol{od}|j,t}\left(\dot{u}^h_{\boldsymbol{d},t+2}\right)^{\delta/\nu}}{\sum_{\boldsymbol{z}\in\mathcal{S}_h(j)}\lambda^h_{\boldsymbol{oz}|j,t}\left(\dot{u}^h_{\boldsymbol{z},t+2}\right)^{\delta/\nu}}, \qquad \forall h,$$

$$\lambda^h_{\boldsymbol{o}j,t+1} = \frac{\lambda^h_{\boldsymbol{o}j,t}\left(D^h_{\boldsymbol{o}j,t}\right)^{\nu/\theta}}{\sum_{\ell=1}^{L}\lambda^h_{\boldsymbol{o}\ell,t}\left(D^h_{\boldsymbol{o}\ell,t}\right)^{\nu/\theta}}, \qquad \forall h,$$

$$\dot{u}^h_{\boldsymbol{o},t+1} = \dot{W}^h_{\boldsymbol{o},t+1}\left(\sum_{\ell=1}^{L}\lambda^h_{\boldsymbol{o}\ell,t}\left(\sum_{\boldsymbol{z}\in\mathcal{S}_h(\ell)}\lambda^h_{\boldsymbol{oz}|\ell,t}\left(\dot{u}^h_{\boldsymbol{z},t+2}\right)^{\delta/\nu}\right)^{\nu/\theta}\right)^{\theta}, \qquad \forall h,$$

$$N^h_{\boldsymbol{d},t+1} = \sum_{\boldsymbol{o}\in\mathcal{S}_h}\lambda^h_{\boldsymbol{od}|j,t}\,\lambda^h_{\boldsymbol{o}j,t}\,N^h_{\boldsymbol{o},t}, \qquad \boldsymbol{d}\in\mathcal{S}_h(j), \quad \forall h,$$

$$M_{j,n,t+1} = N^{s_m}_{j,n,t+1} + \sum_{n_f=1}^{K} N^c_{j,n,n_f,t+1},$$

$$F_{j,n,t+1} = N^{s_f}_{j,n,t+1} + \sum_{n_m=0}^{K}\rho_{j,n_m,n,t+1}N^c_{j,n_m,n,t+1},$$

for all feasible bundles and all sectors at each t, where $D^h_{\boldsymbol{o}\ell,t}$ is defined in equation (C.2) and the sectoral prices, trade shares, and participation objects solve the temporary-equilibrium system in Proposition C.1.

Because the migration costs $\mu_{\boldsymbol{od}}$ are time-invariant, they cancel from all ratios and therefore need not be recovered in levels. The flow term in the couple's continuation value is the only one that contains an additional factor. In particular, because the couple's indirect utility is $W^c_{\boldsymbol{o},t} = B_{\boldsymbol{o},t}\,w^M_{i,k_m,t}/P_{i,t}$, its proportional change is $\dot{B}_{\boldsymbol{o},t+1}\dot{w}^M_{i,k_m,t+1}/\dot{P}_{i,t+1}$. Thus, the relevant term is the *ratio* defined in equation (C.1), rather than the level $B_{\boldsymbol{o},t+1}$.

Algorithm for Solving the Sequential Competitive Equilibrium. Given an initial equilibrium at $t=0$ and an anticipated convergent sequence of sectoral productivity changes $\left\{\dot{A}_{j,n,t}\right\}^{L,K,T}_{j=1,\,n=1,\,t=0}$, the algorithm is as follows:

1. Fix a sufficiently large terminal horizon T and guess a path for the changes in continuation values $\left\{\dot{u}^{h,(0)}_{\boldsymbol{o},t+1}\right\}^{T}_{t=0}$, subject to the terminal condition $\dot{u}^{h,(0)}_{\boldsymbol{o},T+1} = 1$ for every feasible bundle $\boldsymbol{o}$.

2. Decompose the initial migration matrix into location shares $\lambda^h_{\boldsymbol{o}j,0} = \sum_{\boldsymbol{d}\in\mathcal{S}_h(j)}\lambda^h_{\boldsymbol{od},0}$ and within-location shares $\lambda^h_{\boldsymbol{od}|j,0} = \lambda^h_{\boldsymbol{od},0}/\lambda^h_{\boldsymbol{o}j,0}$. Using Proposition C.2, update these objects forward in the following order: the within-location shares, $D^h_{\boldsymbol{o}j,t}$, and then the location shares.

3. Starting from the initial allocation, apply the law of motion to recover the entire path of household masses $\left\{N_{\boldsymbol{d},t}^{s_m,(0)}, N_{\boldsymbol{d},t}^{s_f,(0)}, N_{\boldsymbol{d},t}^{c,(0)}\right\}_{t=0}^{T}$, and aggregate these masses into sectoral male labor supplies $\left\{M_{j,n,t}^{(0)}\right\}_{t=0}^{T}$.

4. Solve the temporary equilibrium sequentially for each period $t \geq 0$:

   (a) Guess the participation rates $\{\rho_{\boldsymbol{d},t+1}^{(0)}\}$ and use them to construct the sectoral female labor supplies $\left\{F_{j,n,t+1}^{(0)}\right\}$.

   (b) Conditional on the labor aggregates and the path of fundamentals, solve the production block for $\left\{\dot{Y}_{j,n,t+1}^{(0)}, \dot{w}_{j,n,t+1}^{M,(0)}, \dot{w}_{j,n,t+1}^{F,(0)}\right\}$, the trade block for $\left\{\dot{p}_{j,n,t+1}^{(0)}, \dot{P}_{j,n,t+1}^{(0)}, \pi_{ik,jn,t+1}^{(0)}\right\}$, and the regional aggregation block for $\left\{\dot{P}_{j,t+1}^{(0)}, \dot{E}_{j,t+1}^{(0)}\right\}$.

   (c) Recover $\left\{\dot{r}_{\boldsymbol{d},t+1}^{(0)}\right\}$, construct the levels $r_{\boldsymbol{d},t+1}^{(0)} = r_{\boldsymbol{d},t}\dot{r}_{\boldsymbol{d},t+1}^{(0)}$, and update $\left\{\rho_{\boldsymbol{d},t+1}^{(1)}, \dot{B}_{\boldsymbol{d},t+1}^{(1)}\right\}$.

   (d) Iterate over steps (a) through (c) until the participation block converges.

5. Given the temporary-equilibrium path and the guessed continuation values, solve backward from T for the updated path $\left\{\dot{u}_{\boldsymbol{o},t+1}^{h,(1)}\right\}_{t=0}^{T}$ using the $\dot{u}^h$ equation in Proposition C.2. This backward solution requires both the location shares and the within-location shares.

6. Use the updated path as the new guess and repeat steps 2 through 5 until the sequential equilibrium converges.

C.2 Solving the No-Shock Equilibrium

The path of sectoral productivity fundamentals in the no-shock economy, $\{A'_{j,n,t}\}_{t\geq 1}$, defines the corresponding productivity hats $\{\hat{A}_{j,n,t}\}_{t\geq 1}$. Under CDP's timing, households do not anticipate the change at $t = 0$. Therefore, $\lambda'^{h}_{\boldsymbol{o}j,0} = \lambda^{h}_{\boldsymbol{o}j,0}$, $\lambda'^{h}_{\boldsymbol{od}|j,0} = \lambda^{h}_{\boldsymbol{od}|j,0}$, and $N'^{h}_{\boldsymbol{o},1} = N^{h}_{\boldsymbol{o},1}$.

Dividing each no-shock migration equation in Proposition C.2 by its with-shock counterpart, and using $\dot{\lambda}^{h}_{\boldsymbol{od}|j,t} = \lambda^{h}_{\boldsymbol{od}|j,t}/\lambda^{h}_{\boldsymbol{od}|j,t-1}$ and $\dot{\lambda}^{h}_{\boldsymbol{o}j,t} = \lambda^{h}_{\boldsymbol{o}j,t}/\lambda^{h}_{\boldsymbol{o}j,t-1}$, yields the following no-shock analogue of equation (C.2):

$$\hat{D}^{h}_{\boldsymbol{o}j,t} \equiv \sum_{\boldsymbol{z}\in\mathcal{S}_h(j)} \lambda'^{h}_{\boldsymbol{oz}|j,t-1}\, \dot{\lambda}^{h}_{\boldsymbol{oz}|j,t}\, \left(\hat{u}^{h}_{\boldsymbol{z},t+1}\right)^{\delta/\nu}. \tag{C.3}$$

Proposition C.3. *Given the with-shock economy and a convergent sequence of productivity changes in the no-shock economy relative to the with-shock path, the no-shock sequential equilibrium can be solved without recovering the fundamental levels. For $t \geq 2$, it is characterized by the following system of nonlinear equations:*

$$\lambda'^{h}_{\boldsymbol{od}|j,t} = \frac{\lambda'^{h}_{\boldsymbol{od}|j,t-1}\,\dot{\lambda}^{h}_{\boldsymbol{od}|j,t}\,\left(\hat{u}^{h}_{\boldsymbol{d},t+1}\right)^{\delta/\nu}}{\hat{D}^{h}_{\boldsymbol{o}j,t}}, \qquad \forall h,$$

$$\lambda'^{h}_{\boldsymbol{o}j,t} = \frac{\lambda'^{h}_{\boldsymbol{o}j,t-1}\,\dot{\lambda}^{h}_{\boldsymbol{o}j,t}\,\left(\hat{D}^{h}_{\boldsymbol{o}j,t}\right)^{\nu/\theta}}{\sum_{\ell=1}^{L}\lambda'^{h}_{\boldsymbol{o}\ell,t-1}\,\dot{\lambda}^{h}_{\boldsymbol{o}\ell,t}\,\left(\hat{D}^{h}_{\boldsymbol{o}\ell,t}\right)^{\nu/\theta}}, \qquad \forall h,$$

$$\hat{u}^{h}_{\boldsymbol{o},t} = \hat{W}^{h}_{\boldsymbol{o},t}\left(\sum_{\ell=1}^{L}\lambda'^{h}_{\boldsymbol{o}\ell,t-1}\,\dot{\lambda}^{h}_{\boldsymbol{o}\ell,t}\left(\sum_{\boldsymbol{z}\in\mathcal{S}_h(\ell)}\lambda'^{h}_{\boldsymbol{oz}|\ell,t-1}\,\dot{\lambda}^{h}_{\boldsymbol{oz}|\ell,t}\,\left(\hat{u}^{h}_{\boldsymbol{z},t+1}\right)^{\delta/\nu}\right)^{\nu/\theta}\right)^{\theta}, \qquad \forall h,$$

$$N'^{h}_{\boldsymbol{d},t+1} = \sum_{\boldsymbol{o}\in\mathcal{S}_h}\lambda'^{h}_{\boldsymbol{od}|j,t}\,\lambda'^{h}_{\boldsymbol{o}j,t}\,N'^{h}_{\boldsymbol{o},t}, \qquad \boldsymbol{d}\in\mathcal{S}_h(j), \quad \forall h,$$

$$M'_{j,n,t+1} = N'^{s_m}_{j,n,t+1} + \sum_{n_f=1}^{K} N'^{c}_{j,n,n_f,t+1},$$

$$F'_{j,n,t+1} = N'^{s_f}_{j,n,t+1} + \sum_{n_m=0}^{K}\rho'_{j,n_m,n,t+1}N'^{c}_{j,n_m,n,t+1}.$$

In period $t = 1$, when households first learn the new path, the migration equations instead become

$$\lambda'^{h}_{\boldsymbol{od}|j,1} = \frac{\upsilon^{h}_{\boldsymbol{od}|j}\left(\hat{u}^{h}_{\boldsymbol{d},2}\right)^{\delta/\nu}}{\sum_{\boldsymbol{z}\in\mathcal{S}_h(j)}\upsilon^{h}_{\boldsymbol{oz}|j}\left(\hat{u}^{h}_{\boldsymbol{z},2}\right)^{\delta/\nu}},$$

$$\lambda'^{h}_{\boldsymbol{o}j,1} = \frac{\lambda^{h}_{\boldsymbol{o}j,1}\left(\sum_{\boldsymbol{z}\in\mathcal{S}_h(j)}\upsilon^{h}_{\boldsymbol{oz}|j}\left(\hat{u}^{h}_{\boldsymbol{z},2}\right)^{\delta/\nu}\right)^{\nu/\theta}}{\sum_{\ell=1}^{L}\lambda^{h}_{\boldsymbol{o}\ell,1}\left(\sum_{\boldsymbol{z}\in\mathcal{S}_h(\ell)}\upsilon^{h}_{\boldsymbol{oz}|\ell}\left(\hat{u}^{h}_{\boldsymbol{z},2}\right)^{\delta/\nu}\right)^{\nu/\theta}},$$

$$\hat{u}^{h}_{\boldsymbol{o},1} = \hat{W}^{h}_{\boldsymbol{o},1}\left(\sum_{\ell=1}^{L}\lambda^{h}_{\boldsymbol{o}\ell,1}\left(\sum_{\boldsymbol{z}\in\mathcal{S}_h(\ell)}\upsilon^{h}_{\boldsymbol{oz}|\ell}\left(\hat{u}^{h}_{\boldsymbol{z},2}\right)^{\delta/\nu}\right)^{\nu/\theta}\right)^{\theta}, \qquad \upsilon^{h}_{\boldsymbol{od}|j} \equiv \lambda^{h}_{\boldsymbol{od}|j,1}\left(\hat{u}^{h}_{\boldsymbol{d},1}\right)^{\delta/\nu}.$$

The no-shock temporary equilibrium is characterized by

$$\hat{w}^{M}_{j,n,t} = \hat{p}_{j,n,t}\hat{A}^{(\sigma-1)/\sigma}_{j,n,t}\hat{M}^{-1/\sigma}_{j,n,t}\hat{Y}^{1/\sigma}_{j,n,t},$$
$$\hat{w}^{F}_{j,n,t} = \hat{p}_{j,n,t}\hat{A}^{(\sigma-1)/\sigma}_{j,n,t}\hat{F}^{-1/\sigma}_{j,n,t}\hat{Y}^{1/\sigma}_{j,n,t},$$
$$\hat{Y}_{j,n,t} = \hat{A}_{j,n,t}\left(s_{j,n,m,t}\hat{M}^{(\sigma-1)/\sigma}_{j,n,t} + s_{j,n,f,t}\hat{F}^{(\sigma-1)/\sigma}_{j,n,t}\right)^{\sigma/(\sigma-1)},$$

$$\hat{P}_{j,n,t} = \left(\sum_{i=1}^{L} \sum_{k=1}^{K} \pi_{ik,jn,t} \left(\hat{p}_{i,k,t} \right)^{1-\eta_n} \right)^{1/(1-\eta_n)},$$

$$\pi'_{ik,jn,t} = \pi_{ik,jn,t} \left(\frac{\hat{p}_{i,k,t}}{\hat{P}_{j,n,t}} \right)^{1-\eta_n},$$

$$\hat{P}_{j,t} = \prod_{n=1}^{K} \hat{P}_{j,n,t}^{\iota_n}, \qquad \hat{E}_{j,t} = \hat{P}_{j,t} \hat{Q}_{j,t},$$

$$\hat{p}_{i,k,t} \hat{Y}_{i,k,t} = \sum_{j=1}^{L} \sum_{n=1}^{K} \varphi_{ik,jn,t} \frac{\pi'_{ik,jn,t}}{\pi_{ik,jn,t}} \hat{E}_{j,t},$$

$$\hat{r}_{\boldsymbol{d},t} = \frac{\hat{w}^F_{j,n_f,t}}{\hat{w}^M_{j,n_m,t}},$$

and the corresponding no-shock level objects satisfy

$$r'_{\boldsymbol{d},t} = r'_{\boldsymbol{d},t-1} \dot{r}_{\boldsymbol{d},t} \hat{r}_{\boldsymbol{d},t}, \qquad \rho'_{\boldsymbol{d},t} = 1 - \chi^{\kappa} \left(1 + r'_{\boldsymbol{d},t} \right)^{-\kappa/\gamma}, \qquad \hat{B}_{\boldsymbol{d},t} = \frac{B'_{\boldsymbol{d},t}}{B_{\boldsymbol{d},t}},$$

with $B'_{\boldsymbol{d},t}$ evaluated at $r'_{\boldsymbol{d},t}$. The with-shock revenue shares $s_{j,n,m,t}$, $s_{j,n,f,t}$ and revenue-allocation shares $\varphi_{ik,jn,t}$ are those of Proposition C.1, taken from the with-shock equilibrium.

Solution Algorithm for the No-Shock Equilibrium. Given the with-shock sequential equilibrium and a convergent sequence of changes in fundamentals, $\left\{ \hat{A}_{j,n,t} \right\}_{t \geq 1}$, that defines the no-shock economy, the no-shock equilibrium is computed as follows:

1. Because the temporary equilibrium and migration shares at $t = 0$ are unchanged, initialize all no-shock objects at their with-shock values.

2. Fix a sufficiently large horizon T and guess a path $\left\{ \hat{u}^{h,(0)}_{\boldsymbol{o},t} \right\}_{t=1}^{T+1}$, imposing $\hat{u}^{h,(0)}_{\boldsymbol{o},T+1} = 1$ for all feasible bundles $\boldsymbol{o}$ and household types h.

3. Use the guessed continuation values to compute the path of no-shock migration shares. At $t = 1$, construct $\upsilon^h_{\boldsymbol{od}|j}$ using the with-shock within-location shares and $\hat{u}^{h,(0)}_{\boldsymbol{d},1}$, and apply the period-1 equations in Proposition C.3. For $t \geq 2$, update, in sequence, the within-location shares, $\hat{D}^h_{\boldsymbol{o}j,t}$ from equation (C.3), and the location shares.

4. Starting from the same initial allocation as in the with-shock economy, use the no-shock shares to compute the implied path of no-shock household populations $\left\{ N'^{s_m,(0)}_{\boldsymbol{d},t}, N'^{s_f,(0)}_{\boldsymbol{d},t}, N'^{c,(0)}_{\boldsymbol{d},t} \right\}_{t=1}^{T+1}$.

5. Given the no-shock household masses, recover the path of no-shock sectoral male labor supply $\left\{M'^{(0)}_{j,n,t}\right\}_{t=1}^{T+1}$.

6. Solve the no-shock temporary equilibrium period by period. For each $t \geq 0$, proceed as follows:

   (a) Guess the no-shock participation rates $\left\{\rho'^{(0)}_{\boldsymbol{d},t+1}\right\}$.

   (b) Use these participation rates to compute no-shock female labor supply $\left\{F'^{(0)}_{j,n,t+1}\right\}$.

   (c) Construct $\left\{\hat{M}^{(0)}_{j,n,t+1}\right\}$ and $\left\{\hat{F}^{(0)}_{j,n,t+1}\right\}$. Given these objects and the changes in fundamentals, recover $\left\{\hat{Y}^{(0)}_{j,n,t+1}\right\}$ from the production block. Next, solve the goods-market-clearing block for $\left\{\hat{p}^{(0)}_{j,n,t+1}, \hat{P}^{(0)}_{j,n,t+1}, \hat{P}^{(0)}_{j,t+1}, \pi'^{(0)}_{ik,jn,t+1}\right\}$ and recover $\left\{\hat{w}^{M,(0)}_{j,n,t+1}, \hat{w}^{F,(0)}_{j,n,t+1}\right\}$ from the firms' first-order conditions. Finally, recover $\left\{\hat{r}^{(0)}_{\boldsymbol{d},t+1}\right\}$.

   (d) Recover $\left\{r'^{(0)}_{\boldsymbol{d},t+1}\right\}$ and update $\left\{\rho'^{(1)}_{\boldsymbol{d},t+1}, B'^{(1)}_{\boldsymbol{d},t+1}\right\}$. If the updated $\rho'^{(1)}_{\boldsymbol{d},t+1}$ differs from its initial guess, return to step (b) and iterate until the participation block converges.

   (e) Repeat steps (a) through (d) sequentially for each period t.

7. Given the no-shock temporary-equilibrium path and the guessed continuation values, solve backward from T for the updated path $\left\{\hat{u}^{h,(1)}_{\boldsymbol{o},t}\right\}_{t=1}^{T+1}$ using the $\hat{u}^h$ equations in Proposition C.3. At $t = 1$, $\hat{u}^h_{\boldsymbol{o},1}$ appears on both sides through θ, so solve this equation as a fixed point.

8. Use the updated path as the new guess and repeat steps 3 through 7 until the continuation-value path converges.

9. Compute welfare using equation (29), with $\hat{\lambda}^h_{\boldsymbol{o}i,s}$ and $\hat{\lambda}^h_{\boldsymbol{oo}|i,s}$ constructed from the no-shock and with-shock location and within-location staying shares.

D. Quantification of the Model

This appendix details three elements of the model's quantification. First, I explain how I assign each non-employed married woman to her potential sector. Then, I describe how I build the migration matrices by household type. Finally, I describe the calibration of the couple's maximization problem.

D.1 Assignment of Non-Employed Wives to Their Potential Sector

The model assigns every wife to a productive sector, but the ACS does not indicate which sector a non-employed married woman would work in. In the ACS, which I use to construct the initial distribution of couples across states and sector pairs, $\{N^c_{\boldsymbol{o},0}\}$, a married woman who is out of the labor force is recorded as non-employed. Thus, I do not observe the counterfactual sector in which she would have worked if the couple had decided that she would participate in the market. In the model, couples choose the wife's productive sector first and her participation second, so she is never assigned to the non-employment sector itself. I therefore need to distribute the non-employed wives across productive sectors.

If I were to distribute them across productive sectors in the same proportions as employed wives, I would ignore a selection problem. The sectors in which married women are more likely to be employed are also the ones in which they are less likely to be non-employed. Following the distribution of employed wives would therefore assign a larger share of non-employed wives to the sectors that are least likely to be their potential sector.

To recover the potential sector, I return to the model. I drop the time index and write the labor supply of married women in state j and sector n as

$$F^c_{j,n} = \rho_{j,n} N^c_{j,n}.$$

The mass of non-employed wives in the same cell, which I denote by Non-$F^c_{j,n}$, is

$$\text{Non-}F^c_{j,n} = (1 - \rho_{j,n}) N^c_{j,n}.$$

The ratio of non-employed to employed wives is therefore

$$\frac{\text{Non-}F^c_{j,n}}{F^c_{j,n}} = \frac{(1 - \rho_{j,n})}{\rho_{j,n}},$$

which implies that the mass of non-employed wives in that cell is

$$\text{Non-}F^c_{j,n} \;=\; \underbrace{F^c_{j,n}}_{\text{observed}} \times \underbrace{\frac{1-\rho_{j,n}}{\rho_{j,n}}}_{\text{the model's odds of non-employment}} .$$

Dividing by $\rho_{j,n}$ therefore corrects the selection problem and assigns each non-employed wife to her potential productive sector.

These cell-level masses are in shares rather than headcounts, so nothing in the model's odds forces them to sum to the number of non-employed married women actually observed in state j. I therefore normalize them within each state, rescaling the $\{\text{Non-}F^c_{j,n}\}_n$ proportionally so that they add up to the observed state total.

D.2 Building the Migration Matrix

Since I solve the model with Dynamic Hat Algebra, the time-invariant mobility costs $\mu_{\boldsymbol{od}}$ cancel from the recursion that governs the choice probabilities (Proposition C.2). The recursion requires only two initial objects for each household type: the location shares $\lambda^h_{\boldsymbol{o}j,0}$ of equation (17) and the within-location shares $\lambda^h_{\boldsymbol{od}|j,0}$ of equation (18).

Following CDP's Appendix E, I build the initial migration matrix as the product of an interstate probability from the ACS and a sectoral transition probability from the Current Population Survey (CPS), where an interstate mover's sectoral transition is imputed from the destination state's sector matrix:

$$\lambda^h_{(i,k)(j,n),0} = \underbrace{f^h_{ij}}_{\text{ACS}} \times \underbrace{P^h_j(n \mid k)}_{\text{CPS}} .$$

Summing over the destination sectors of location j gives $\lambda^h_{\boldsymbol{o}j,0} = f^h_{ij}$, and dividing by that sum gives $\lambda^h_{\boldsymbol{od}|j,0} = P^h_j(n \mid k)$.[36]

I build the matrices in three steps: interstate flows from the ACS, sectoral transitions from the CPS, and a mapping of both onto the model's full grid. First, I derive interstate flows from the ACS migration question for 2001–2007, which records the state of residence one year earlier. The resulting state-to-state matrix is therefore annual, whereas the model period is quarterly. Following CDP's Appendix E, I assume that interstate moves are evenly distributed throughout the year and that a household makes at most one such move annually, so the quarterly matrix retains one-fourth of the moves and imputes the remaining three-

[36] Where an origin puts no mass on a location, the within-location share is undefined. I set it to zero rather than to a uniform distribution, since the two factors only ever enter as a product, and a uniform distribution would place mass on destinations that the data indicate are unreachable once the location share moves away from zero.

fourths as non-moves:

$$f^q_{i\to j} = \tfrac{1}{4} f_{i\to j} \quad (j \neq i), \qquad f^q_{i\to i} = \tfrac{1}{4} f_{i\to i} + \tfrac{3}{4}.$$

I apply this adjustment to every quarter from 2000 to 2007. The adjustment is quantitatively important: combining the *annual* interstate probability with the *quarterly* sector matrix without it would inflate mobility by roughly a factor of six, implying a quarterly own-state retention rate of 0.959 for single men against 0.994 in CDP. The error would fall entirely on the location margin, which is calibrated to the population's response to the shock.

Next, I estimate sectoral transitions from the CPS on fewer sectors than the model's 22 sectors plus non-employment, because disaggregating by household type would otherwise leave too few observations per cell. I therefore pool the 12 manufacturing sub-sectors into a single manufacturing sector and the 8 service sub-sectors into a single service sector, leaving an aggregated five-sector structure: non-employment, manufacturing, retail, construction, and services. I estimate transitions on this aggregation separately for single men, single women, and couples, with couples moving on the joint space of both spouses' groups.

Combining the first two steps gives a transition matrix across five sectors and 50 states, by household type, which I then map onto the model's grid of 22 productive sectors. Let $\tilde{\boldsymbol{o}}$ denote the origin bundle in the five-sector grid and $\tilde{\boldsymbol{d}}$ denote the corresponding destination bundle. The production side considers 22 productive sectors across 50 states. Applying the same sectoral aggregation to the production side would lose the dispersion across manufacturing sectors in the China Trade Shock. I map the matrices onto the full grid by factorizing each transition probability into a group-level move and a within-group destination. For an individual of gender $g \in \{m, f\}$ in a household of type h,

$$\underbrace{P_{h,g}(\boldsymbol{d} \mid \boldsymbol{o})}_{\text{model's grid}} = \underbrace{P_h(\tilde{\boldsymbol{d}} \mid \tilde{\boldsymbol{o}})}_{\text{ACS + CPS, by household type}} \times \underbrace{P_g(\boldsymbol{d} \mid \tilde{\boldsymbol{d}},\ \boldsymbol{o})}_{\text{within-group composition, by gender}},$$

so that household type governs *which* broad sector, and which state, a household moves to, and the within-group composition governs *where within* that sector it lands. Because this factorizes a single conditional distribution, the rows of the resulting matrix sum to one by construction. The within-group composition only redistributes mass among sectors of the same group *in the same state*, so it leaves the location shares unchanged and acts only on the within-location shares.

To build the within-group composition, I split the CPS sample by sex and compute, within manufacturing and within services, the share of men and of women going to each sub-sector. This step matters: men and women sort differently within both groups, with men concentrated in heavy and durable manufacturing and women in lighter, non-durable

goods. Without it, the model would miss one of the channels through which the shock affected men and women differently.

D.3 Calibration of the Couple's Maximization Problem

This subsection documents the calibration of the couple's maximization problem in equation (20), with $\beta \sim \text{Pareto}(\chi, \kappa)$. Four parameters govern it: the wife's weight in the household objective, α; the public-good expenditure share, γ; and the lower bound and shape of the home-production shock, χ and κ. Because I follow Chiappori and Mazzocco (2017)'s method to solve the couple's problem, I identify the four parameters *sequentially*, each from its own source and each mapped onto a distinct piece of variation.

D.3.1 α and γ from Expenditure Data

The optimal allocations for private and public consumption in equation (21) are Cobb-Douglas shares, so both parameters map exactly onto expenditure shares. The parameter α maps onto the share of female private consumption in total private consumption, and γ maps onto the share of public expenditure in total expenditure:

$$\alpha = \frac{c_f^*}{c_f^* + c_m^*}, \qquad \gamma = \frac{q^*}{q^* + c_f^* + c_m^*}.$$

Data. I use the Consumer Expenditure Interview Survey (CEX), 2000–2007, to construct three inputs.[37] First, the *assignable private goods* c_m and c_f are adult apparel and footwear.[38] Second, the *public good* q is the market input to home production: because $Q = \beta^e q$, the object I need is what the household combines with the wife's home time. It comprises shelter, fuels and utilities, food at home, cable and internet services, do-it-yourself repair materials, and household operations such as domestic services, lawn care, and appliance repair. Household operations are the *market substitutes* for home time, so omitting them would measure q selectively on precisely the margin the model is about. Third, I construct *total expenditure* Y as consumption only.[39]

The sample consists of household-year pairs with complete twelve-month expenditure records. Couples are intact married couples with both spouses present, aged 25–65; singles are one-person consumer units in the same age band. I drop households containing a 16- or 17-year-old, because the CEX books apparel for ages 16 and over under the adult men's and

[37] The complete list of expenditure codes entering each of the three objects is available upon request.

[38] Children's, boys', girls', and infants' apparel are excluded, since the model has no children.

[39] The CEX expenditure file also carries asset stocks, debt balances, loans, principal repayments, and asset-sale receipts. All of these are excluded, as are transfers out of the household and gifts.

women's codes, so a teenager's clothing would otherwise be read as a spouse's assignable private consumption. The couples and singles samples are disjoint by construction.

Estimating α. I estimate the wife's weight in the household objective, α, from apparel, the only assignable good in the CEX, correcting for the fact that women spend more on apparel than men. Apparel is only a *subset* of private consumption, so the raw apparel ratio, 0.60 in this sample, is not α. Taken at face value, the raw ratio would imply that the wife has the larger Pareto weight. To recover α, I follow Browning et al. (2013): I estimate an Engel curve for apparel among *singles* of each gender and recover the resource shares ϱ_f and ϱ_m such that, evaluated at each spouse's own resources $\varrho_g Y$, these Engel curves reproduce the aggregate apparel spending observed in couples. Using both ϱ_f and ϱ_m, I recover $\alpha = \varrho_f/(\varrho_f + \varrho_m)$. The correction is substantial: it moves α from 0.60 to 0.47, which *reverses* which spouse has the larger bargaining power.

Voena (2015) estimates a value of 0.3, which is one-third smaller than the one I estimate. The two numbers do not align because α captures different objects in the two papers. In Voena (2015)'s world, α prices the wife's outside option given changes in divorce laws across US states, while in my model, α reflects her observed consumption. In addition, she estimates α using data from the PSID between 1980 and 1990, years when female labor force participation was lower than in the 2000s, and women's bargaining power is expected to rise as they enter the labor force.

Estimating γ. I measure the public-good expenditure share, γ, directly from expenditure data rather than as the residual implied by the resource shares. One alternative would be to set γ equal to the residual $1 - \varrho_f - \varrho_m$. The residual is 0.13, implying that the couple spends roughly \$5,400 a year on public goods, while couples are *observed* spending \$22,300 a year on shelter, utilities, and food alone. The residual is this small because the apparel Engel curves are essentially flat ($R^2 = 0.004$). Browning et al. (2013)'s method allows me to recover the *ratio*, but it carries almost no information about the *level* of resources, and the residual absorbs the error. I therefore measure γ directly as the share of total spending devoted to market inputs for home production, including shelter, utilities, food at home, and market substitutes for home time, such as domestic services and appliance repair. This yields $\gamma = 0.4887$.

Del Boca and Flinn (1995) estimate γ to be 0.204. The two values differ because 0.204 is an estimate based on divorced parents, whose only joint consumption concerns their children; they cannot enjoy economies of scale in housing, utilities, or transportation, among others. For couples who live together, joint consumption is therefore expected to account for a

significant share of total consumption.

Standard errors. The CEX is a stratified, multistage, clustered sample, and the BLS does not publish the design variables needed for an analytic variance. Thus, following the BLS recommendation, I re-run the *entire* estimator on each of the 44 replicate weight vectors and take the spread across replicates as the sampling variance. The resulting standard errors are 0.021 for α and 0.006 for γ.

D.3.2 κ/γ from the China Trade Shock in the PSID

As shown in Section IV, the participation rule satisfies

$$\log\big(1-\rho(r)\big) = \underbrace{\kappa \log \chi}_{\text{intercept}} - \underbrace{(\kappa/\gamma)}_{\text{slope}} \log(1+r). \tag{D.1}$$

This equation is a structural relation, not a regression. Cross-sectional variation in the relative wage r is not exogenous: r is an equilibrium object. In addition, I observe r only when the wife participates in the market. I therefore use the empirical design in Section II as the identification strategy for the participation block.

In particular, I identify κ/γ from the experiment in which the *husband's* industry is exposed. The wife is not directly displaced, so the shock affects her participation decision only through r, which is exactly the exclusion restriction required by equation (D.1). Because the equation is linear, the parameter is a Wald ratio:

$$\frac{\kappa}{\gamma} = -\frac{\Delta \log(1-\rho)}{\Delta \log(1+r)}. \tag{D.2}$$

To construct the numerator of equation (D.2), I use the estimates in Panel B of Table 3. Wives of exposed husbands raise labor-force participation by 4.2 p.p. from a baseline of 0.78, so ρ increases from 0.78 to 0.822. Then, the numerator is:

$$\Delta \log(1-\rho) = \log(1-0.822) - \log(1-0.78) = -0.212.$$

To construct the denominator, I estimate the static version of equation (5) on log(Hourly Wage) for each spouse. The sample consists of individuals who have at least one observation with positive wages before and during the shock. Table D.1 shows that when the husband is exposed to the shock, his hourly wage decreases by 16.4 log points while her wage decreases by 4.2 log points. Since the husband's wage falls by more than his wife's, the relative wage she faces rises. Using the estimates in Table D.1, r increases by 12.2 log points, which implies an increase of 13%. The r implied by the baseline mean is 0.88.

However, this number includes only employed individuals, while the numerator considers all individuals in the sample. Then, the correct value for r_0 is 0.607. Thus, the denominator of equation (D.2) is

$$\Delta \log(1+r) = \log(1+0.686) - \log(1+0.607) = 0.0477.$$

Substituting the numerator and the denominator into equation (D.2),

$$\frac{\kappa}{\gamma} = -\frac{-0.212}{+0.048} = 4.44.$$

With $\gamma = 0.4887$, I obtain $\kappa = \gamma \cdot \frac{\kappa}{\gamma} = 2.17$, which satisfies the integrability restriction $\kappa > \gamma$.

Table D.1: Husband's Exposure to the China Trade Shock and log(Hourly Wage)

	(1) Husband	(2) Wife
Exposed Husband=1	-0.078*	0.150***
	(0.042)	(0.052)
(Exposed Husband=1)x(Post=1)	-0.164*	-0.042
	(0.086)	(0.070)
Observations	12,023	9,979
R^2	0.35	0.31
Baseline Mean	3.31	2.91

Notes. Robust standard errors are two-way clustered at the husband's industry and individual levels. Given that I restrict the sample to individuals who have at least one observation with positive wages before and during the shock, the number of observations in this table does not match the number of observations in Fact 3. * for p<0.10, ** for p<0.05, *** for p<0.01.

D.3.3 χ as a Level Restriction

The participation rule in equation (D.1) is log-linear in $1-\rho$: its *slope* in $\log(1+r)$ identifies κ/γ, and its *intercept* identifies χ. With κ/γ estimated above and $\kappa = \gamma \cdot (\kappa/\gamma)$, χ is the single remaining free parameter, and I set it so that the model reproduces the observed *level* of married women's participation in 2000Q1, $\rho_0 = 0.621$. Married women's participation is therefore a *targeted* moment and is not reported as an untargeted success of the model.

I calibrate χ only on couples in which the husband is employed in a productive sector, because the model misses the participation of wives of non-employed husbands. In the data, wives of non-employed husbands work *less*, whereas the model predicts that they work *more*. Since ρ is monotone in r, and a non-employed husband's income is the benefit b_j, the ratio $r = w_f/b_j$ is mechanically the largest anywhere in the model, inducing wives to work. The data instead reflect strong positive assortative matching in the marriage market (Browning et al., 2014), which the model abstracts from. I denote the bundles in which the husband is employed by $\bar{\boldsymbol{o}}$.

Let $m_{\bar{\boldsymbol{o}},0}$ be the mass of couples in bundle $\bar{\boldsymbol{o}}$ at $t = 0$ and $r_{\bar{\boldsymbol{o}},0} = w^F_{\bar{\boldsymbol{o}},0}/w^M_{\bar{\boldsymbol{o}},0}$ its wage ratio. I then set χ such that the mass-weighted average participation rate the model generates equals the one in the data:

$$\sum_{\bar{\boldsymbol{o}}} m_{\bar{\boldsymbol{o}},0}\, \rho(r_{\bar{\boldsymbol{o}},0}; \chi) \;=\; \rho_0. \tag{D.3}$$

Since ρ is strictly decreasing in χ, equation (D.3) has a unique root, and it yields $\chi = 1.92$. This satisfies the support restriction $\chi \geq 1$ and leaves the participation margin interior in every cell.

E. Additional Results in the Quantitative Exercises

E.1 The Same Trade Shock in Every Economy

The household block changes who bears the China Trade Shock, but not the size of the shock where trade operates. In Table E.1, I compare CDP with the baseline economy, the all-singles economy, the all-singles economy without migration, and the three economies in which I shut down migration, partially shut down participation, or both. Manufacturing jobs per capita fall by between 0.197 and 0.206 p.p. in every economy I study, within 4.2% of the baseline. Welfare in the rest of the world does not move either: the mean gain of the other 36 regions is between 0.156% and 0.158% in every economy, and their gains correlate with those in CDP at 0.997. The sectors that drive the shock do not change either. I compute each sector's contribution to the change in employment at $t = 199$, as in Figures 2 and 5 of Caliendo et al. (2019). Across manufacturing industries, the contributions to the manufacturing employment decline in the baseline and the all-singles economy have a rank correlation of 0.94, and across non-manufacturing sectors, the contributions to the non-manufacturing employment gain have a rank correlation of 0.98. In both economies, computers, electronics, and furniture lead the decline, and other services and health lead the gain.

Table E.1: Same Shock in Every Economy

	CDP	Baseline	All-singles	All-singles, no migration	No migration	No participation	No migration & no participation
Manufacturing, $t = 199$							
Manuf. jobs per capita (p.p.)	−0.228	−0.206	−0.202	−0.200	−0.202	−0.200	−0.197
% change vs Baseline	+11.0	—	−1.6	−2.9	−1.7	−2.6	−4.2
Rest of the world (36 regions)							
Mean welfare gain (%)	0.158	0.156	0.158	0.157	0.156	0.156	0.156
Corr. with CDP	—	0.997	0.997	0.997	0.997	0.997	0.997

Notes. Responses at $t = 199$. Manufacturing jobs per capita is manufacturing employment divided by population, expressed in p.p.; % change vs. baseline is $100 \times (X/\text{baseline} - 1)$. CDP is the economy of Caliendo et al. (2019). Rest of the world: consumption-equivalent welfare gains of the 36 regions (35 countries and the rest of the world; China excluded), computed as in Caliendo et al. (2019) and summed over $t = 2, \ldots, 199$. The economies are defined in Table 7.

E.2 Effects of Shutting Down Migration

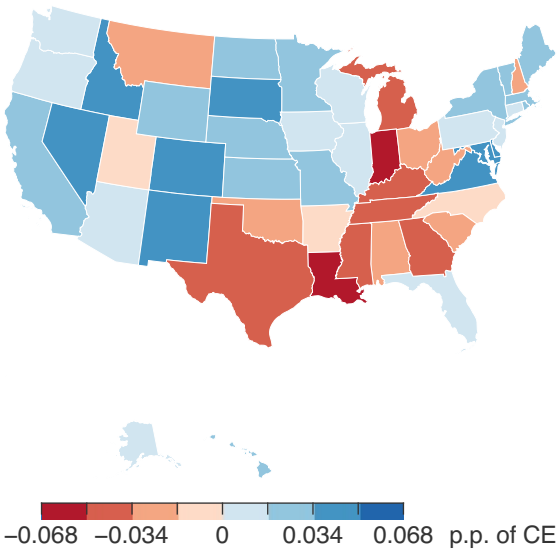
Shutting down migration barely moves aggregate welfare (+1.0%; Table 9), even though the employment response is 12% smaller. Moving is how people reach jobs, not how they

capture gains, and since sector switching is still allowed, a displaced worker still has their primary adjustment margin. This near-null, however, hides large spatial effects: welfare becomes 61% more spatially unequal (the coefficient of variation of state welfare gains rises from 0.28 to 0.45; Panel B of Table E.4), and the correlation between a state's welfare change in this exercise and its population response in the baseline is +0.98. Migration redistributes gains across states: the losers are the exposed states that were losing population, led by Indiana, Louisiana, and Kentucky; the winners are the destination states that were gaining population, which no longer have their gains diluted by arrivals (Figure E.1). Couples make this redistribution larger: without migration, the welfare gain of the people living in states losing people falls by 0.037 p.p. in the baseline economy and by 0.026 p.p. in the all-singles economy (Panel B of Table E.4).

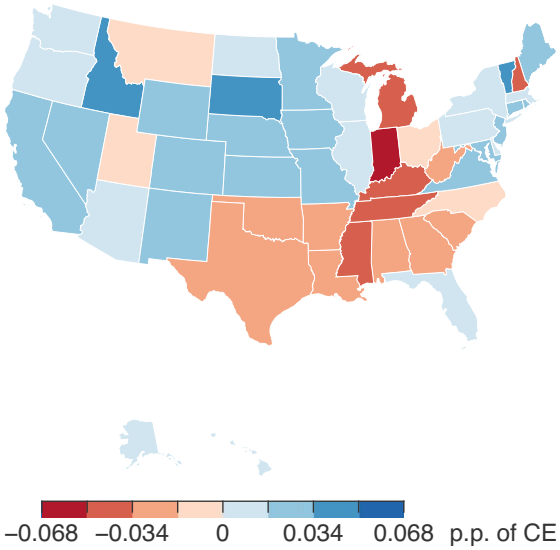
Unexposed couples benefit from shutting down migration (their gain rises by 2%; Table E.4) because they do not have to bear the costs of mobility decisions made by exposed households. When the exposed relocate, they disproportionately move to areas where the unexposed already live; their arrival increases local labor supply and pushes local wages down. With migration shut down, nobody arrives, and the unexposed keep a tighter local market.

Figure E.1: Change in State Welfare from Shutting Down Migration

(a) Baseline: no migration minus baseline



(b) All-singles: no migration minus all-singles



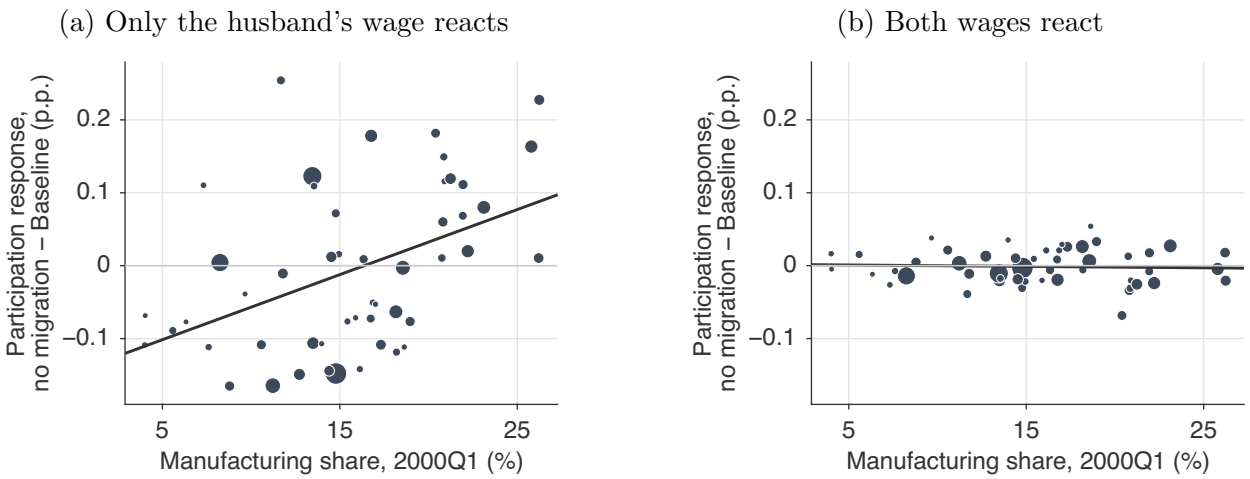
Notes. Change in state money-metric welfare, in p.p., for the people living in the state in 2000Q1, weighting every economy by the baseline money metrics. One common scale.

Participation does not compensate for the lack of spatial mobility: the participation

response of married women is 3.4% smaller in the counterfactual economy without migration (Panel C of Table E.3). Although this result may seem counterintuitive, it is driven by how couples decide on her participation. Given that ρ is homogeneous of degree zero in wages, it depends only on $r = w_f/w_m$ and not on the level, and this counterfactual moves both spouses' wages by nearly the same amount, leaving r intact to the third decimal. The participation response falls the least for couples in which the husband is exposed (-2.8%, -3.0%) and the most for unexposed couples (-3.5%). In states that lose people, shutting down migration reduces the shock's effect on men's employment rate from 0.10 to 0.03 p.p., whereas the effect on women's employment falls by a smaller amount, from 0.15 to 0.12 p.p.

Wives do not step in when migration is shut down, because general equilibrium undoes the partial-equilibrium incentive. Figure E.2 plots, state by state, the change in married women's participation response when migration is shut down. If only the husband's wage reacted, wives' participation would rise most where manufacturing was concentrated (correlation $+0.45$, panel (a)). Once both spouses' wages react, this relationship disappears (-0.05, panel (b)), and the change follows the population response to migration in the state ($+0.69$): participation falls most in the states people would have left, where both spouses' wages fall together.

Figure E.2: Why Wives Do Not Step In When Migration Is Shut Down

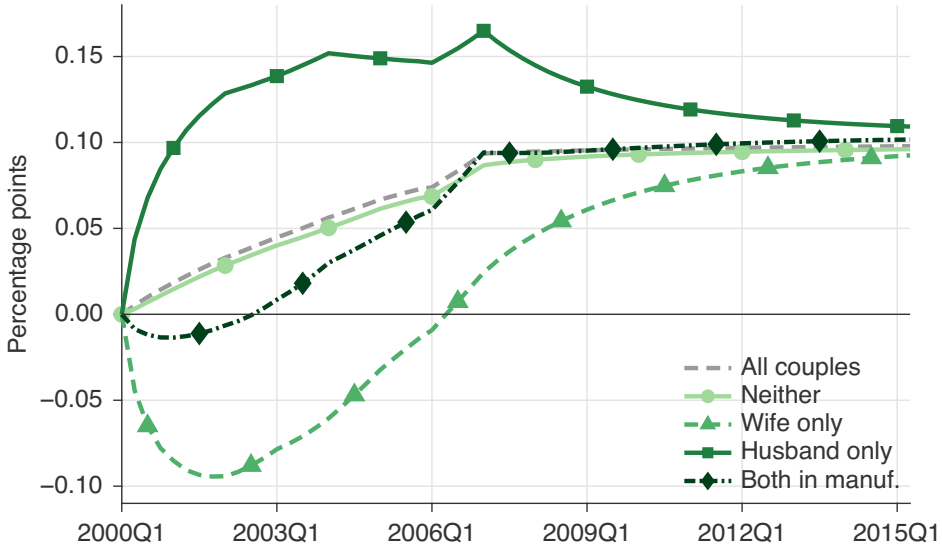


Notes. Each dot is a state, sized by its number of couples. The vertical axis is the change in married women's participation response at $t = 199$, in p.p., when migration is shut down, for the state's own married women. In panel (a), it is a partial-equilibrium calculation in which only the husband's wage takes its value from the economy without migration. The horizontal axis is the state's manufacturing share of employment in 2000Q1.

E.3 Participation as a Mitigation Mechanism

Married women's participation response differs by the couple's exposure in the short run, as discussed in Section V.2. Figure E.3 plots it over the first 62 quarters; its long-run values are in Table 6.

Figure E.3: Married Women's Short-Run Participation Response by Couples' Exposure



Notes. Baseline economy. Change in married women's quarterly participation rate (with-shock minus no-shock economy), in p.p., over the first 62 quarters (2000Q1–2015Q2), for the couples that started in each cell at $t = 1$. Couples are tagged by both spouses' sectors at $t = 1$; a non-working wife is tagged by her potential sector

Employment and migration when participation is partially shut down. Damping participation has the largest employment effect of the two margins (Table E.3): while the manufacturing decline is only 3% smaller in the counterfactual, the employment response is 24% smaller. If wives cannot respond to the shock by increasing their participation, the overall employment effect is lower. In addition, through general equilibrium effects, the employment response is lower for married men (Panel B of Table E.3).

When participation is damped, couples with an exposed husband move slightly more: the couples' migration response rises from 0.0141 to 0.0146 p.p. when only the husband is exposed and from 0.0078 to 0.0080 p.p. when both are, whereas all other couples move less (Panel D of Table E.3). Moving does not make up for the loss: shutting down migration costs couples in which only the husband is exposed 3.4% of their gain, whether or not wives' participation is damped (Panel B of Table 10). Conversely, wives do not work more when migration is shut down: their participation response is 3.4% smaller without migration, and an additional 2.7% smaller when participation is also damped. For couples with an exposed

husband, the two margins are therefore asymmetric substitutes: they move more when wives cannot respond, but wives do not work more when they cannot move.

Valuing wives' time at home. In the baseline economy, a non-working wife's time is valued as home production, through $B_{\boldsymbol{o},t}$ in the couple's payoff

$$\frac{w^{M}_{i,k_m,t}}{P_{i,t}} \cdot B_{\boldsymbol{o},t}$$

To measure how much of the gains from the added-worker effect this valuation absorbs, I recompute couples' welfare with every decision held fixed and two alternative values of $B_{\boldsymbol{o},t}$. Under *market income only*, a non-working wife's time is worth nothing,

$$B^{\text{Market}}_{\boldsymbol{o},t} = \mathcal{A}\,(1 + \rho_{\boldsymbol{o},t}\, r_{\boldsymbol{o},t}).$$

Under *valued like singles*, a non-working wife receives the non-employment benefit, as a non-employed single does,

$$B^{\text{Singles}}_{\boldsymbol{o},t} = \mathcal{A}\left[1 + \rho_{\boldsymbol{o},t}\, r_{\boldsymbol{o},t} + (1 - \rho_{\boldsymbol{o},t})\,\frac{b_i}{w^{M}_{i,k_m,t}}\right],$$

with b_i held fixed in real terms.

Valuing wives' time at home like singles' raises the baseline economy's aggregate gain, such that it would be 8.5% higher than the all-singles economy (Table 9). The valuation also reorders couples: couples in which only the wife is exposed gain 0.022% rather than 0.161%, and those in which only the husband is exposed gain 0.320% rather than 0.140%. When the wife's wage falls, home production insures the couple; when the husband's falls, the couple gives up home production as she enters the market.

The two margins are weak complements. Table E.2 compares the welfare change from shutting down both margins with the sum of shutting down each one alone. For every group, shutting down both costs slightly less than the sum of the two, by between 0.0 and 0.4 percentage points; participation is worth 4.6% of the baseline gain with migration and 4.4% without it.

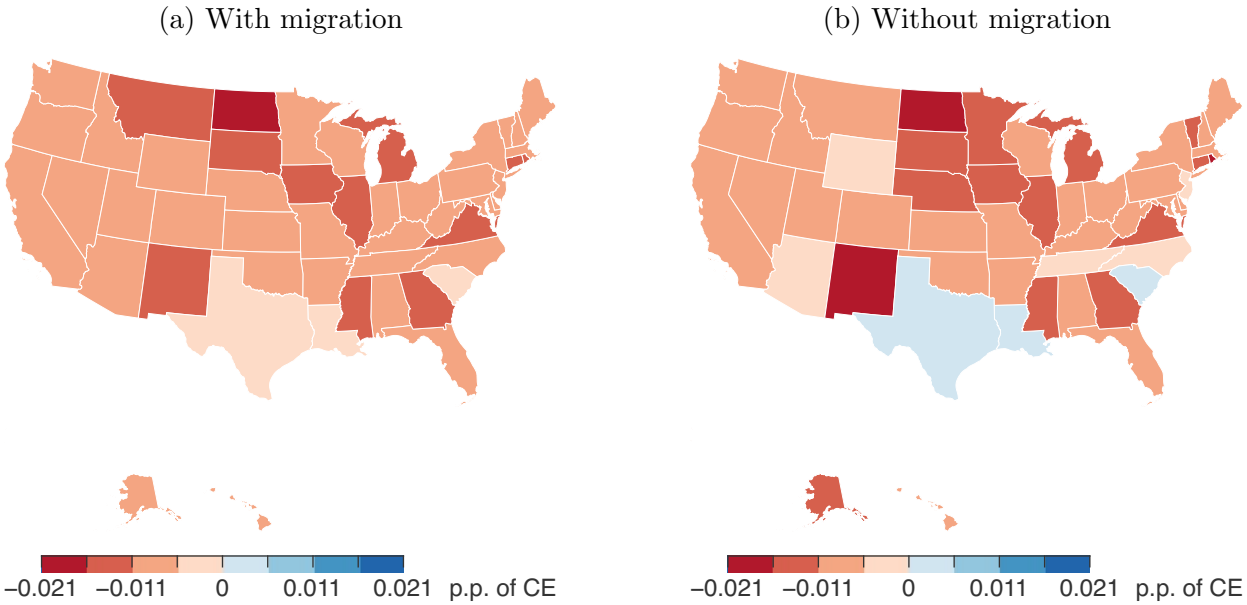
Damping participation lowers welfare in almost every state, with or without migration. Figure E.4 maps the change in state welfare from damping participation, with migration (no participation minus baseline) and without it (no migration & no participation minus no migration), on a common scale. With migration, all 50 states lose; without migration, 47 do, and Louisiana, South Carolina, and Texas gain slightly. The two maps correlate at 0.98 across states.

Table E.2: Welfare Effects of Shutting Down Migration and Participation

	Baseline (% CE)	No migration (%)	No participation (%)	No migration & no participation (%)	Sum of the two (%)	Interaction (p.p.)
Aggregate	0.175	+1.0	−4.6	−3.4	−3.7	+0.2
Single men	0.157	+2.9	−2.4	+0.7	+0.6	+0.1
Single women	0.234	+0.9	+8.8	+9.7	+9.7	+0.0
Couples	0.170	+0.7	−7.2	−6.2	−6.5	+0.3
Neither	0.180	+1.6	−6.9	−5.0	−5.2	+0.3
Wife only	0.161	+0.7	−5.0	−4.0	−4.3	+0.3
Husband only	0.140	−3.4	−10.2	−13.2	−13.6	+0.3
Both in manuf.	0.125	−2.7	−7.6	−9.9	−10.3	+0.4

Notes. The first column reports the baseline economy's money-metric consumption-equivalent gain (%); the next three columns report 100 × (economy/baseline − 1), weighting every economy by the baseline money metric at 2000Q1. The sum of the two is the sum of the no migration and no participation columns, and the interaction is the no migration & no participation column minus that sum, in points of the percentage change.

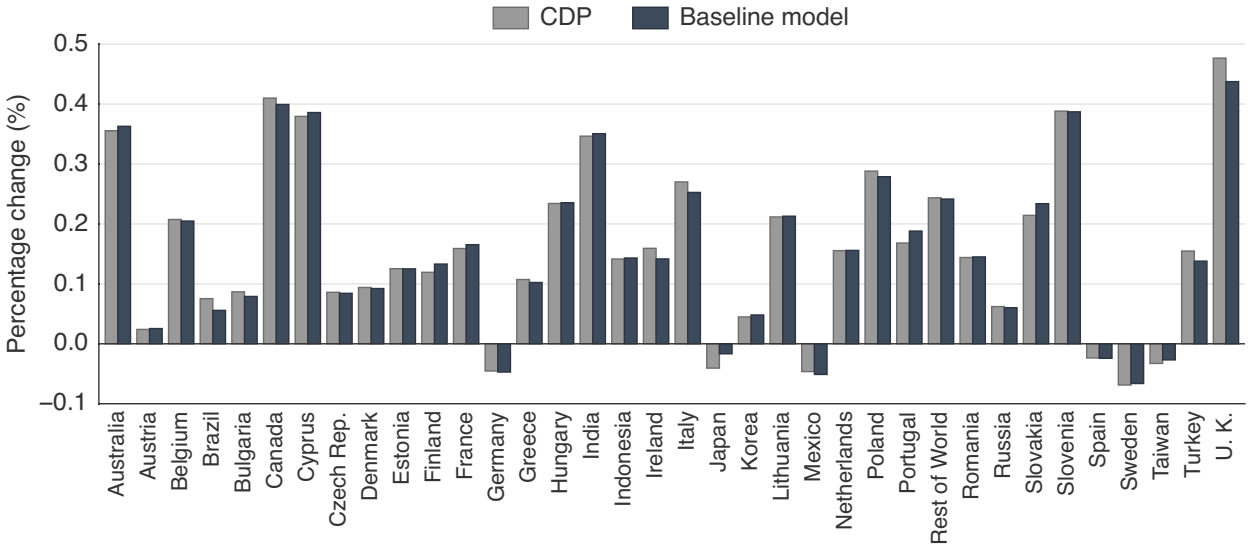
Figure E.4: Change in State Welfare from Damping Participation



Notes. Change in state money-metric welfare, in p.p., for the people living in the state in 2000Q1, weighting every economy by the baseline money metrics. Panel (a) is no participation minus baseline; panel (b) is no migration & no participation minus no migration. One common scale.

E.4 Additional Figures

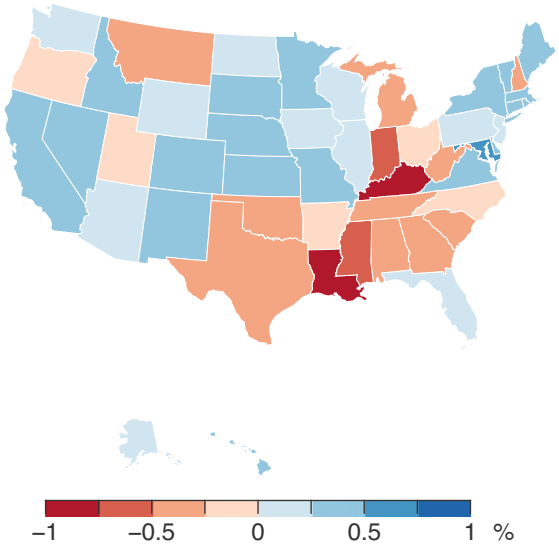
Figure E.5: Comparison of Welfare Effects in the Rest of the World



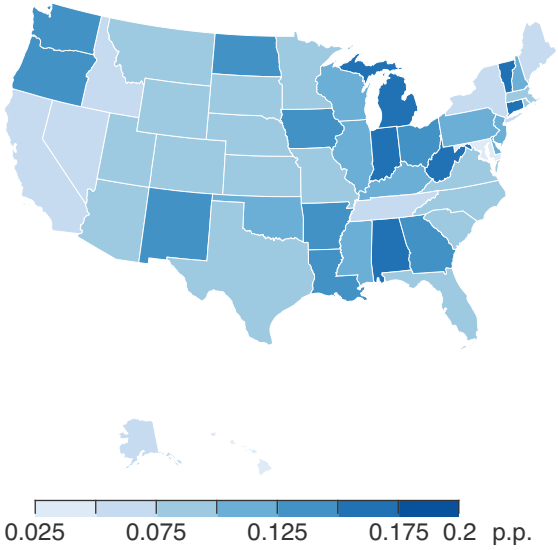
Notes. Consumption-equivalent welfare gains of the 36 regions (35 countries and the rest of the world; China excluded), calculated in both models as in Caliendo et al. (2019) and summed over $t = 2, \ldots, 199$.

Figure E.6: The China Trade Shock across States

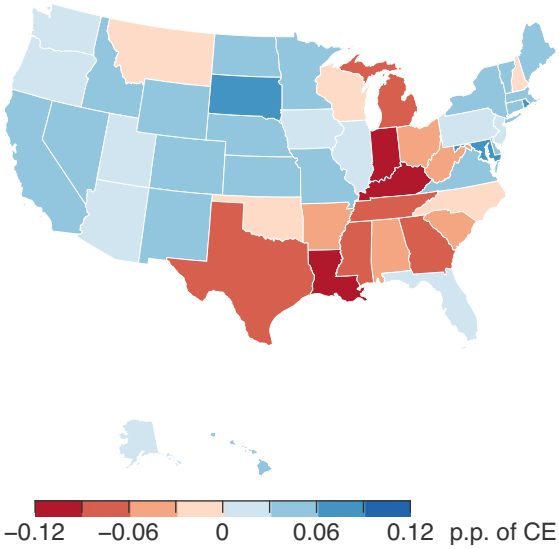
(a) Population change (%)



(b) Married women's participation (p.p.)



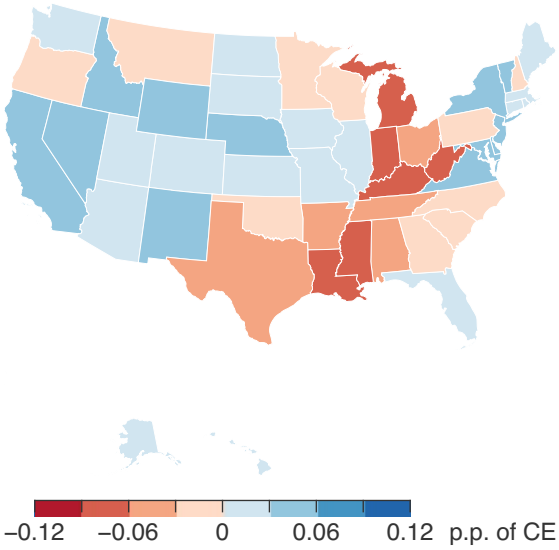
(c) State welfare minus national (p.p.)



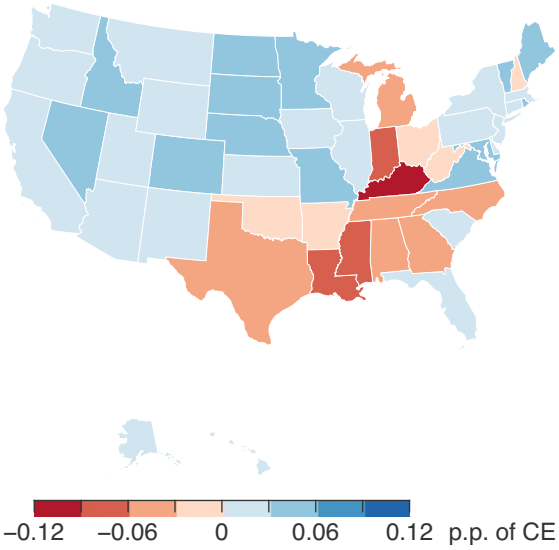
Notes. Panels (a) to (c) are long-run responses ($t = 199$): the change in the state's population as a percentage of its no-shock level, which sums to zero across states because the national population is fixed; the change in married women's participation rate, in p.p., averaged over the couples living in the state; and the state's money-metric welfare gain minus the national gain (0.175), in p.p., for the people living in the state in 2000Q1.

Figure E.7: Distribution of Welfare Effects by Household Type

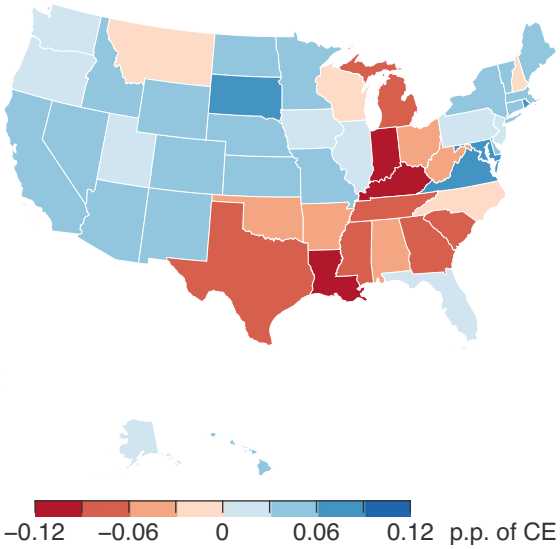
(a) Single men



(b) Single women

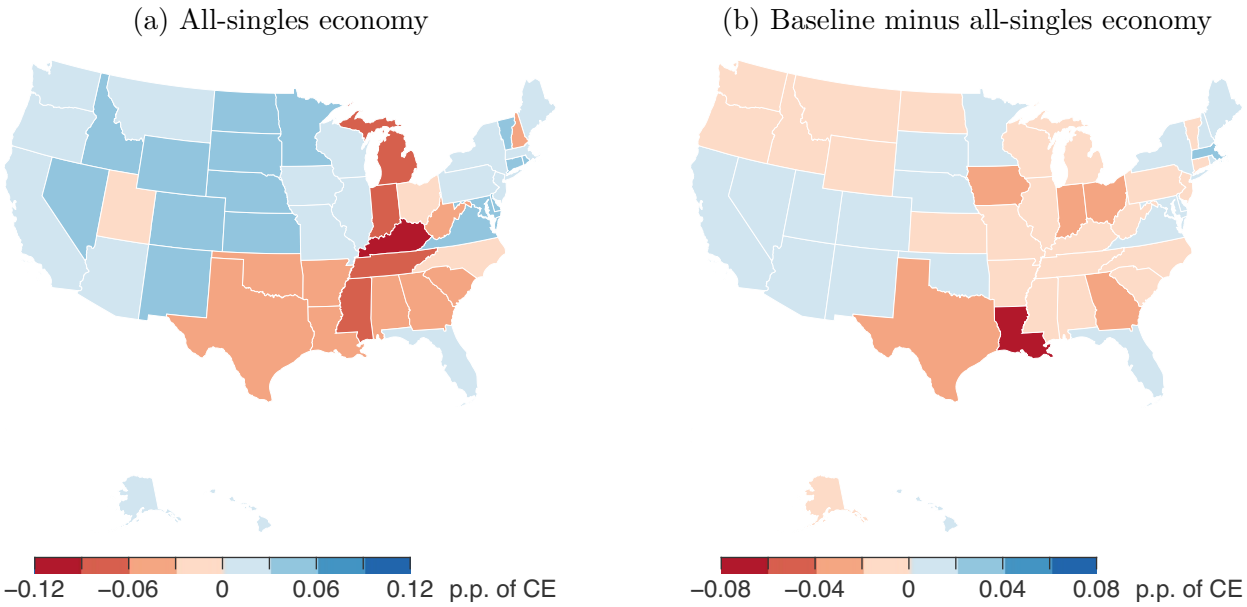


(c) Couples



Notes. State welfare minus the group's own national welfare, money-metric consumption-equivalent, in p.p., for the people living in the state in 2000Q1. The population-weighted coefficient of variation of state welfare gains is 0.23 for single men, 0.16 for single women, and 0.31 for couples.

Figure E.8: Distribution of Welfare Effects in the All-Singles Economy



Notes. Money-metric consumption-equivalent welfare gains, in p.p., for the same people located at their 2000Q1 state, weighting both economies by the baseline money metrics. Panel (a) is state welfare minus the national gain of the all-singles economy (0.178), on the same scale as Figure E.6c. Panel (b) is the baseline economy's state welfare minus the all-singles economy's.

E.5 Additional Tables

Table E.3: Responses to the China Trade Shock across Economies: All Groups

	Baseline	All-singles	All-singles, no migration	No migration	No participation	No migration & no participation
Panel A. Manufacturing						
Manuf. share of employment (p.p.)	−0.293	−0.291	−0.286	−0.287	−0.285	−0.279
Panel B. Employment rate (p.p.)						
Aggregate	0.112	0.132	0.119	0.099	0.085	0.073
Men	0.099	0.092	0.074	0.079	0.090	0.072
Women	0.124	0.170	0.161	0.117	0.080	0.075
Single men	0.106	0.092	0.075	0.082	0.104	0.081
Single women	0.158	0.170	0.161	0.143	0.171	0.157
Married men	0.096	0.093	0.074	0.079	0.086	0.069
Married women	0.111	0.170	0.161	0.107	0.043	0.041
Panel C. Married women's participation (p.p.)						
All couples	0.101			0.097	0.041	0.040
Neither	0.100			0.096	0.041	0.040
Wife only	0.101			0.097	0.041	0.040
Husband only	0.105			0.102	0.042	0.042
Both in manuf.	0.103			0.100	0.042	0.041
Panel D. Migration						
Migration response of the exposed (p.p.)	+0.0166	+0.0194	—	—	+0.0170	—
Share living in a different state because of the shock (%)						
Aggregate	0.150	0.130	—	—	0.148	—
Single men	0.151	0.150	—	—	0.150	—
Single women	0.127	0.116	—	—	0.123	—
Married men	0.159	0.151	—	—	0.158	—
Married women	0.159	0.116	—	—	0.158	—
Couples' migration response (p.p.)						
All couples	−0.0066			—	−0.0071	—
Neither	−0.0109			—	−0.0116	—
Wife only	−0.0062			—	−0.0066	—
Husband only	+0.0141			—	+0.0146	—
Both in manuf.	+0.0078			—	+0.0080	—
Panel E. Wages and labor supply						
Female real-wage response (%)	0.405	0.362	0.365	0.405	0.442	0.442
Male real-wage response (%)	0.236	0.258	0.265	0.243	0.230	0.237
Relative wage r response (%)	0.169	0.104	0.100	0.162	0.212	0.205
Female labor-supply elasticity	0.452	0.682	0.644	0.426	0.276	0.258

Notes. Responses at $t = 199$, in levels. Panel A is in p.p. of employment and Panel B in p.p. of population. Panel C is the change in married women's quarterly participation rate, in p.p., by the couple's exposure at $t = 1$. In Panel D, the migration response of the exposed is the difference between the shares of manufacturing and non-manufacturing workers who have changed state at least once by $t = 199$, in p.p., counting each person once. Panel E reports employment-weighted log changes in real wages (×100); the female labor-supply elasticity is the log female employment response over the log female real-wage response.

Table E.4: Welfare Effects of the China Trade Shock across Economies: All Groups

	Baseline economy		Counterfactual economies				
	Home production	Wives' time valued as singles'	All-singles	All-singles, no migration	No migration	No participation	No migration & no participation
Panel A. Welfare gain (% CE)							
Aggregate	0.175	0.194	0.178	0.180	0.176	0.167	0.169
Single men	0.157	0.157	0.166	0.170	0.162	0.153	0.158
Single women	0.234	0.234	0.194	0.195	0.236	0.255	0.257
Couples	0.170	0.193	0.178	0.180	0.171	0.157	0.159
Married men	0.170	0.193	0.164	0.167	0.171	0.157	0.159
Married women	0.170	0.193	0.193	0.194	0.171	0.157	0.159
Couples, by exposure							
Neither	0.180	0.180	0.184	0.186	0.182	0.167	0.171
Wife only	0.161	0.022	0.174	0.175	0.162	0.153	0.155
Husband only	0.140	0.320	0.161	0.159	0.135	0.126	0.121
Both in manuf.	0.125	0.136	0.155	0.153	0.122	0.115	0.113
Panel B. Spatial							
CV of state welfare	0.278		0.204	0.327	0.448	0.289	0.462

Notes. Money-metric consumption-equivalent gains (%), summed over $t = 2, \ldots, 199$. Each row represents the same people across all economies, and each economy weights each person by their baseline money metric in 2000Q1. In the economies with couples, married men and women carry the couple's welfare; in the all-singles economies, they are the same people as singles, and couples' rows average the two former spouses. The first two columns are the baseline economy: the first values a non-working wife's time as home production, as in the model, and the second values it as a single, at the non-employment benefit b_j. In Panel B, CV is the population-weighted coefficient of variation of state welfare gains. Weighting the all-singles economy by its own wages instead, its aggregate gain is 0.177.